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Winden

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(45) **Date of Patent:** ***Aug. 26, 2025**

(54) **HAND TRUCK AUTOMATED LOAD
UNLOAD**

(56) **References Cited**

U.S. PATENT DOCUMENTS

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(72) Inventor: **Steve Winden**, Mesa, AZ (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 870 days.

This patent is subject to a terminal disclaimer.

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(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation-in-part of application No. 16/727,981, filed on Dec. 27, 2019, now Pat. No. 11,305,802.

(60) Provisional application No. 62/848,021, filed on May 15, 2019.

(51) **Int. Cl.**
B62B 5/00 (2006.01)
B60R 9/06 (2006.01)

(52) **U.S. Cl.**
CPC **B62B 5/0003** (2013.01); **B60R 9/06** (2013.01)

(58) **Field of Classification Search**
CPC B62B 5/0003; B62B 1/12; B62B 1/10; B60D 1/065; B60D 1/52; B60D 2001/005; B60R 9/06; B60R 2011/004; B60R 2011/0078; B60R 11/00
See application file for complete search history.

5,181,822 A	1/1993	Allsop et al.
5,845,831 A	12/1998	Nusbaum et al.
6,155,588 A	12/2000	Maxey
6,237,823 B1	5/2001	Stewart et al.
6,314,891 B1	11/2001	Larson
7,641,235 B1	1/2010	Anduss
8,123,238 B1	2/2012	Burgess
8,210,559 B2	7/2012	Russell
8,480,149 B2	7/2013	Durand
8,505,951 B2	8/2013	Bohse
9,096,160 B2	8/2015	Le Anna
9,216,698 B2	12/2015	Rhodes
10,059,276 B2	8/2018	Phillips
10,112,523 B2	10/2018	McConn et al.
10,189,419 B2	1/2019	Billard
10,315,474 B2	6/2019	Reynolds

(Continued)

Primary Examiner — Valentin Neacsu

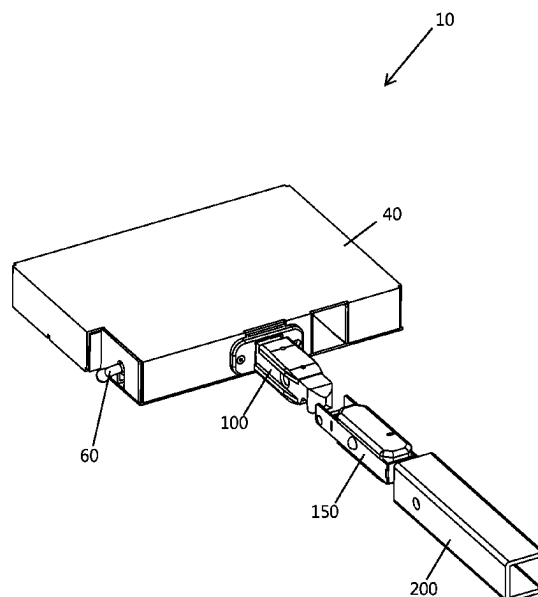
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(57) **ABSTRACT**

An apparatus for interconnecting a hand truck or dolly to a vehicle receiver or hitch while hauling a heavy load. The apparatus includes a base, an extendable lever arm and a separable intermediary glide. The base couples with the hand truck, the lever arm is extendable from the base, and the intermediary glide interconnects the lever arm to a receiver attached to a vehicle. In accordance with certain aspects of embodiments of the invention the hand truck and glide may be automated to automatically stabilize the hand truck, lift it into alignment with the receiver, and then lock the lever arm into place relative to the receiver and vehicle. Further, the system in accordance with aspects of the invention may automatically unload the hand truck from engagement with the vehicle.

20 Claims, 57 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

11,305,802	B2 *	4/2022	Winden	B60D 1/52
2004/0173654	A1	9/2004	McAlister	
2008/0101899	A1	5/2008	Slonecker	
2010/0066069	A1	3/2010	Bradshaw	
2019/0168555	A1	6/2019	Axelsson, Jr.	

* cited by examiner

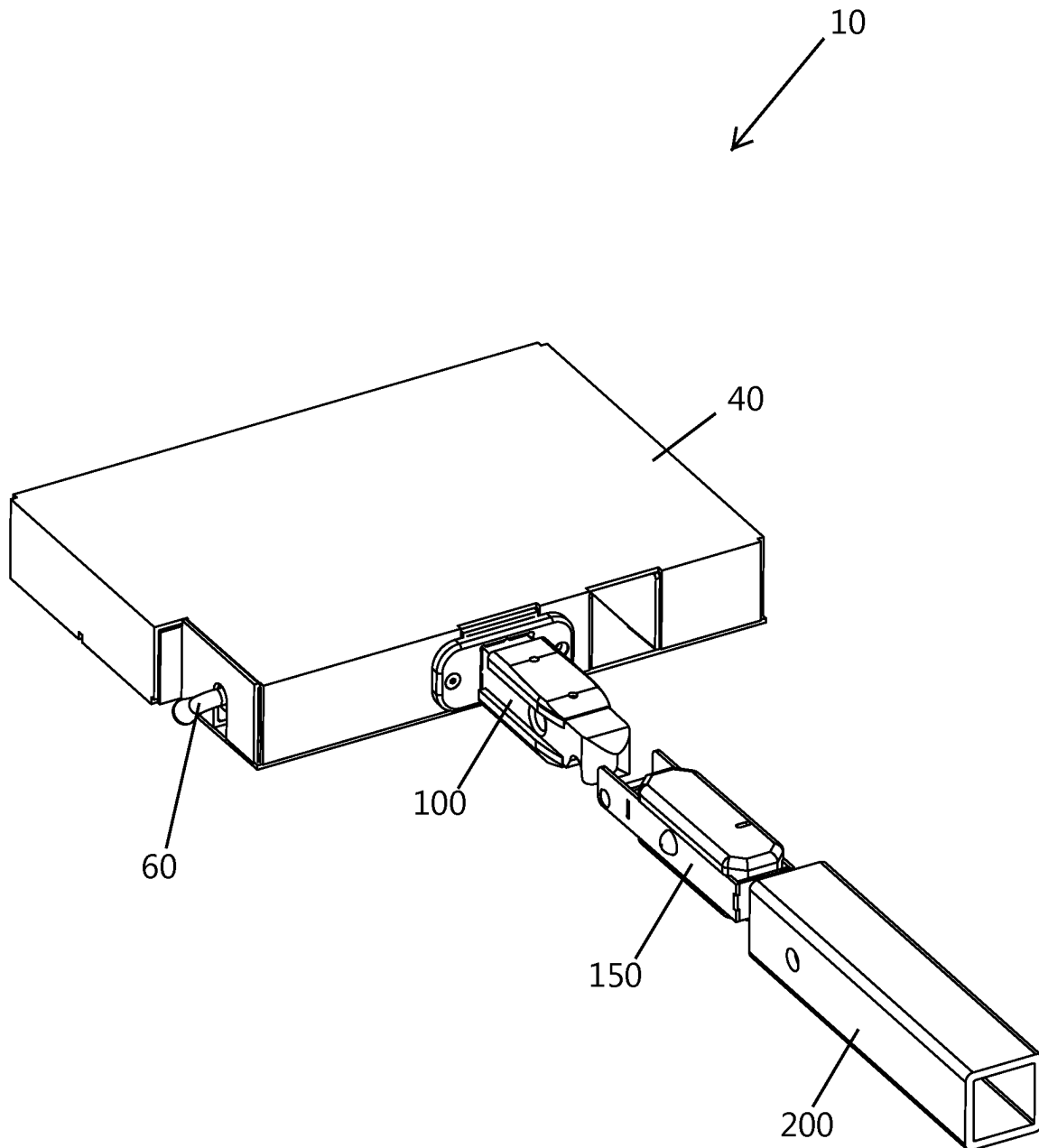


FIG. 1

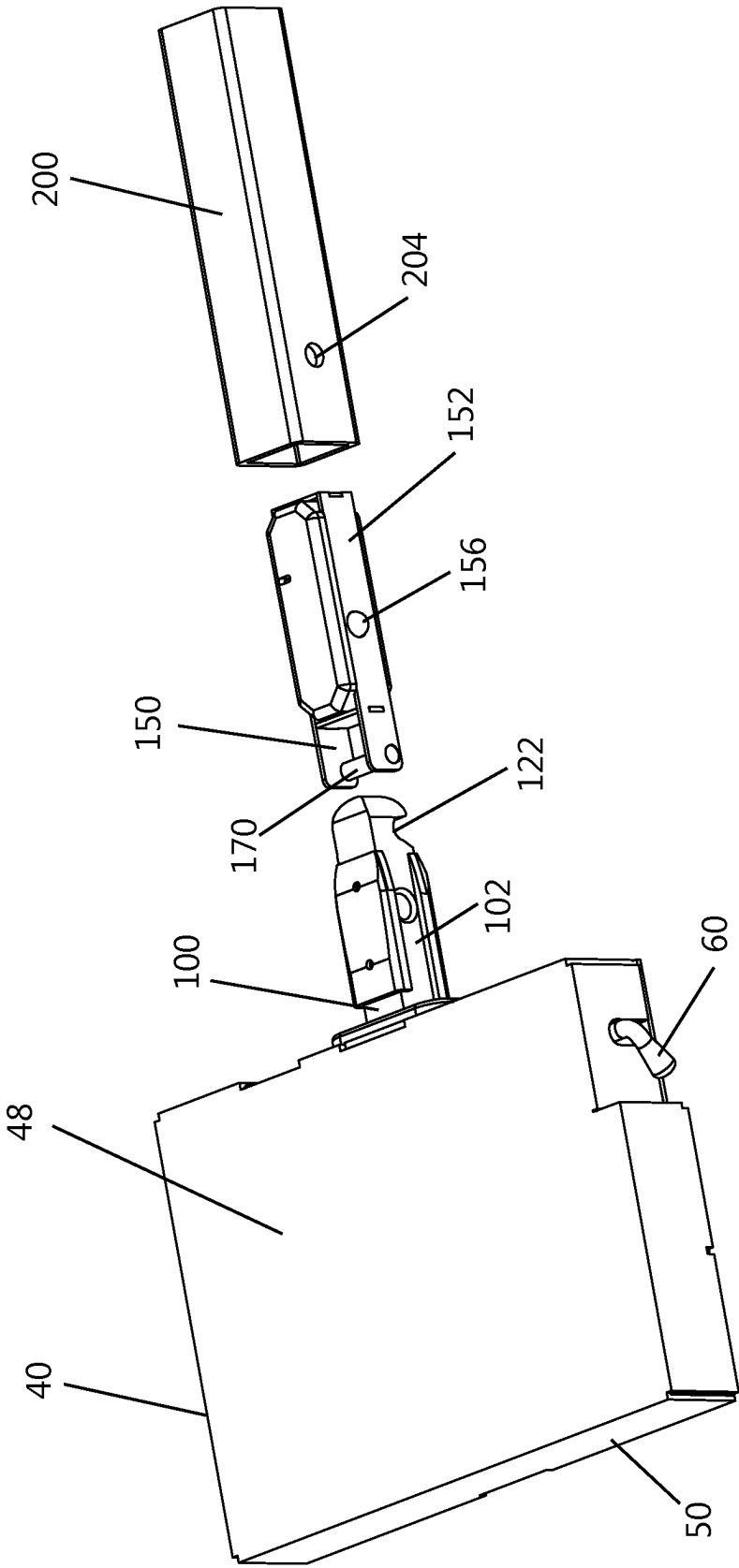
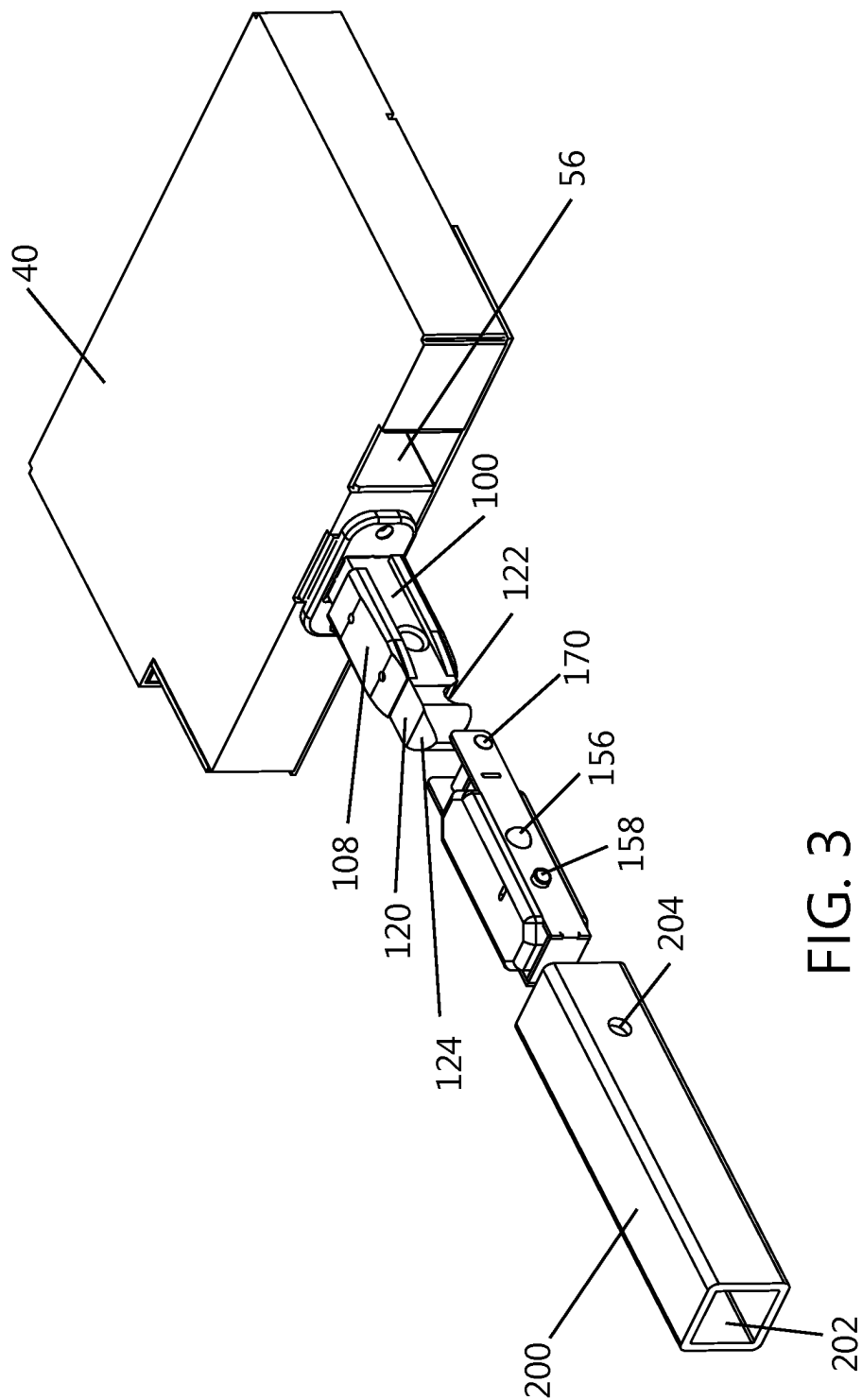


FIG. 2



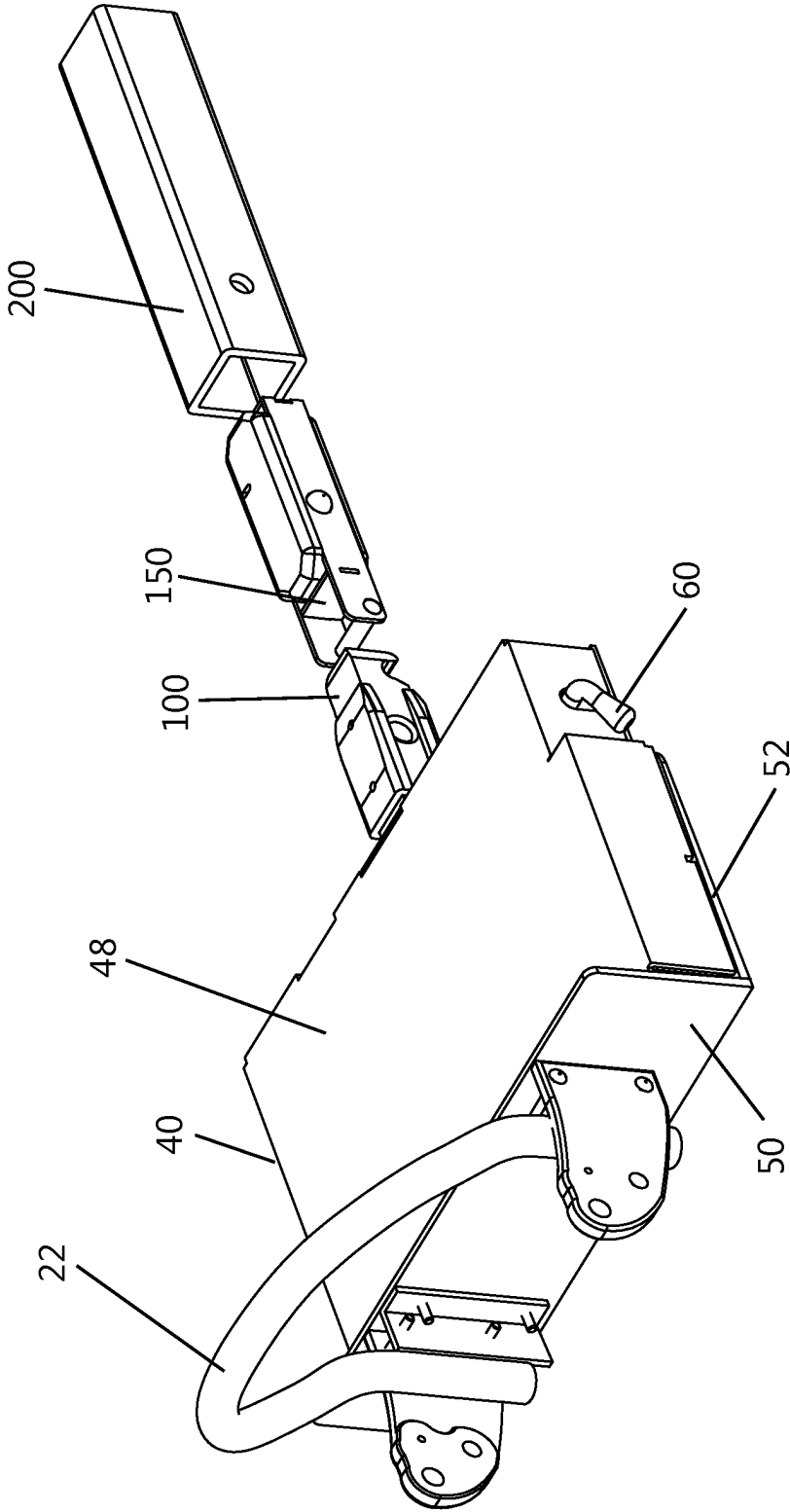


FIG. 4

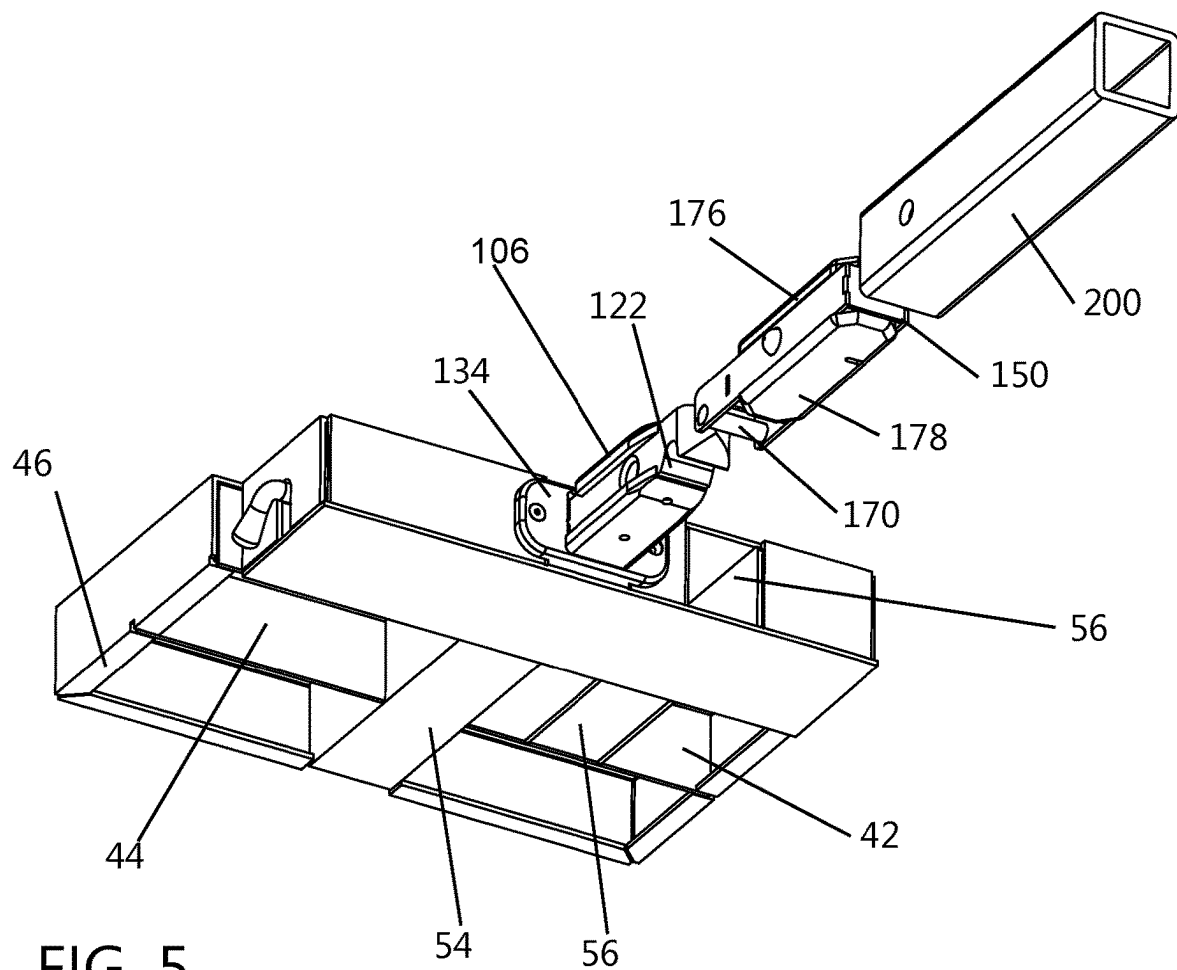


FIG. 5

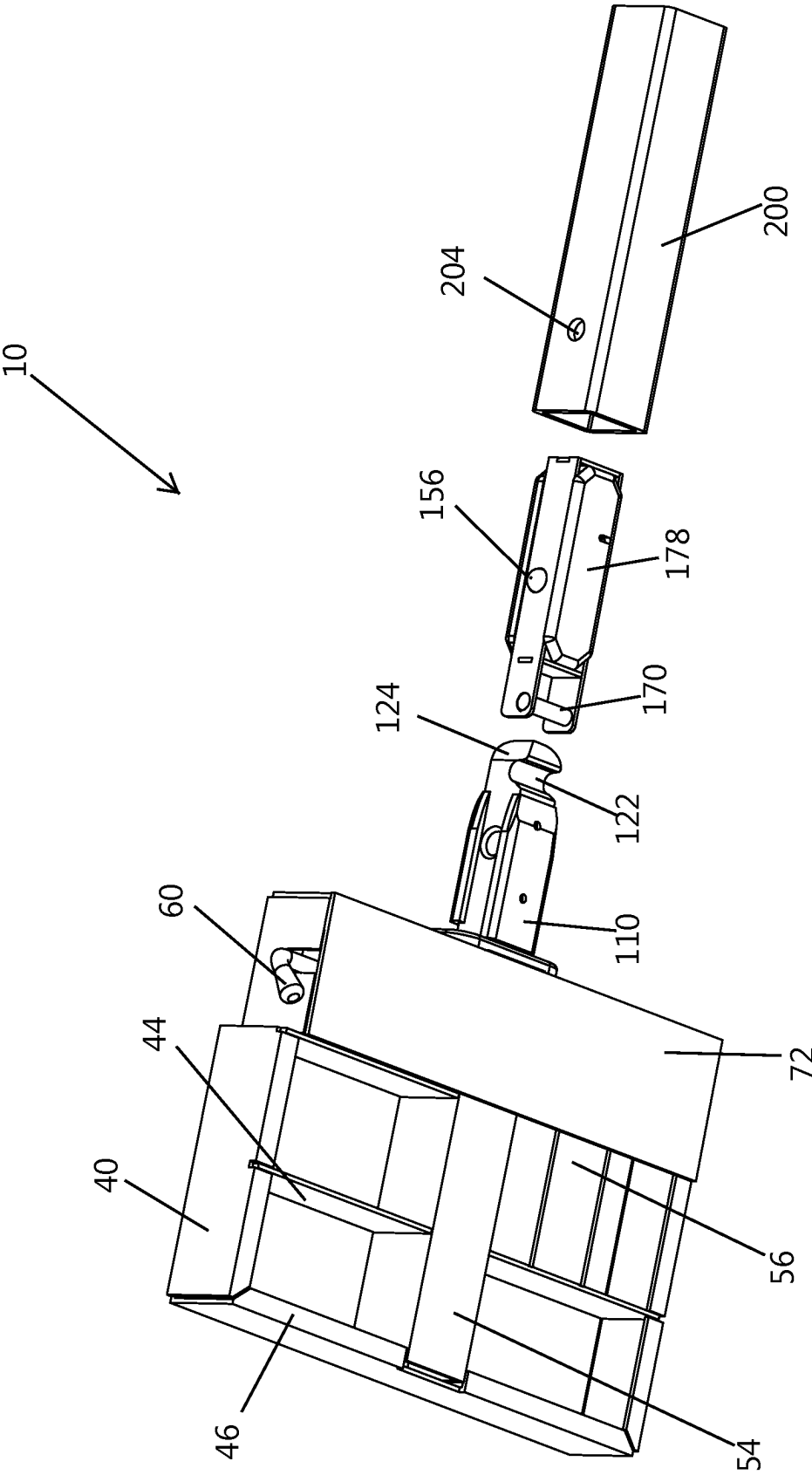
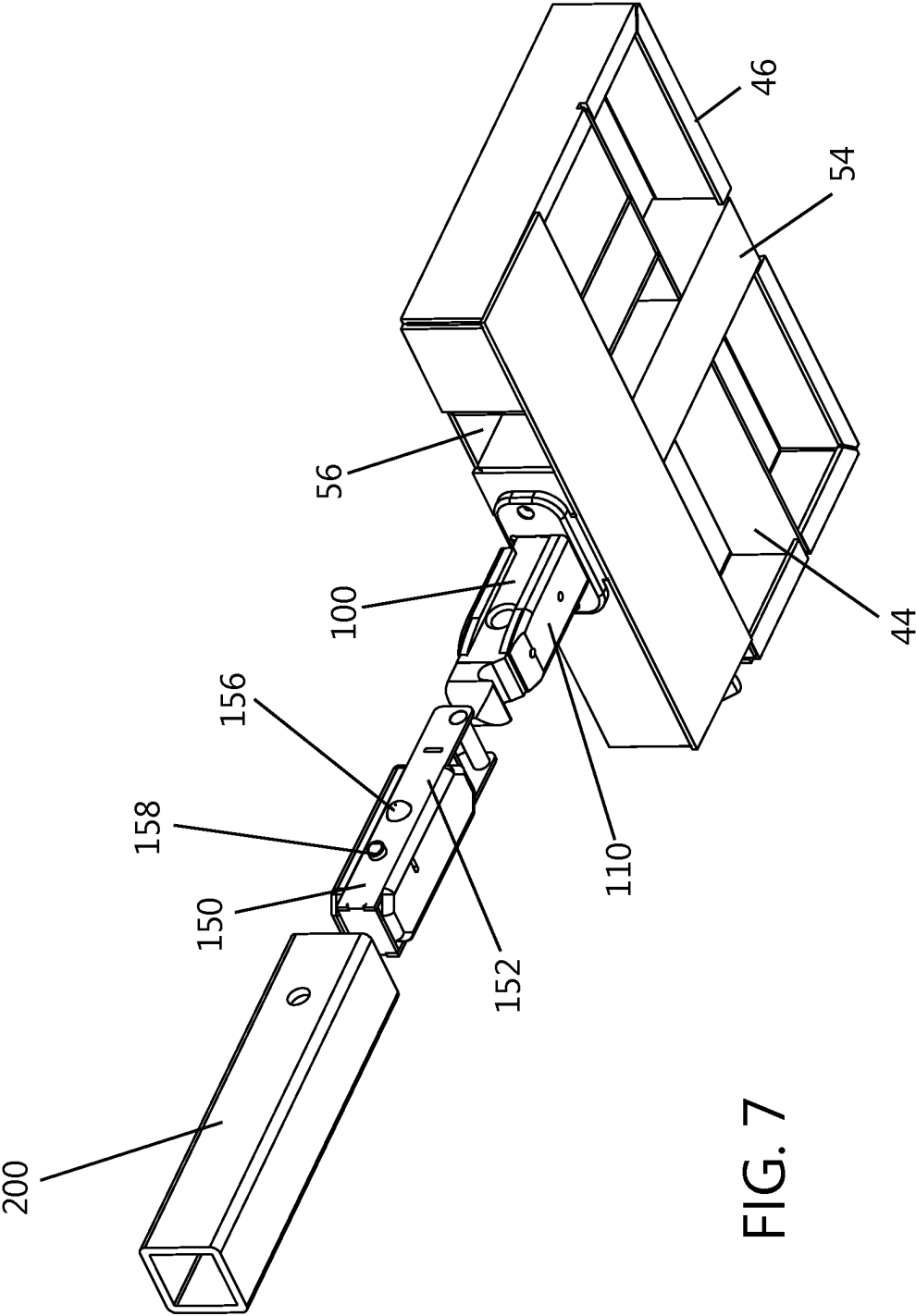


FIG. 6



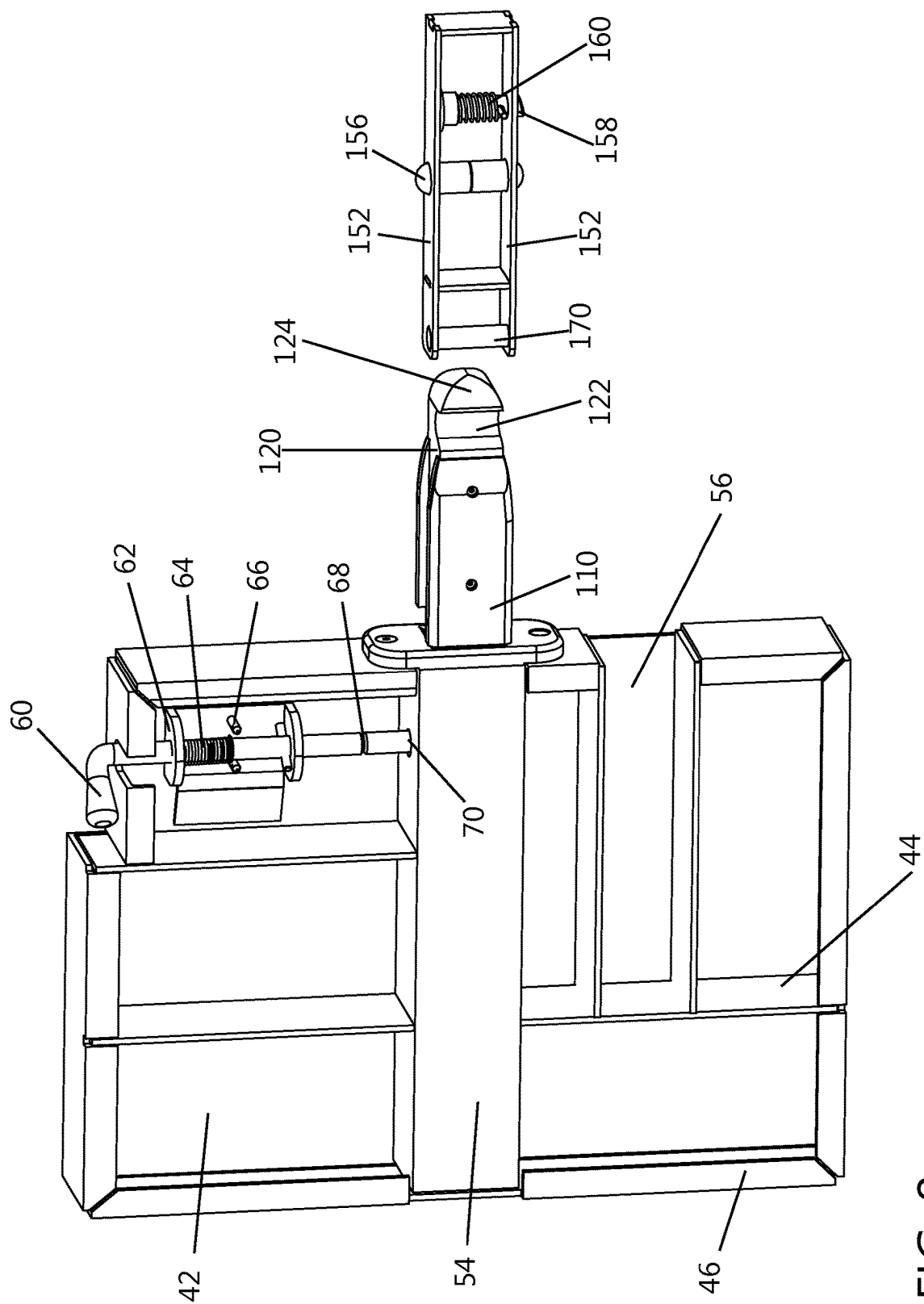


FIG. 8

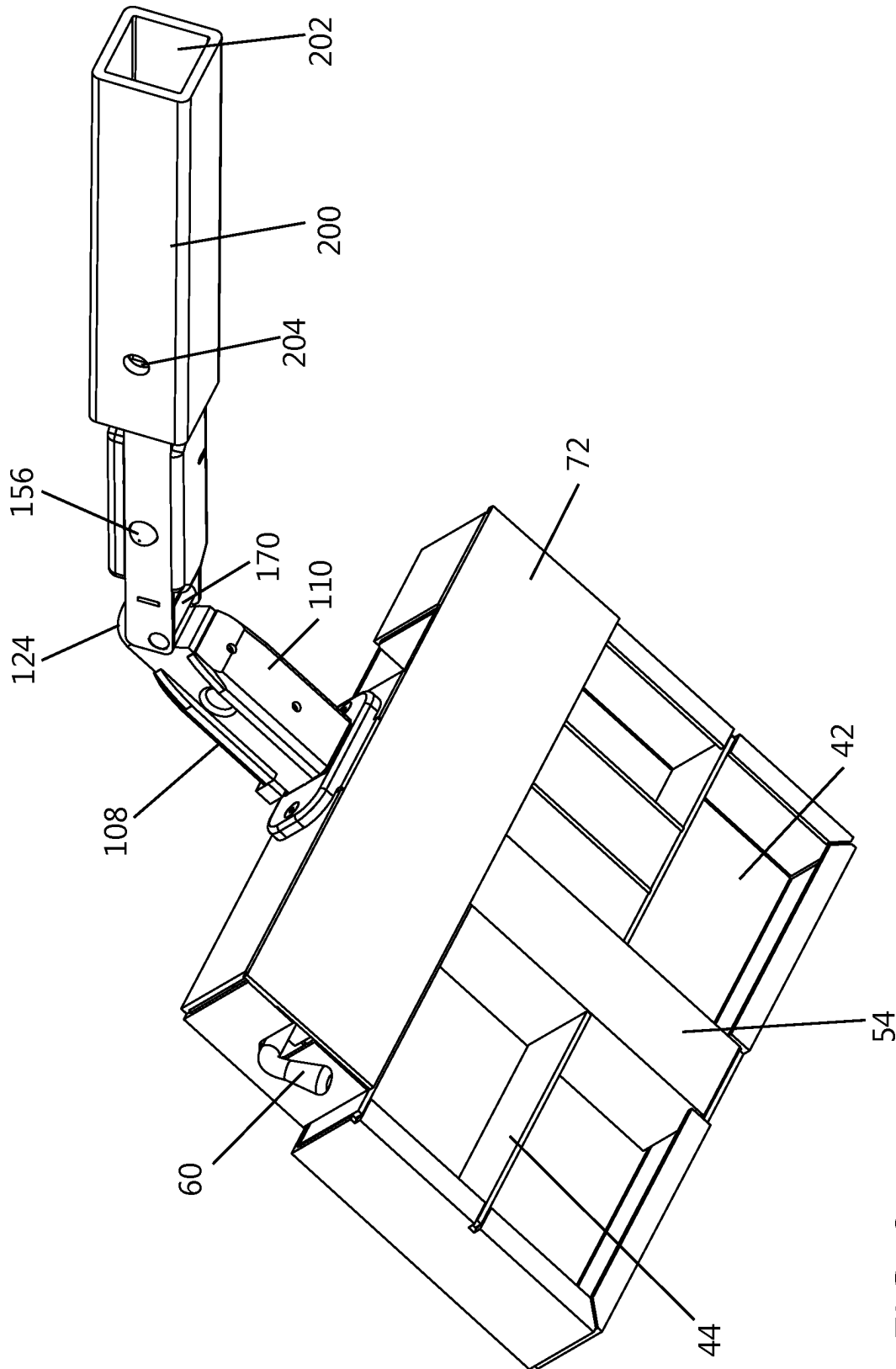
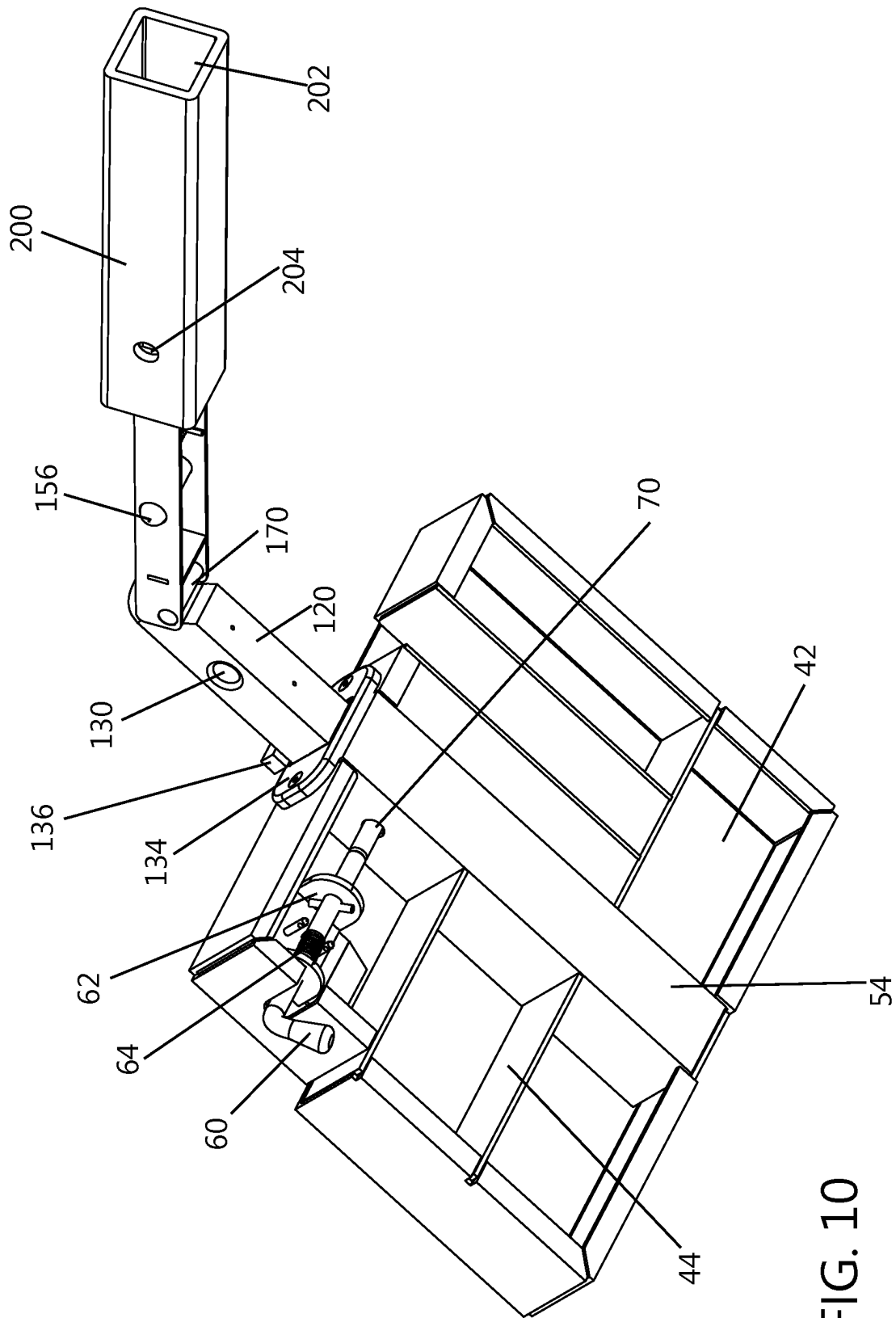


FIG. 9



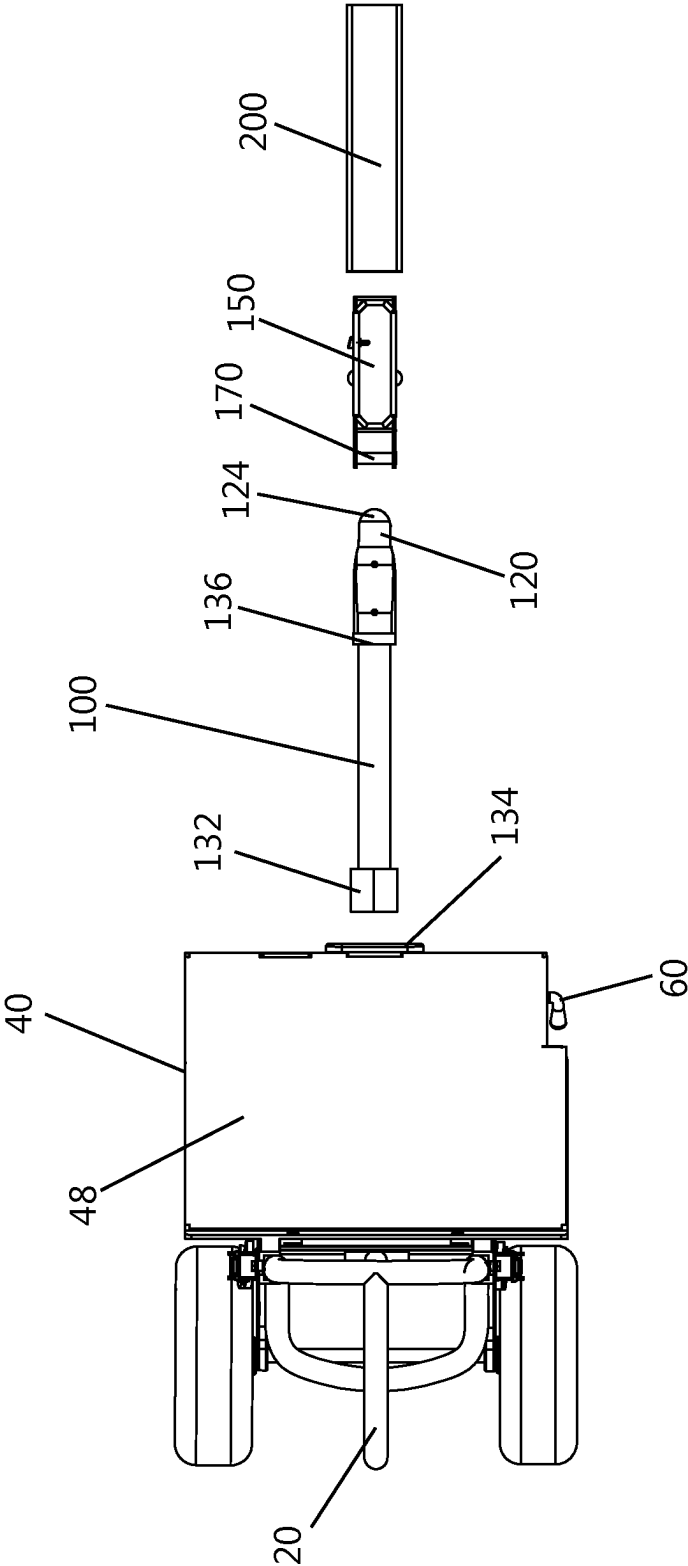


FIG. 11

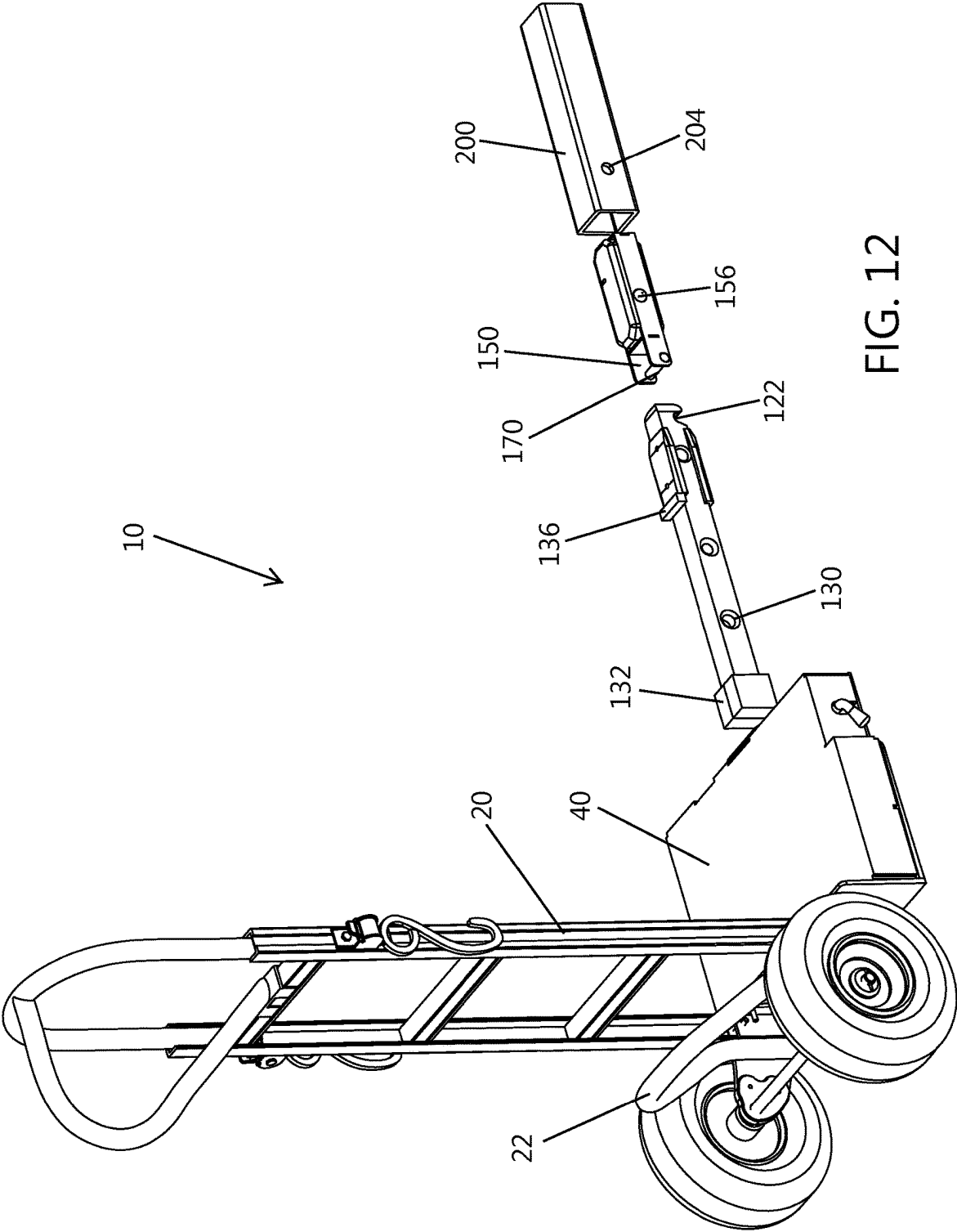
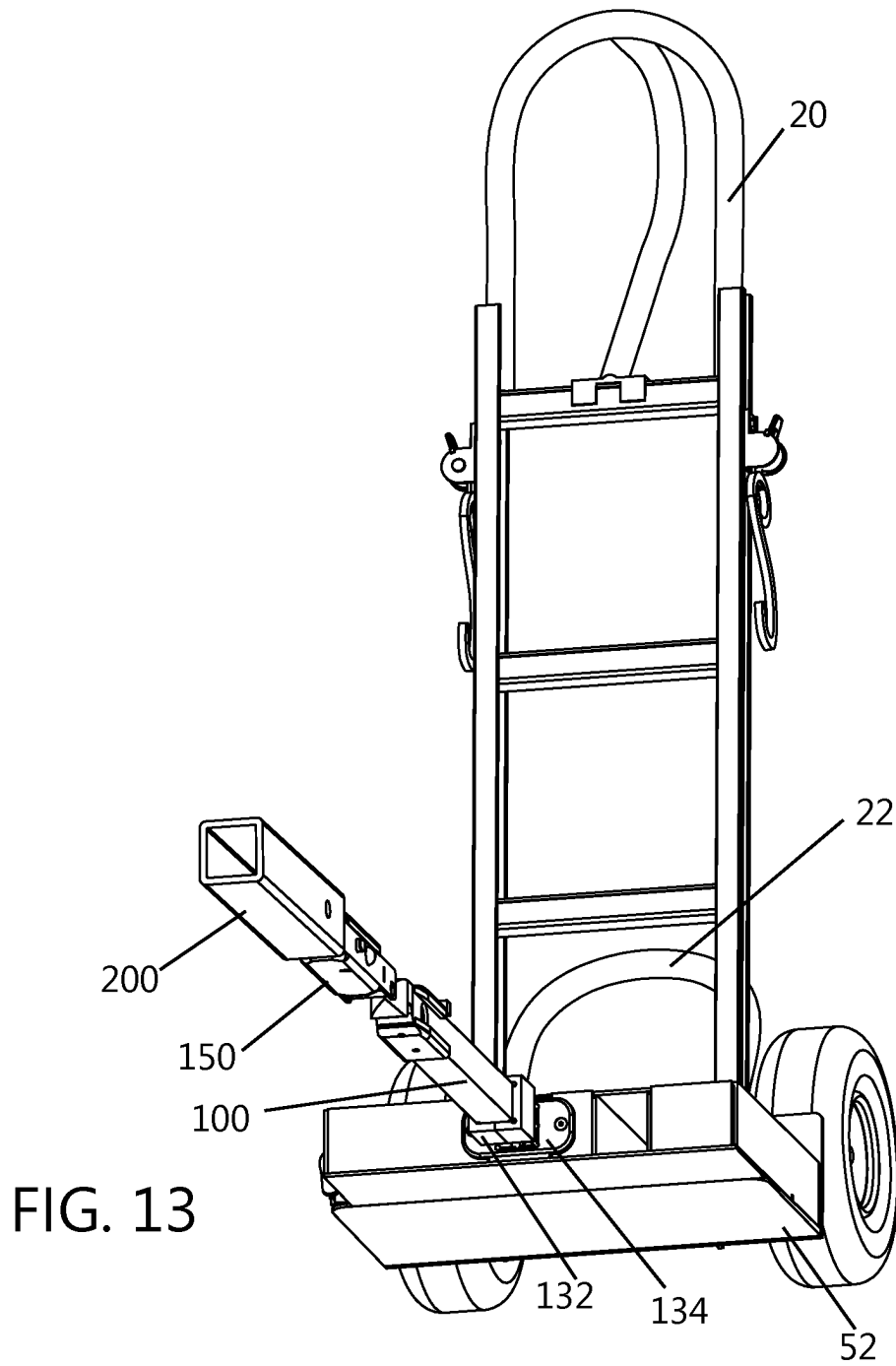
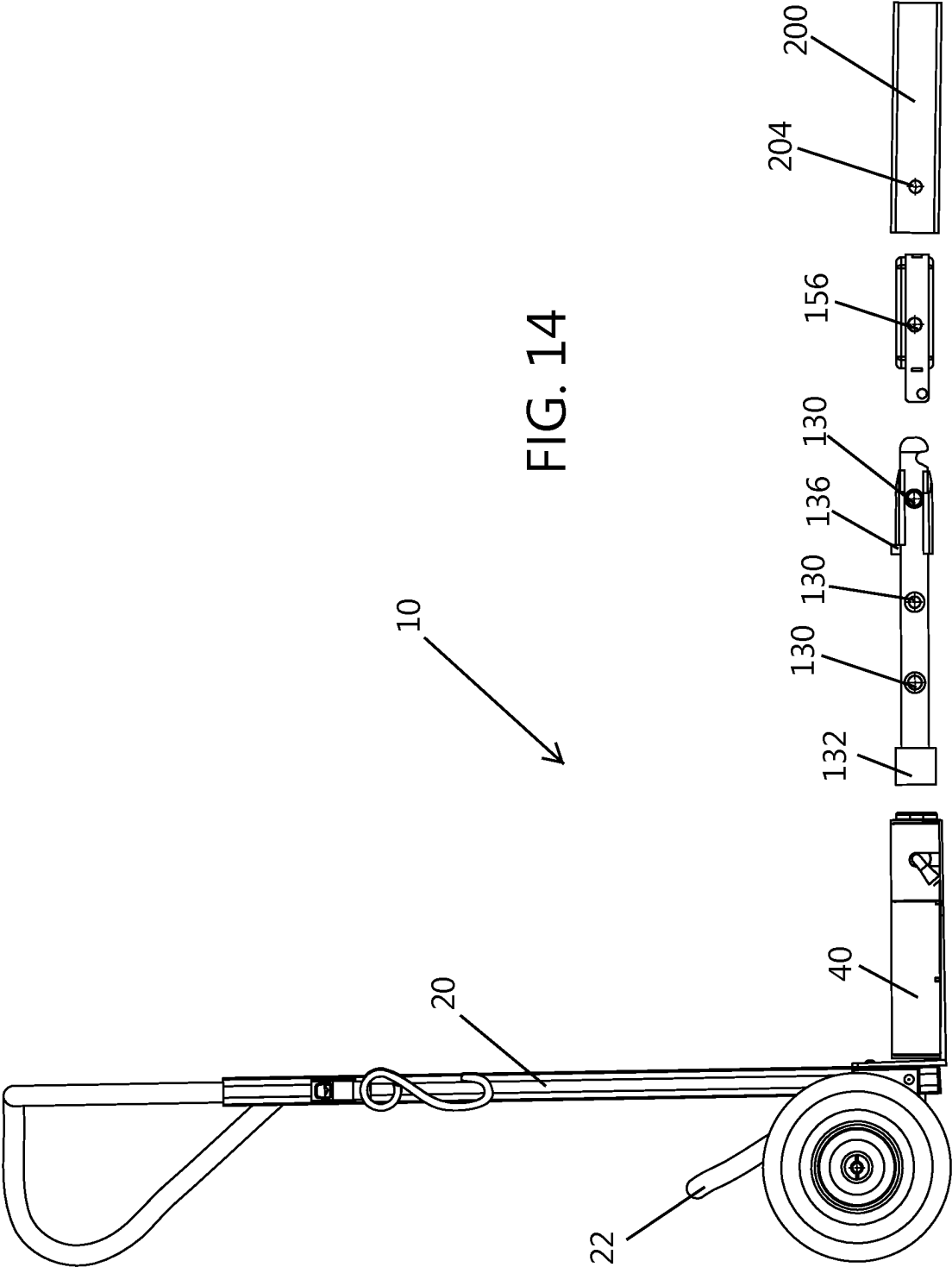


FIG. 12





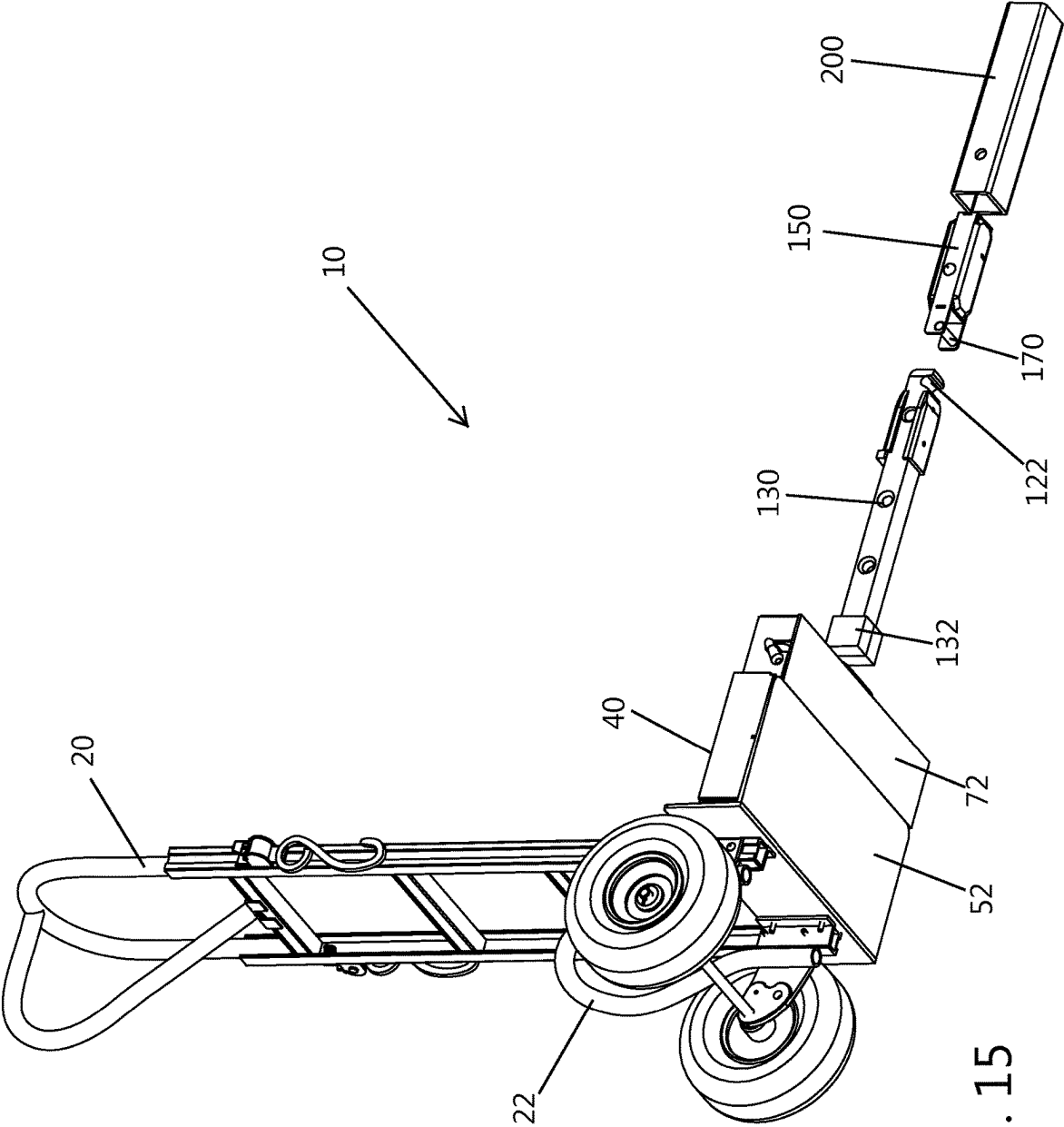


FIG. 15

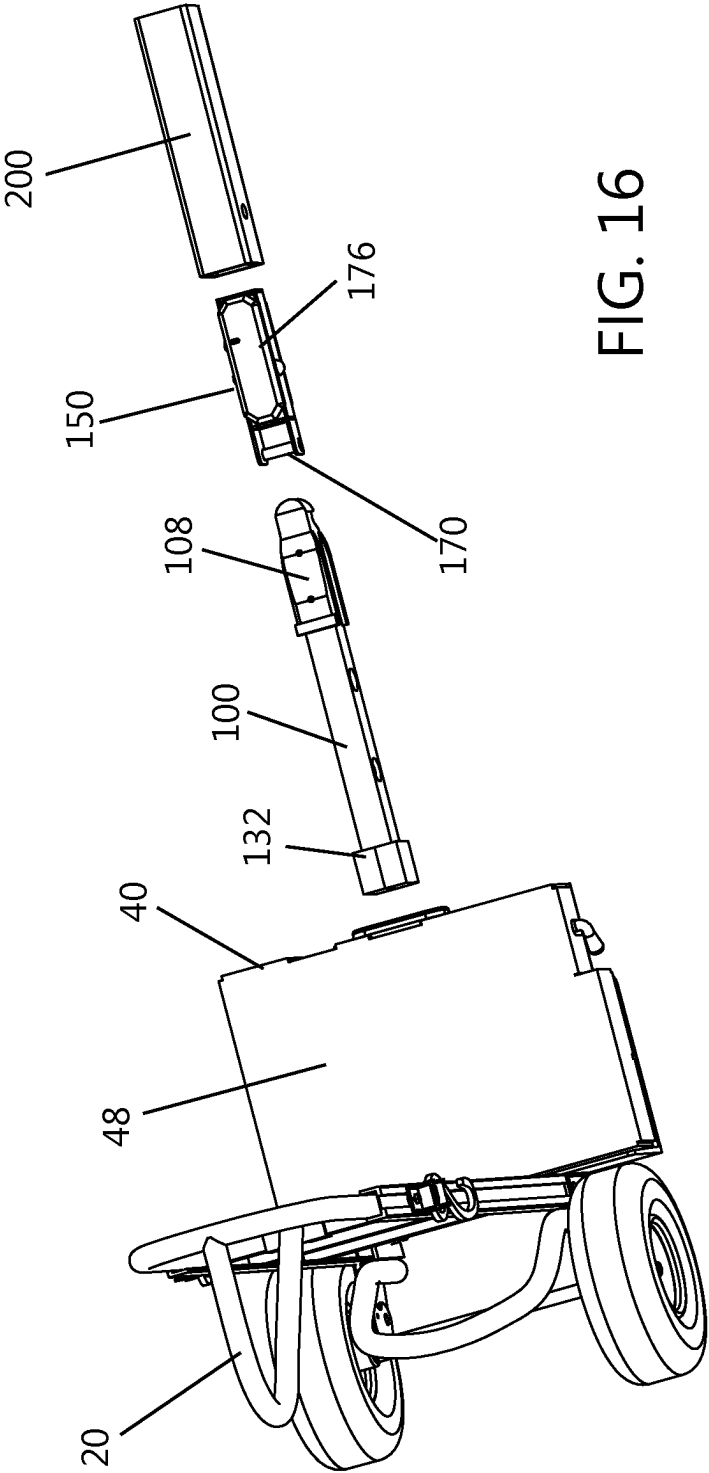
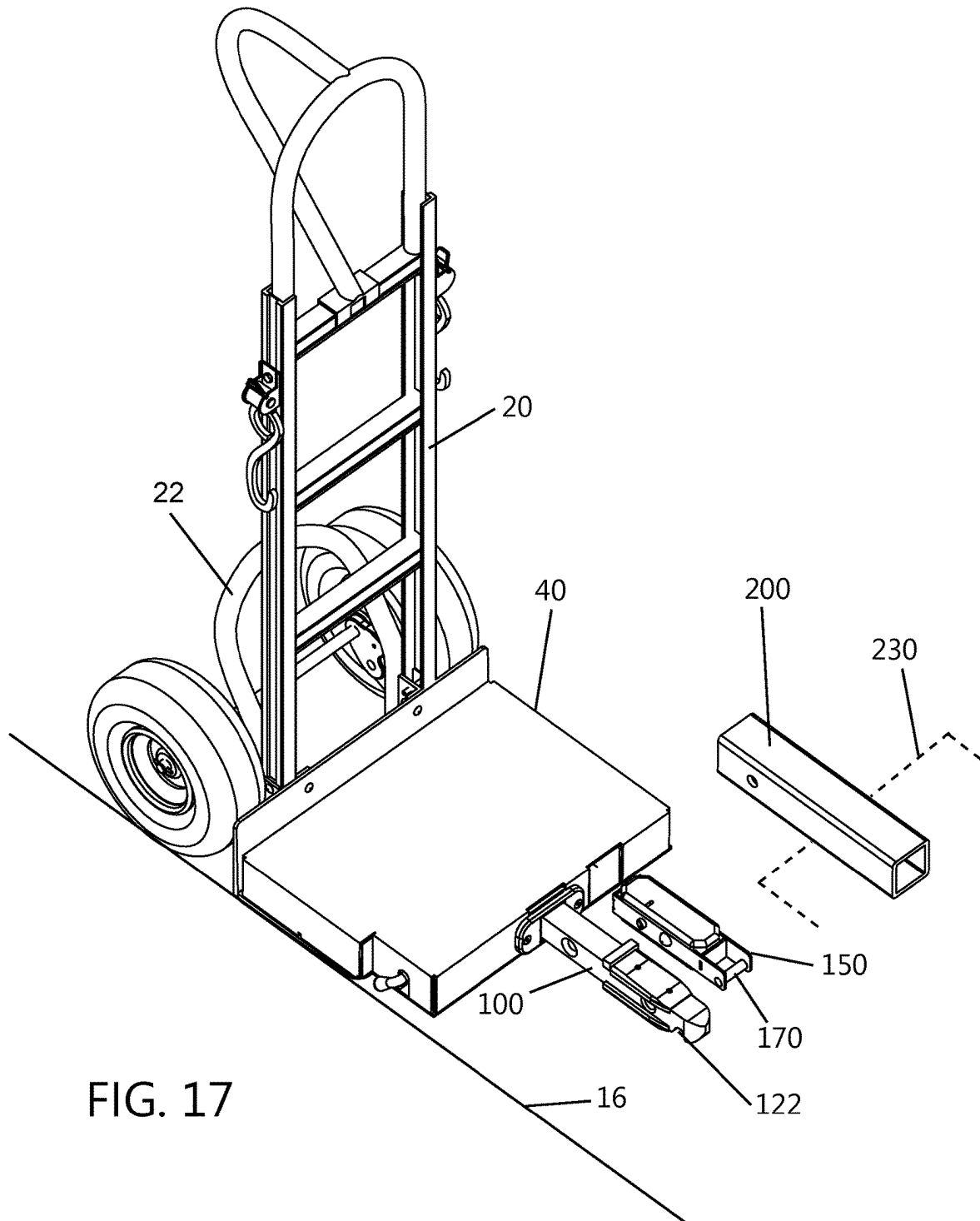
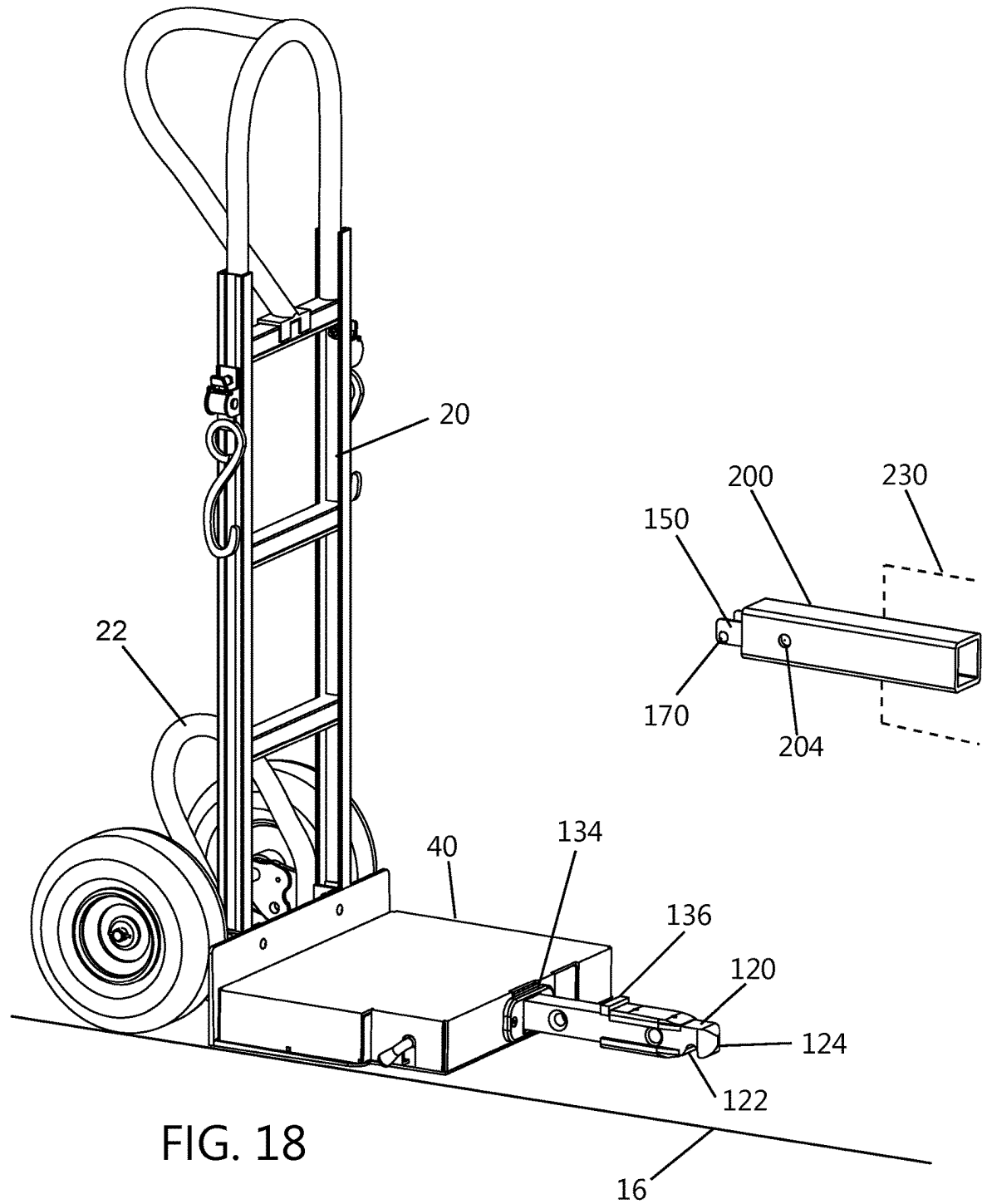
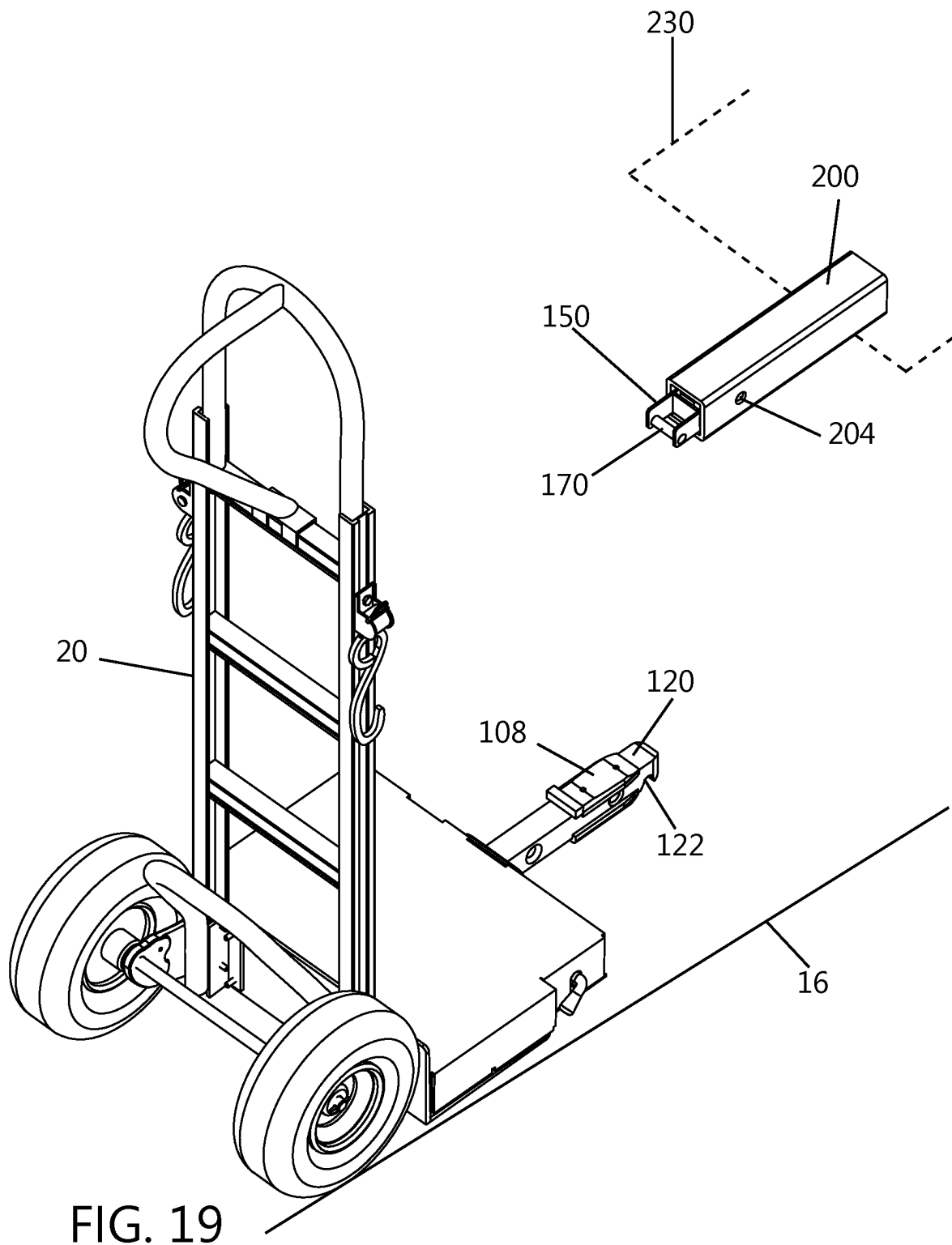
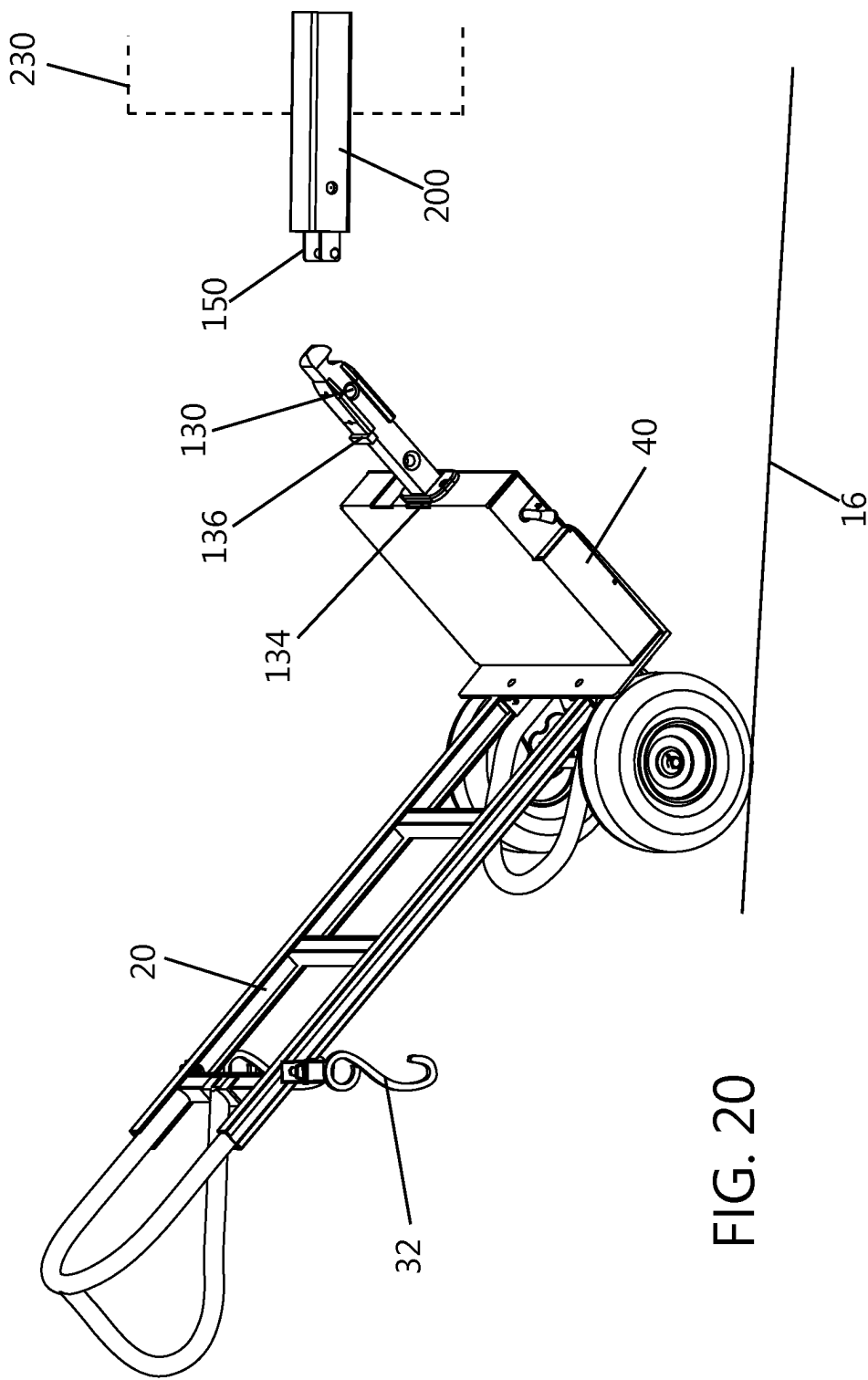


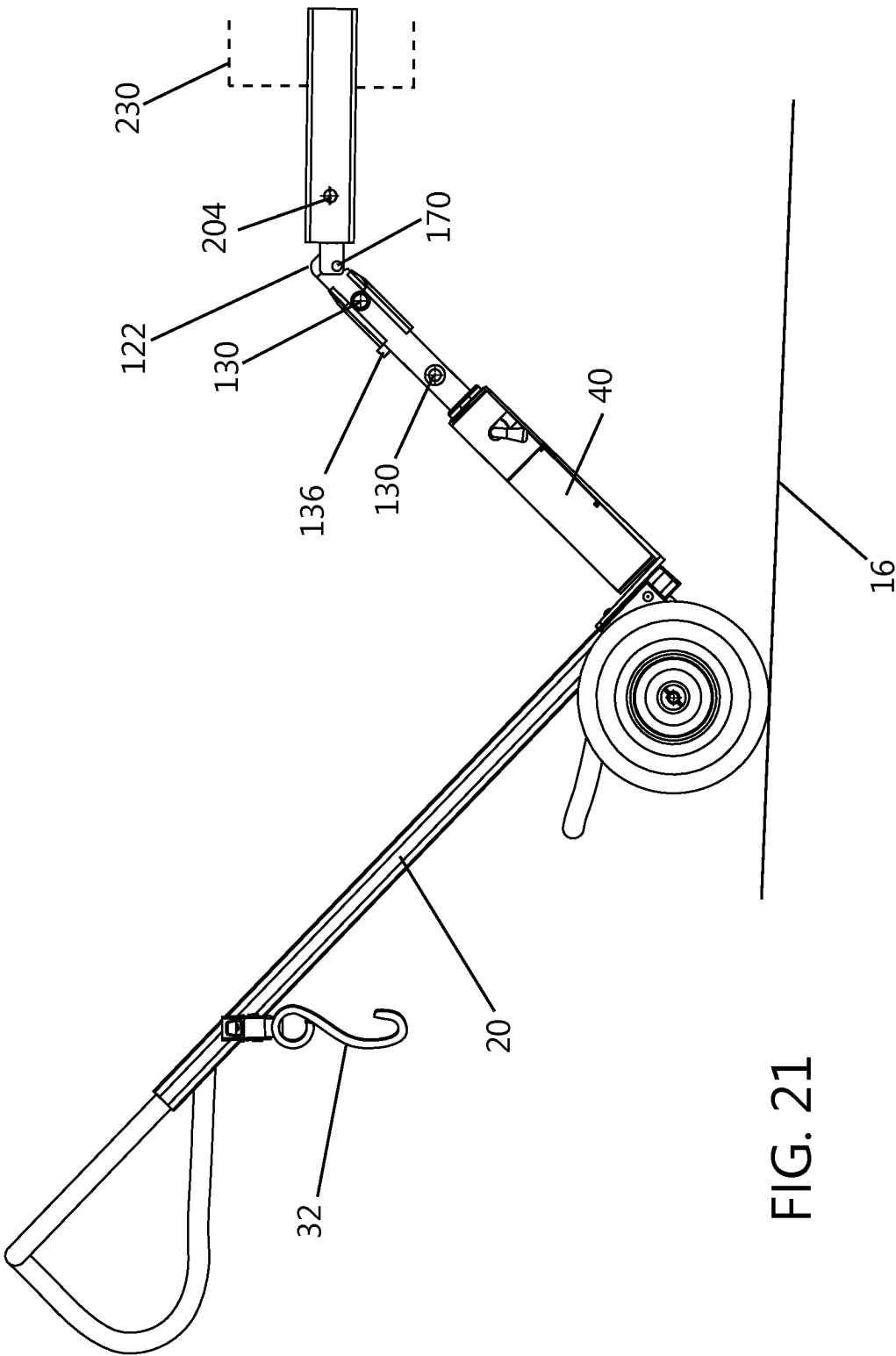
FIG. 16











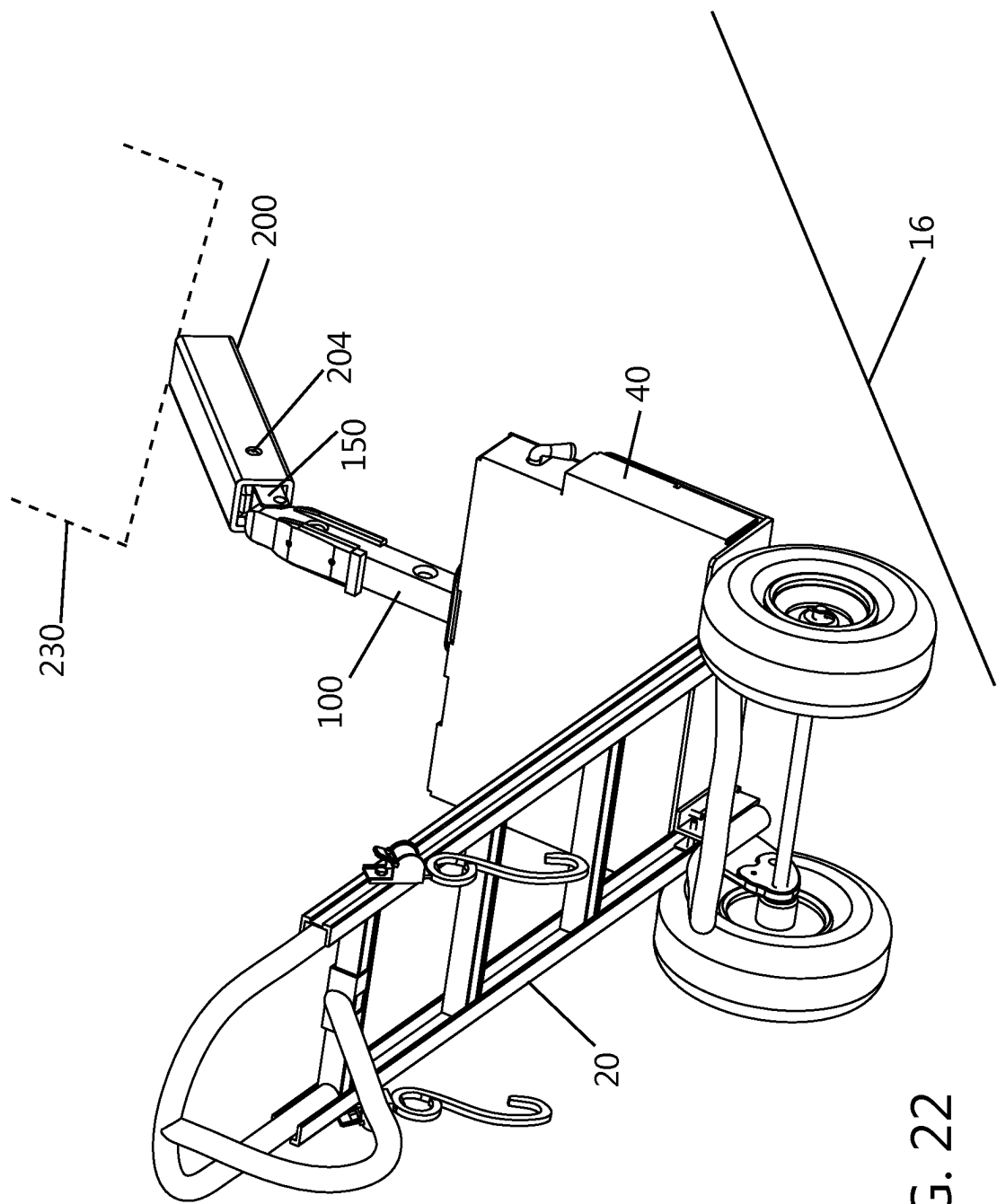
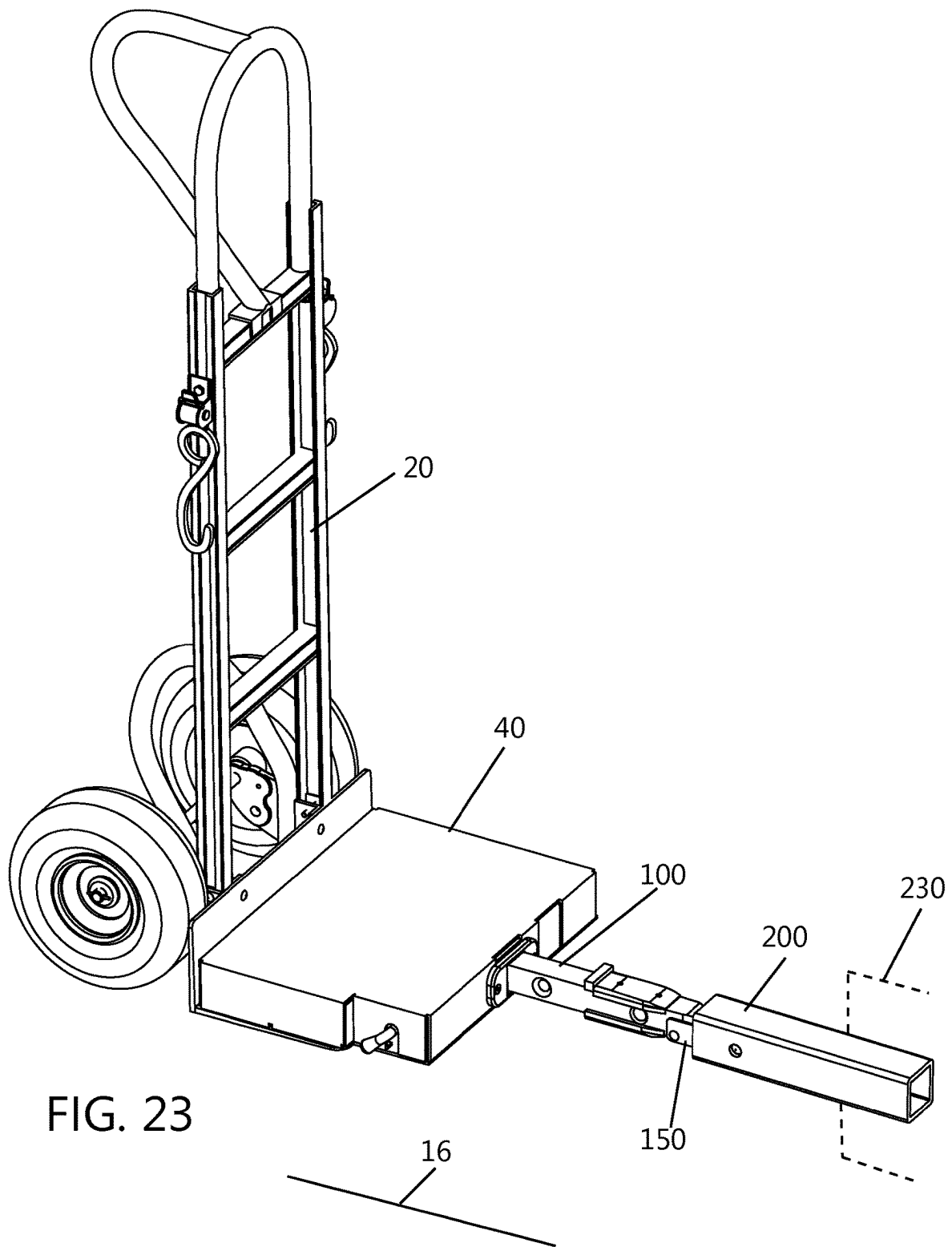
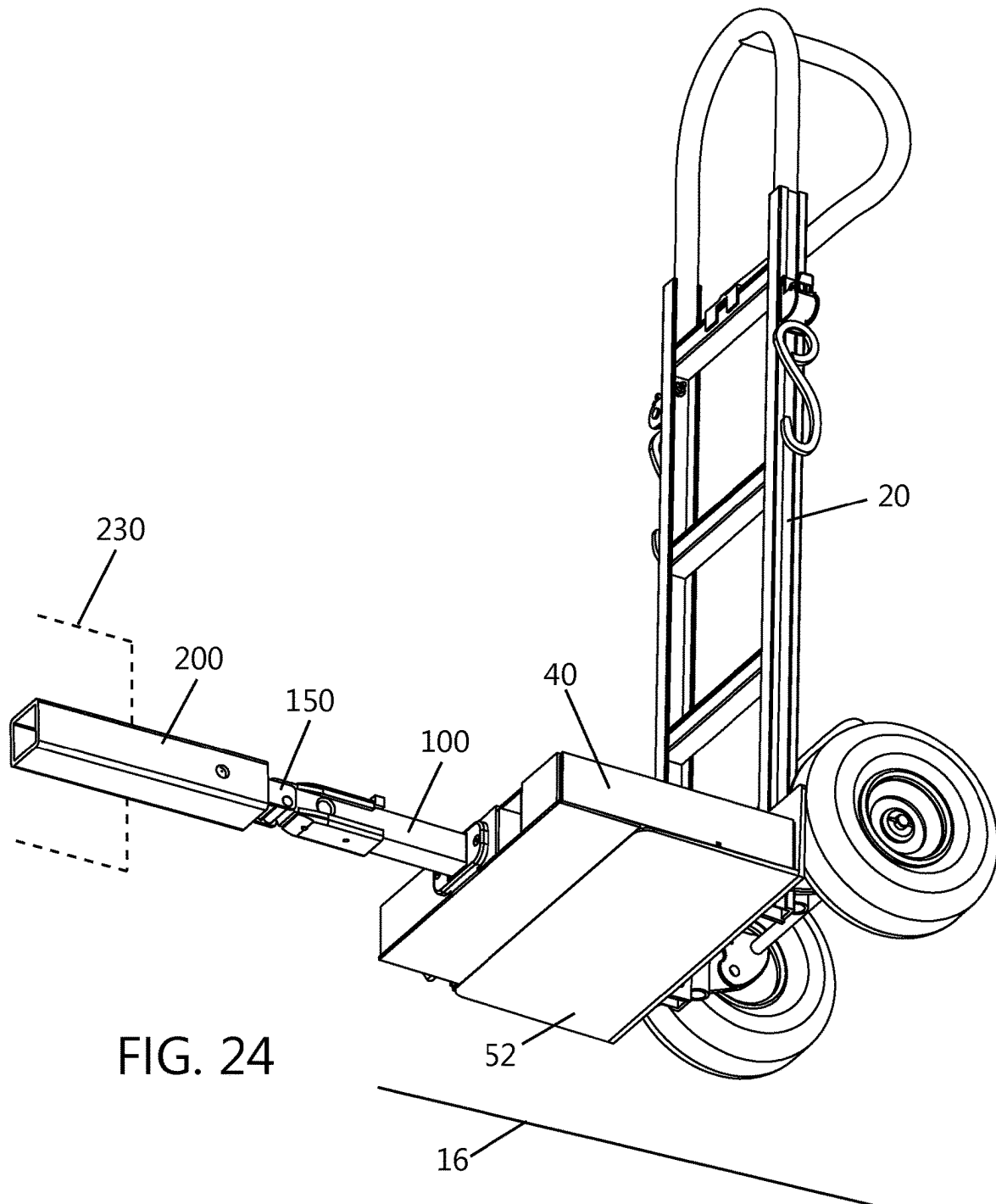
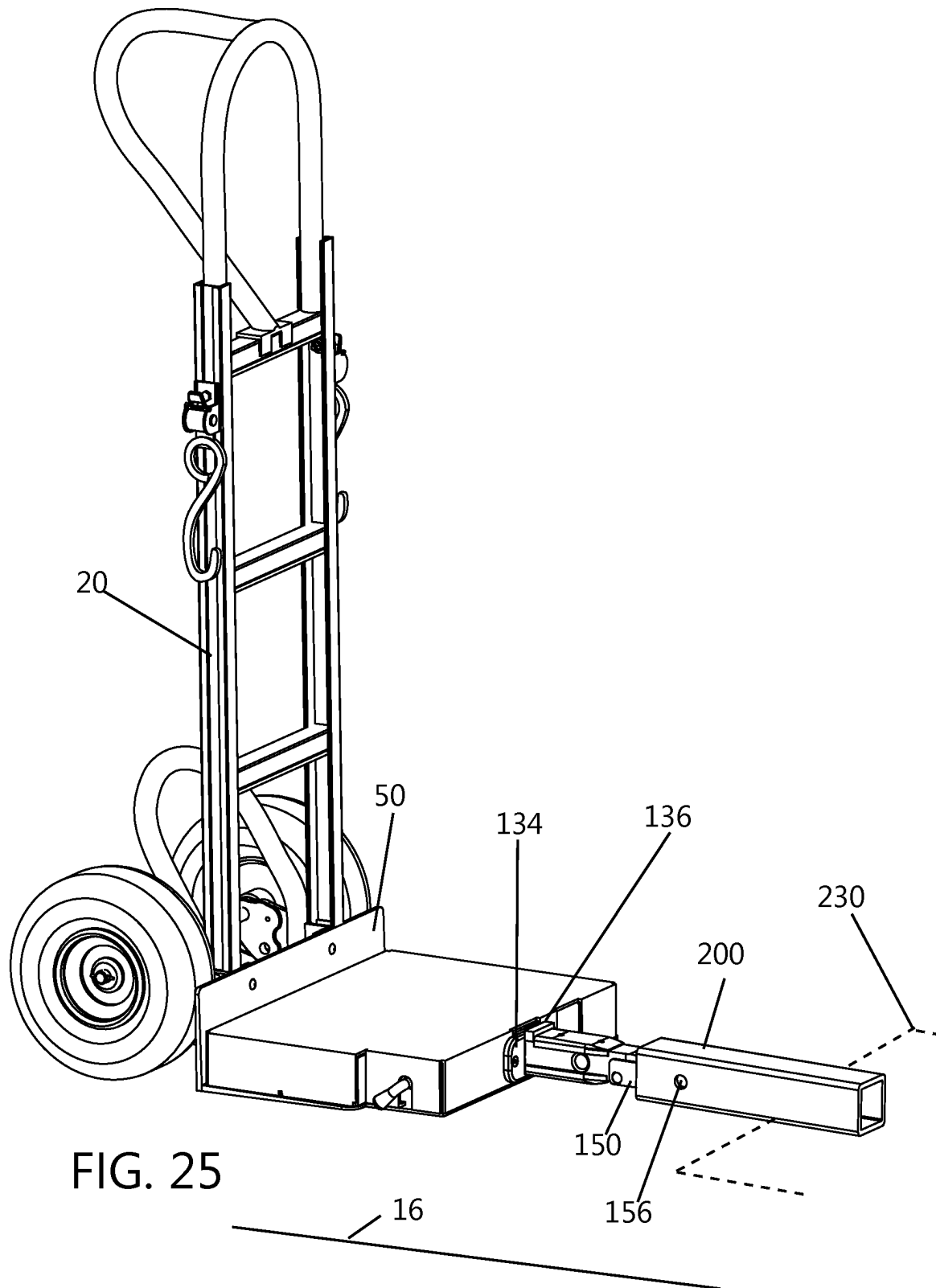
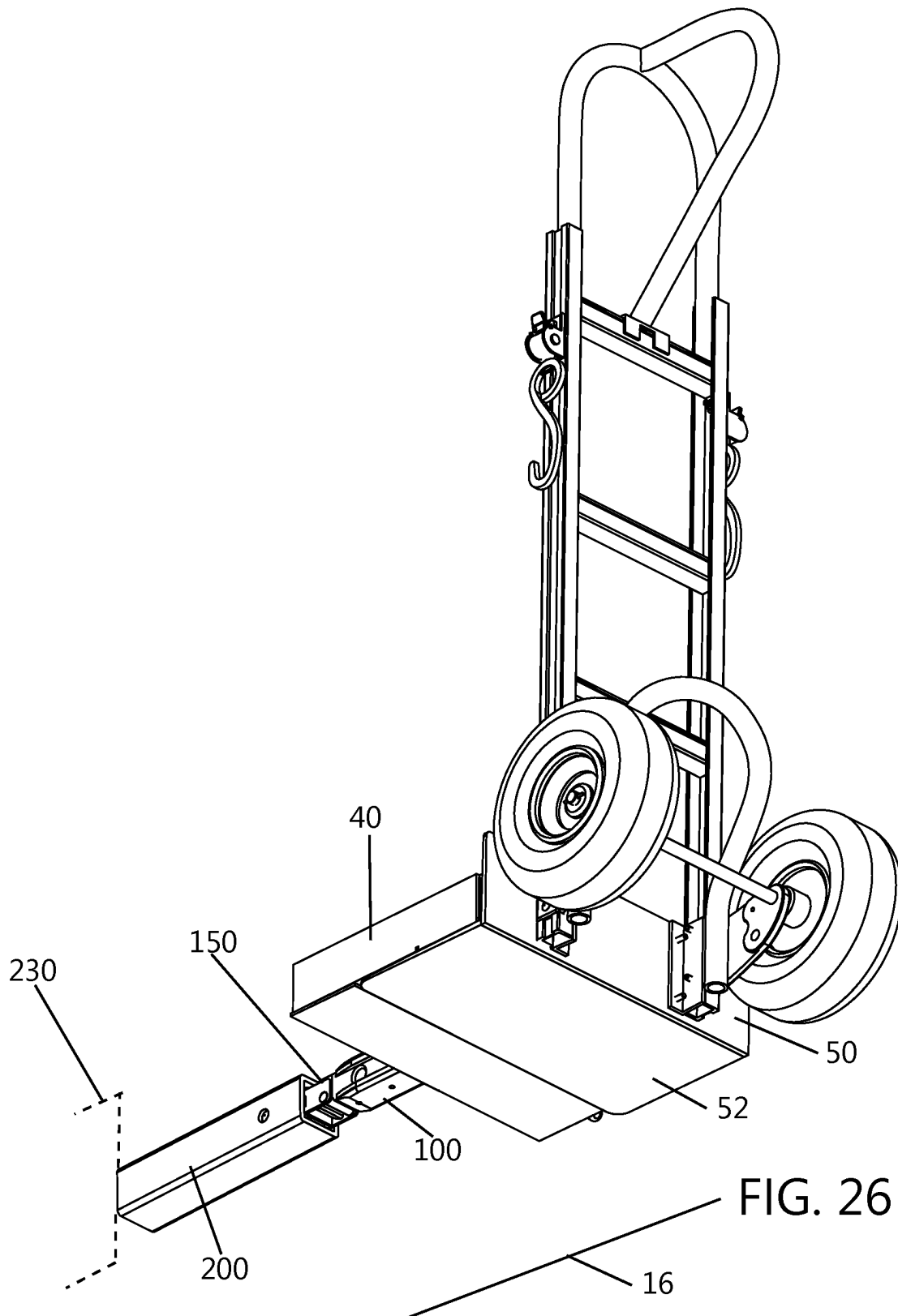


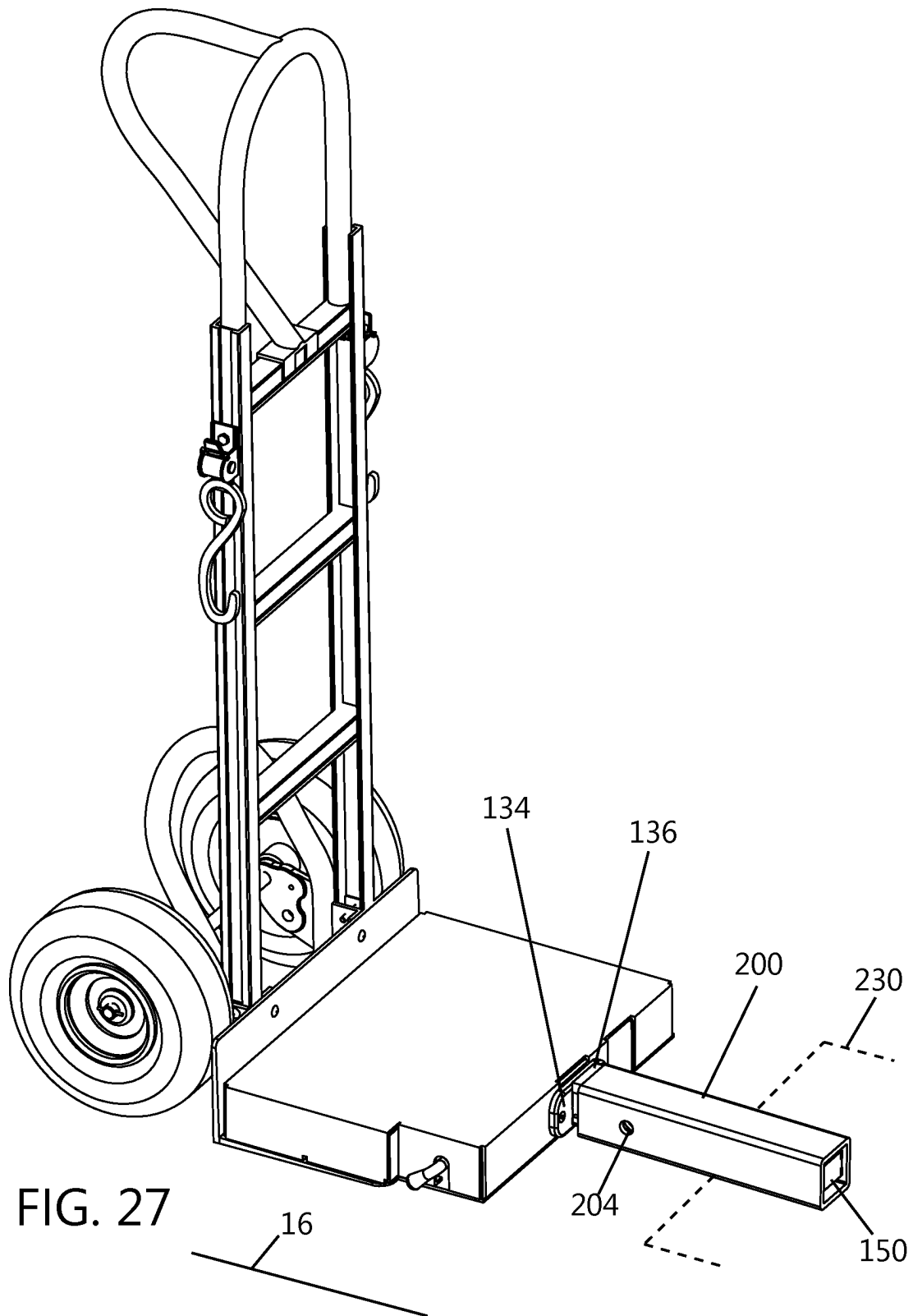
FIG. 22

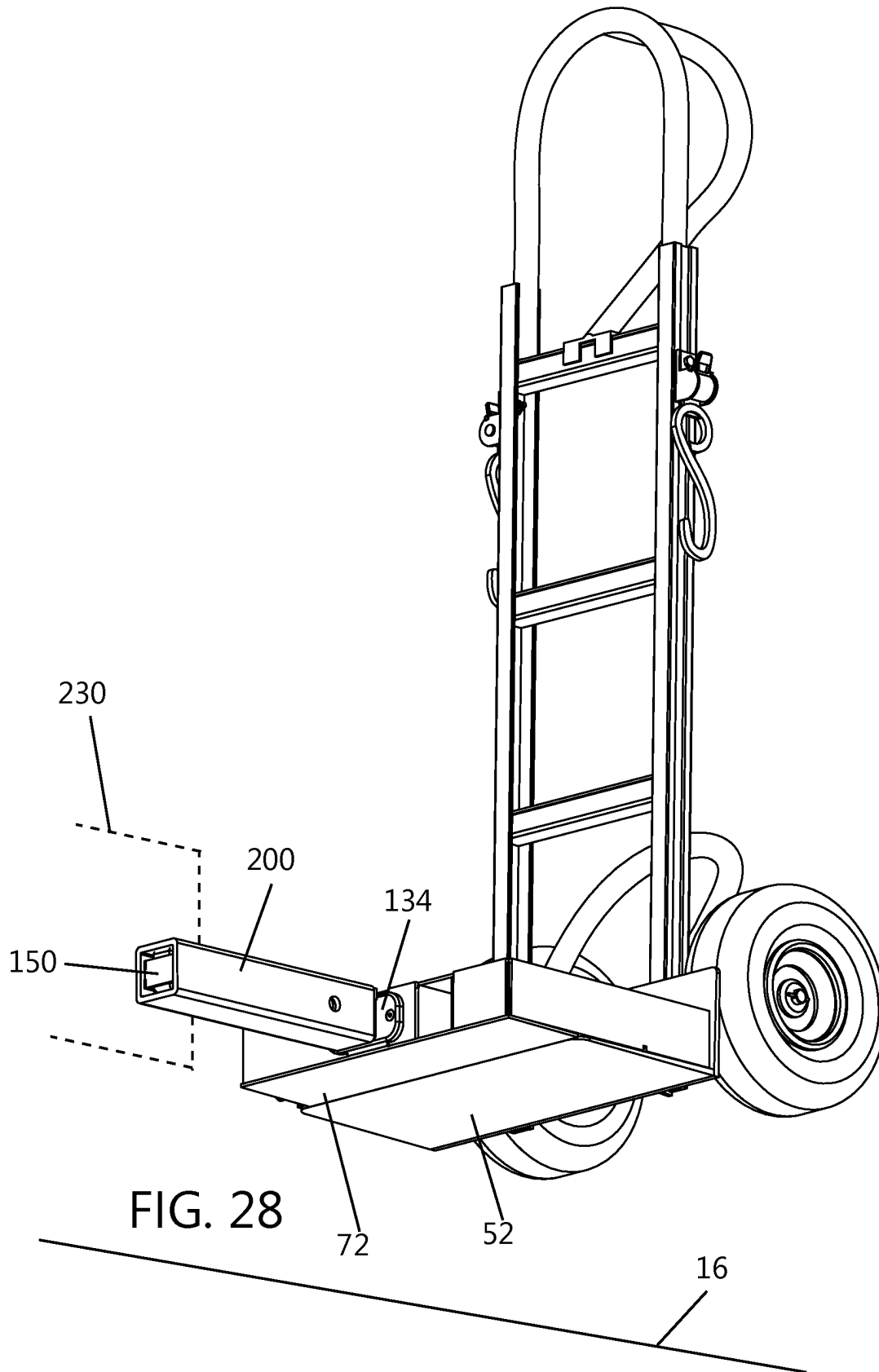












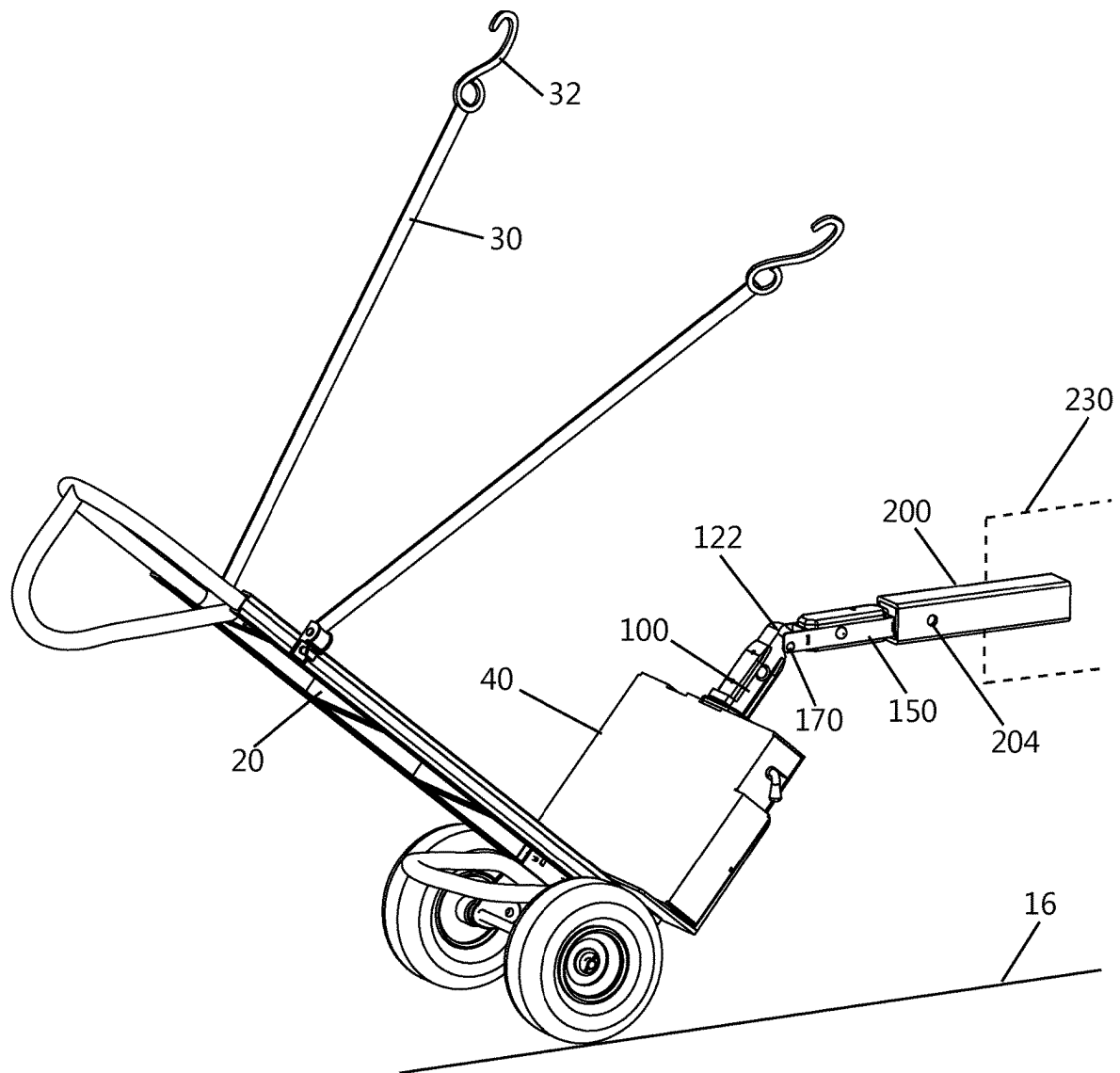


FIG. 29

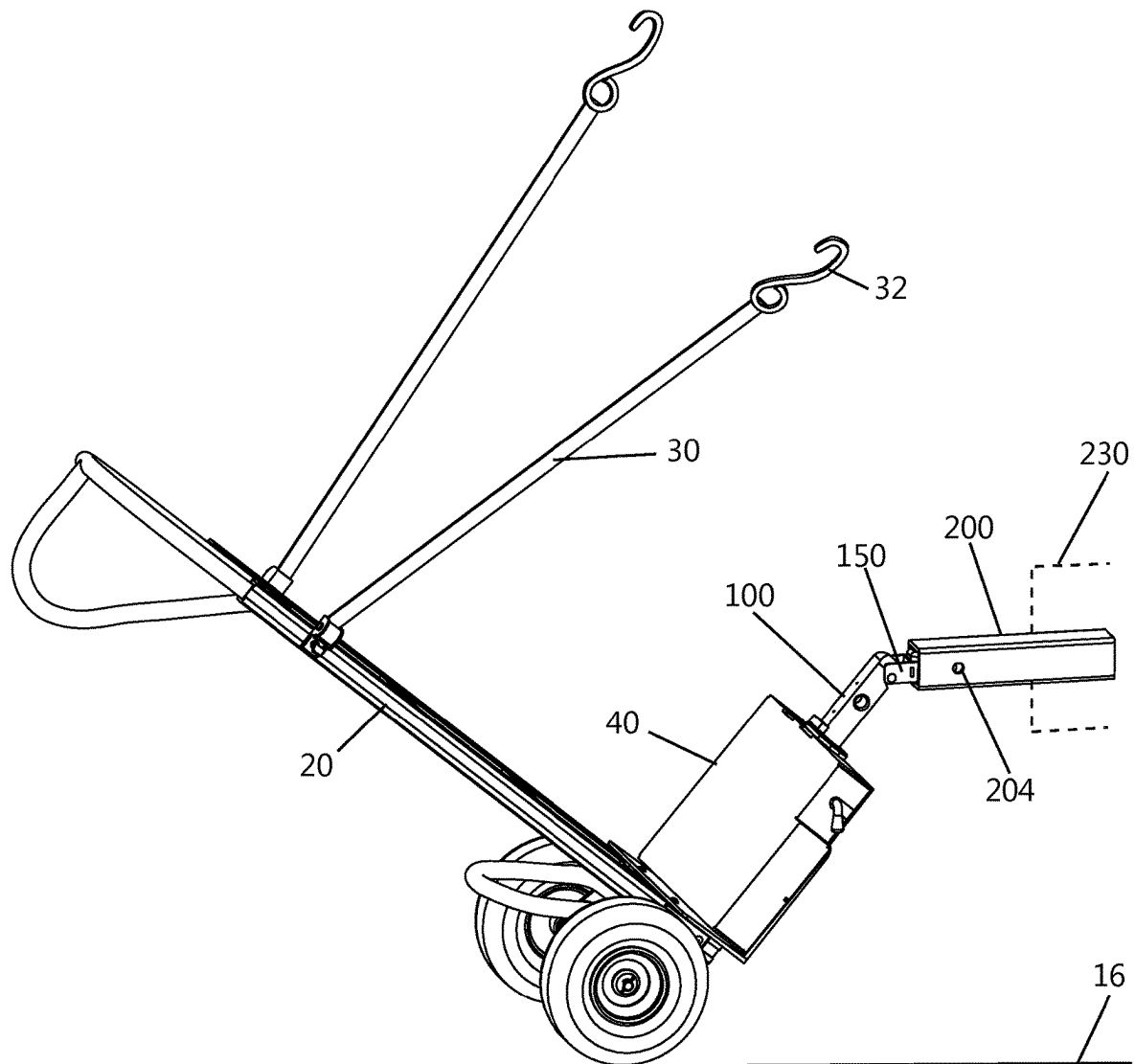
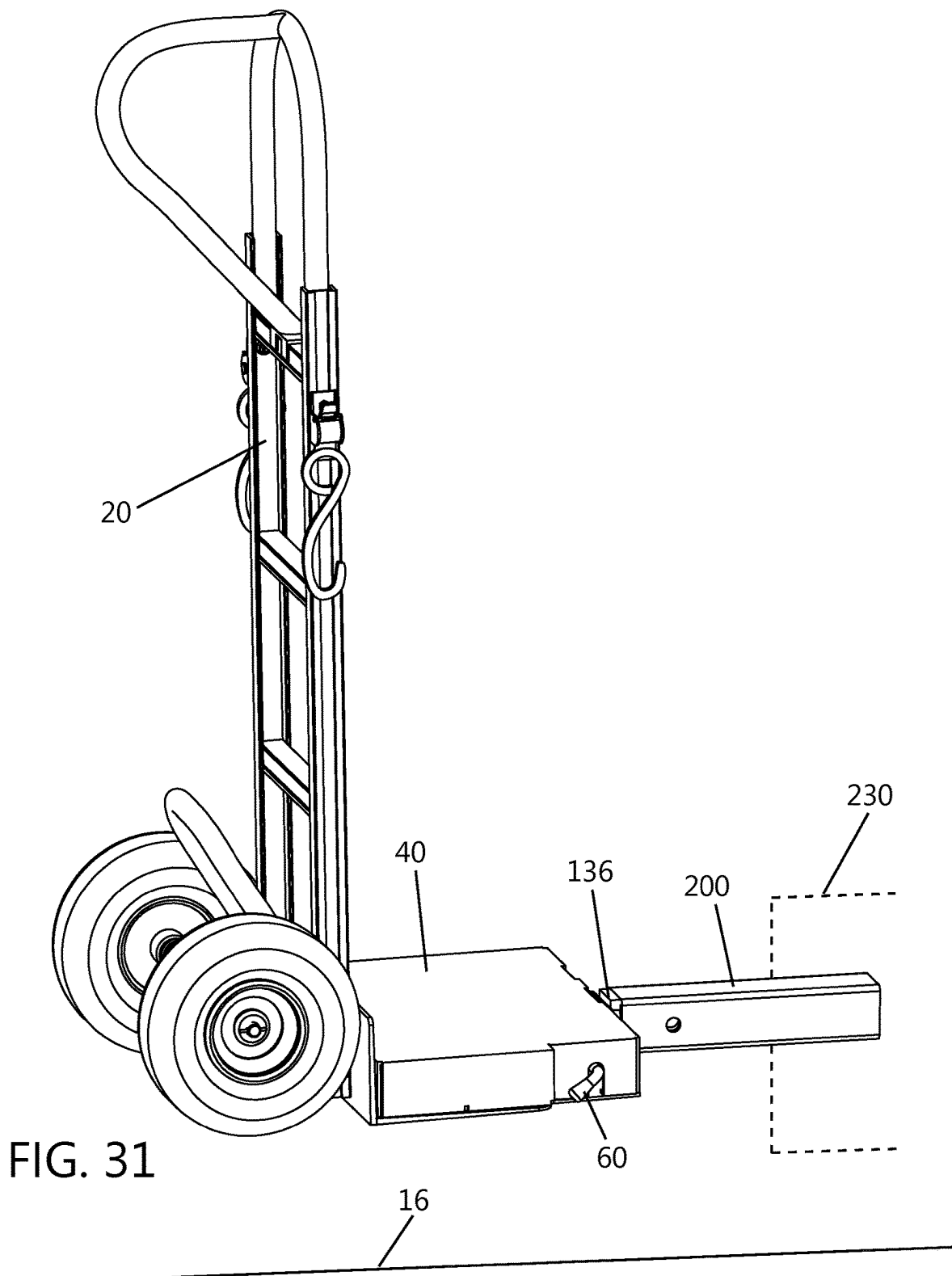
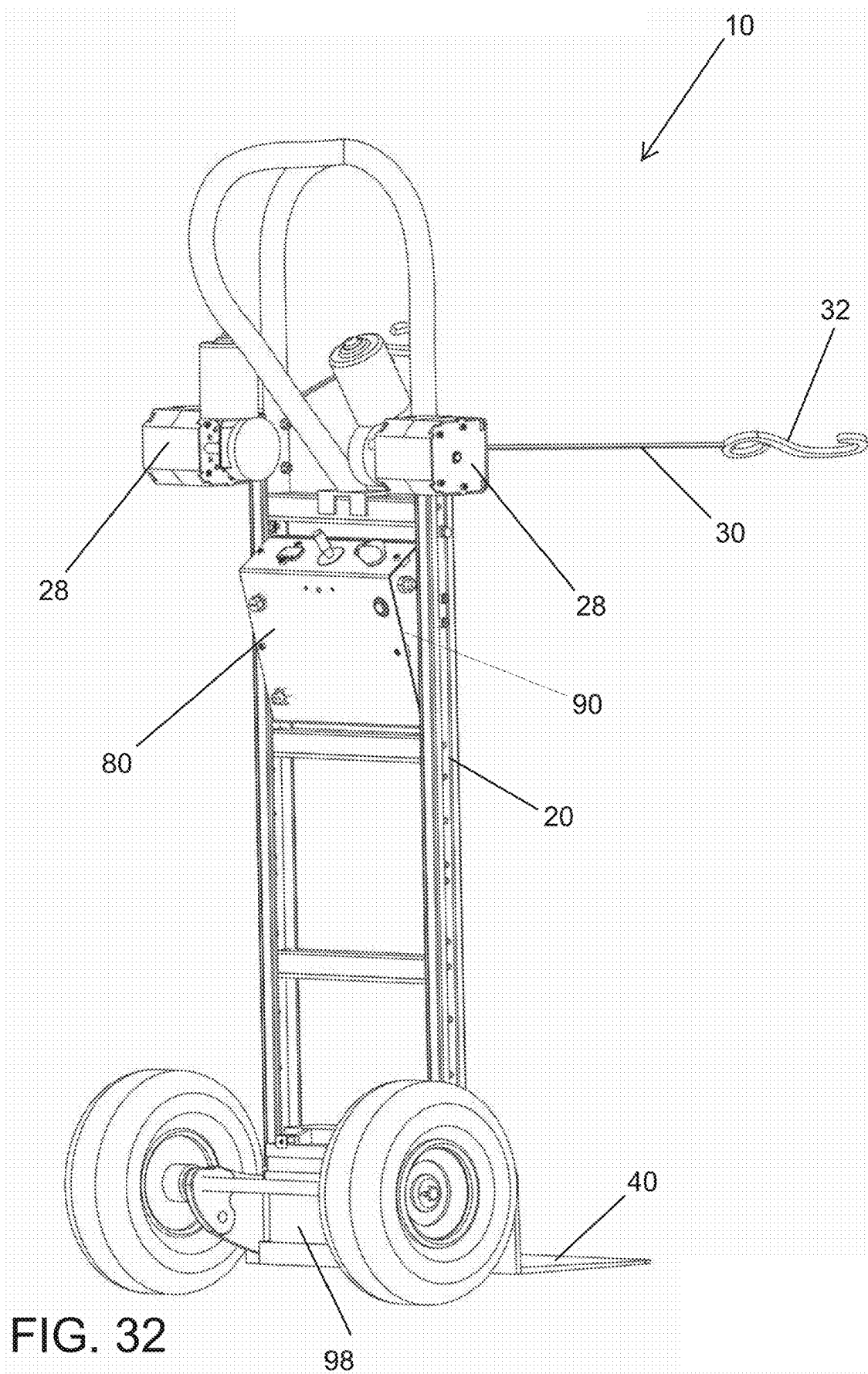


FIG. 30





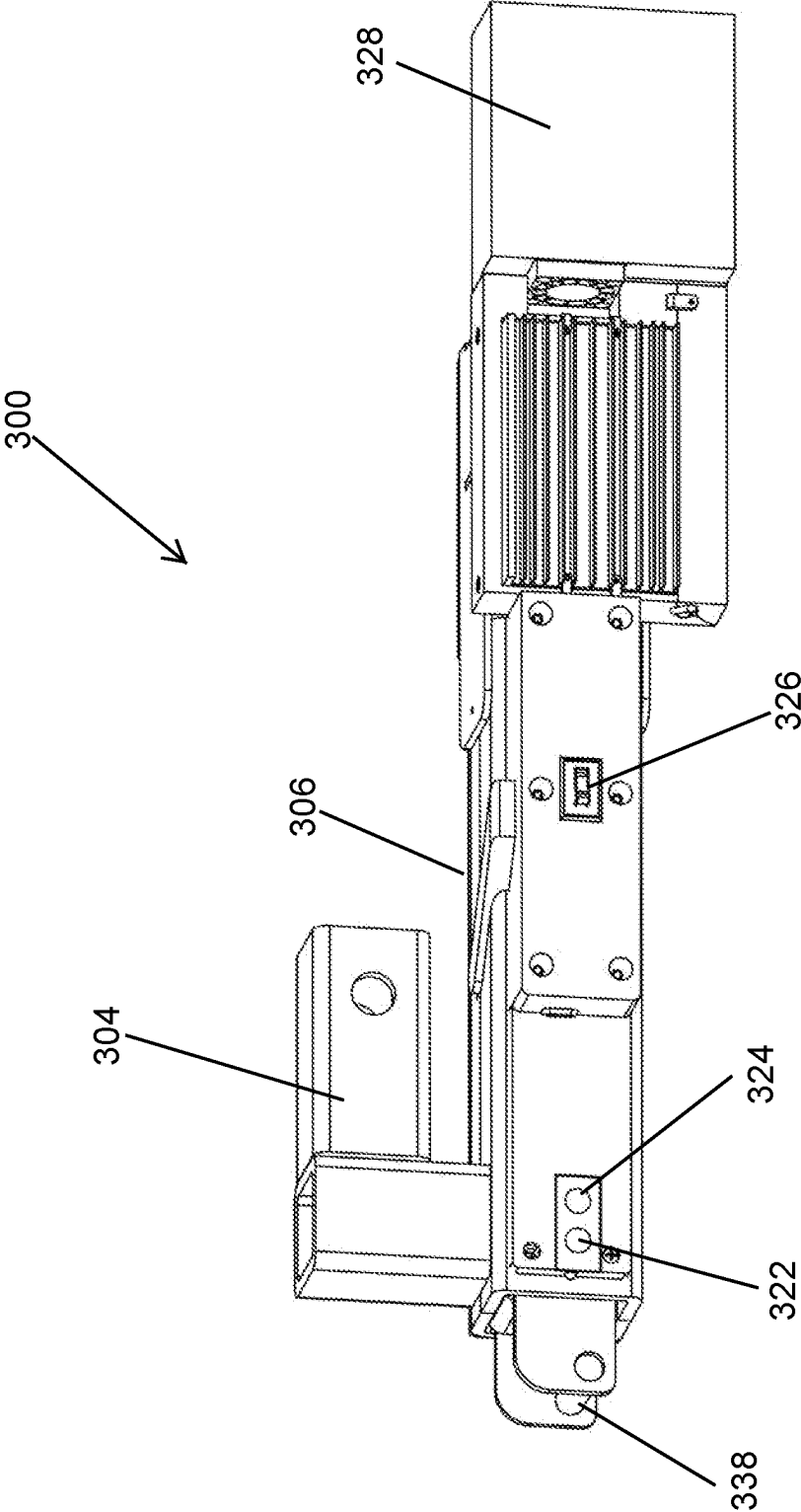
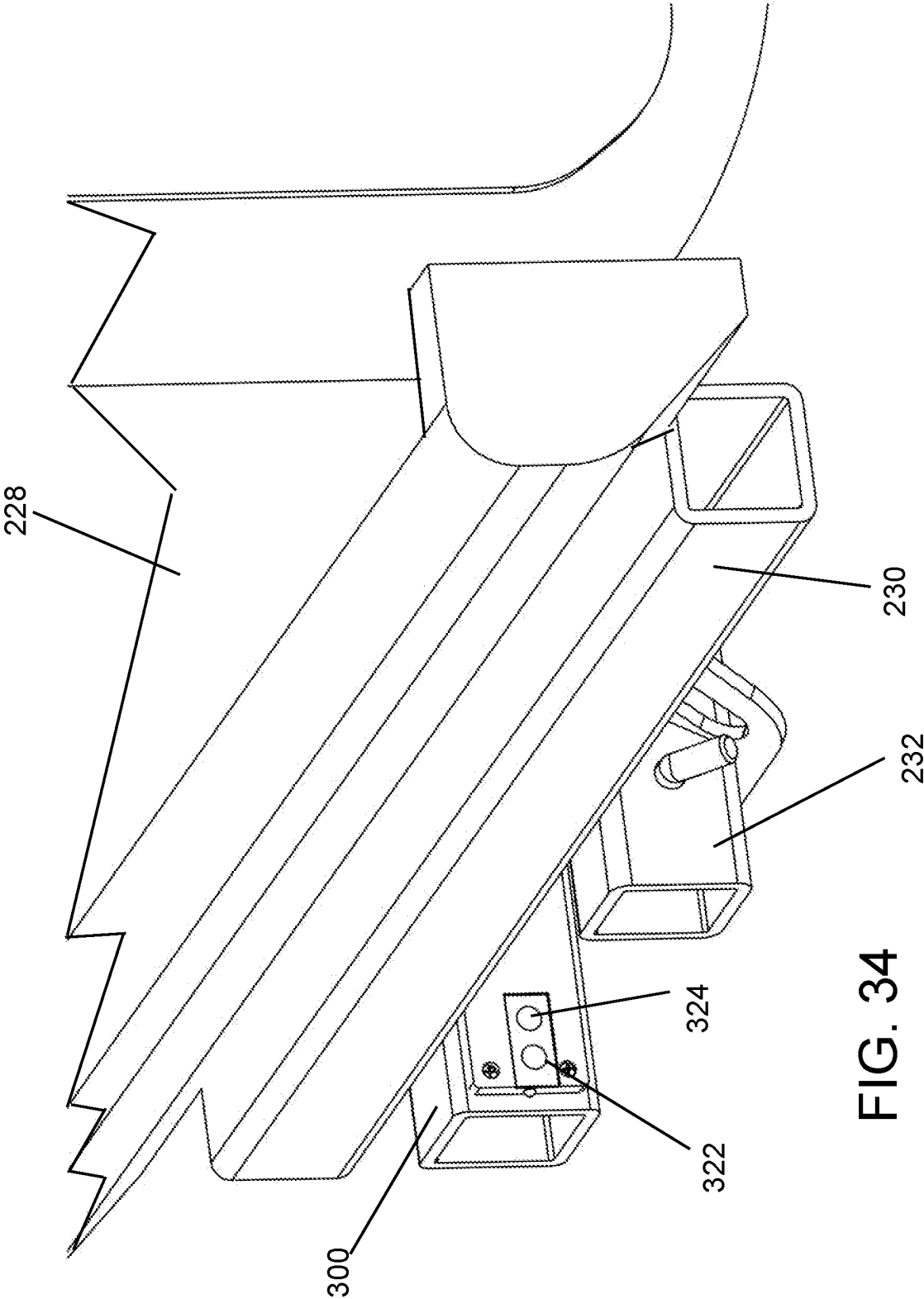


FIG. 33



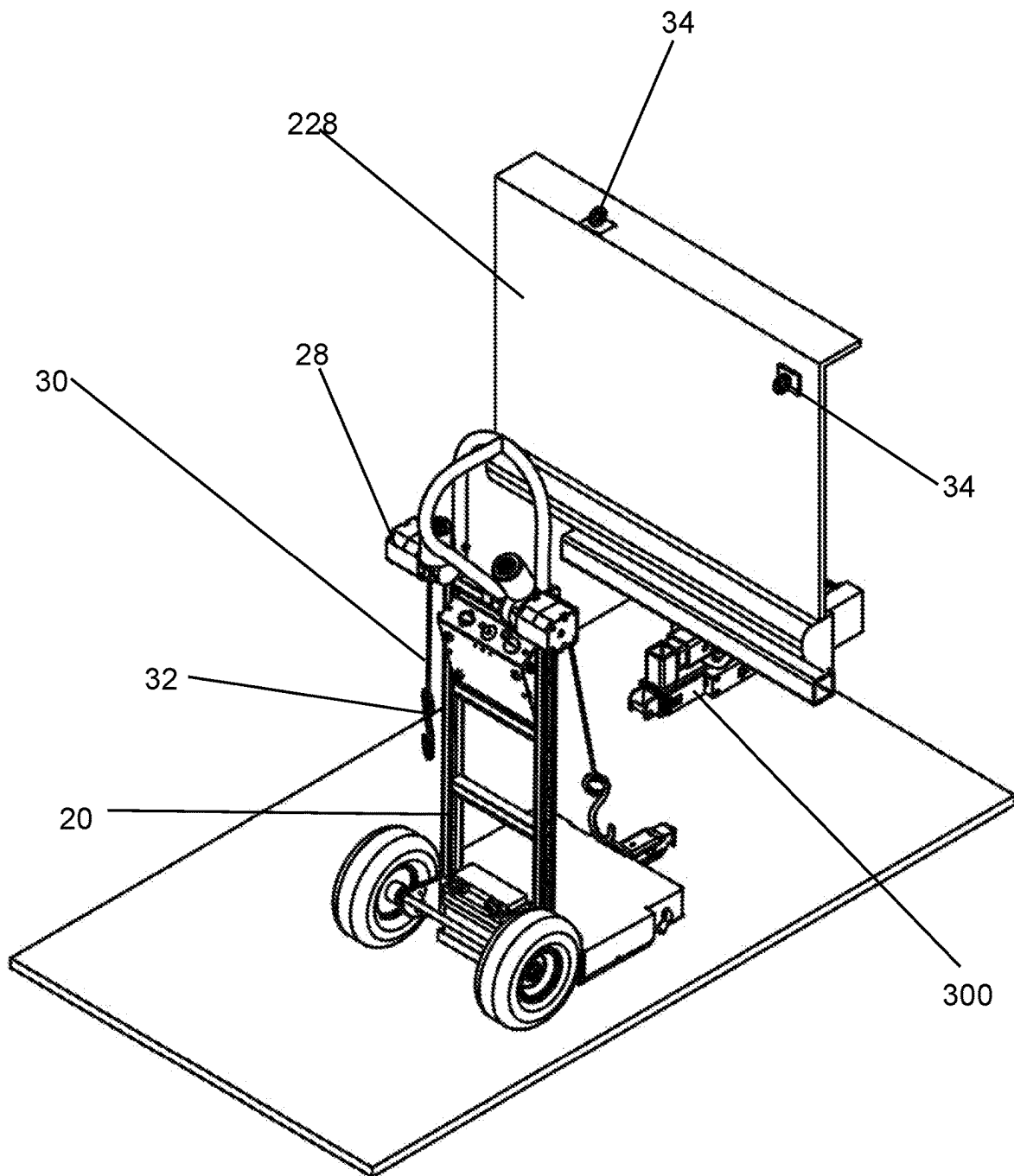


FIG. 35

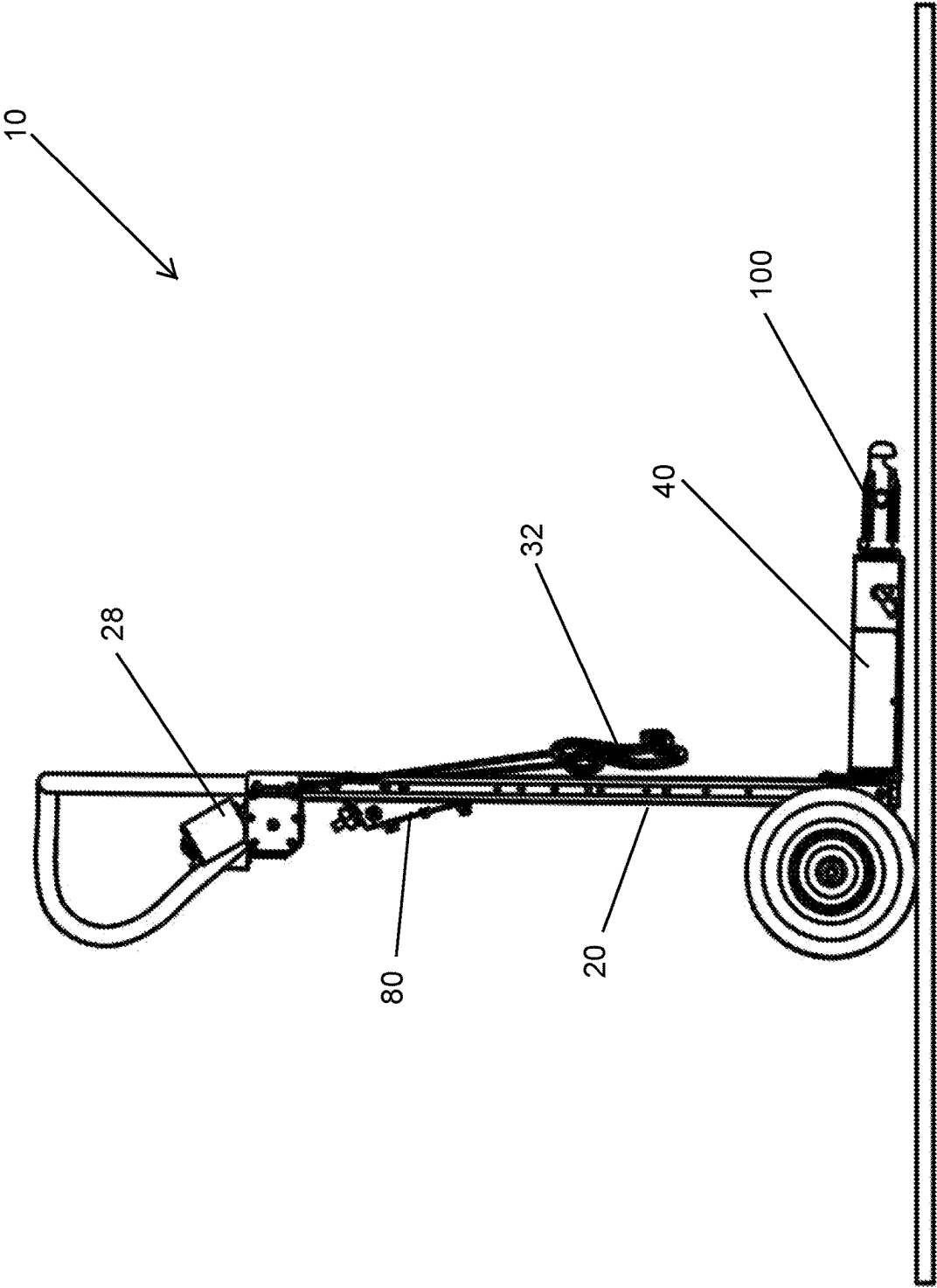


FIG. 36

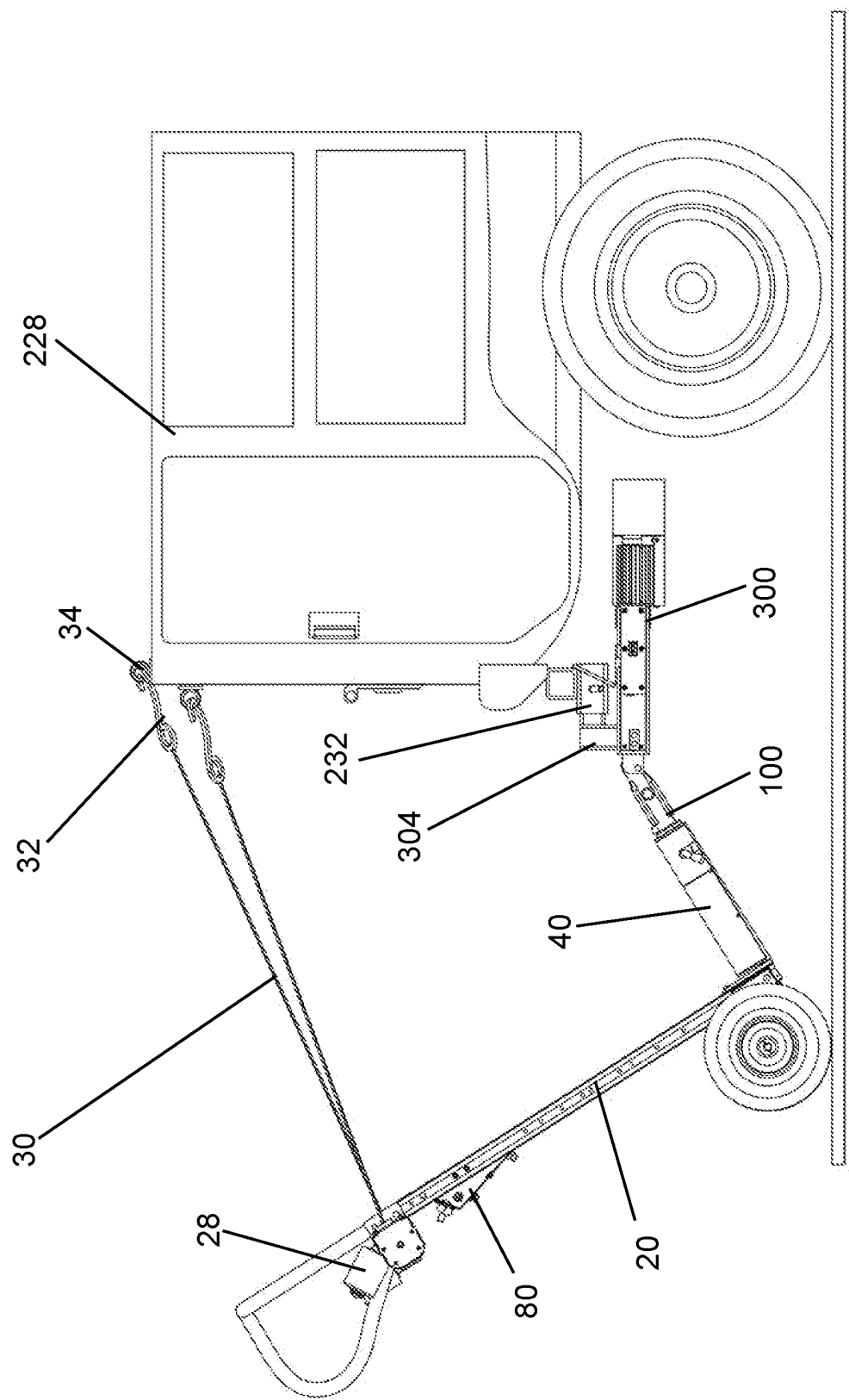


FIG. 37

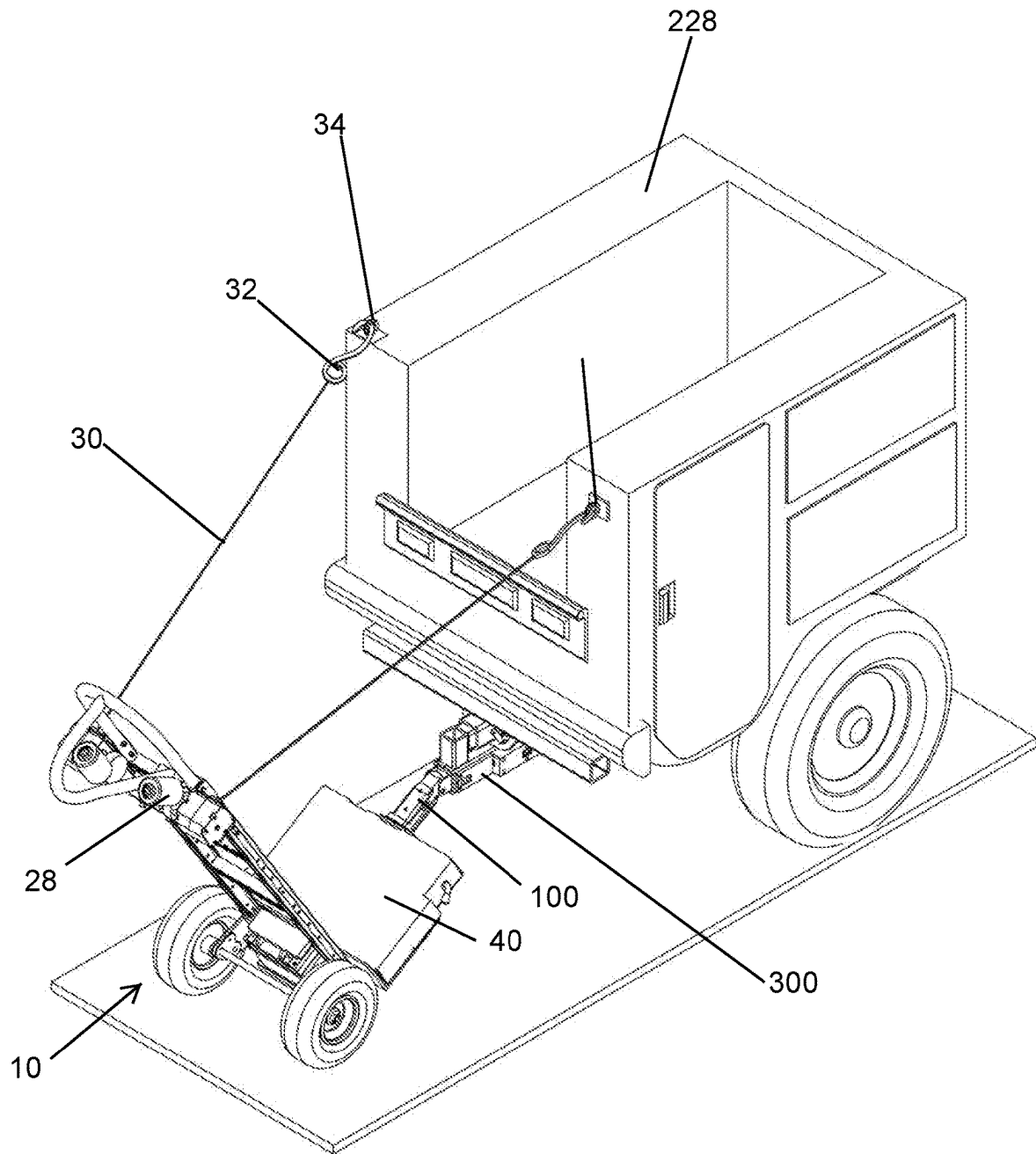
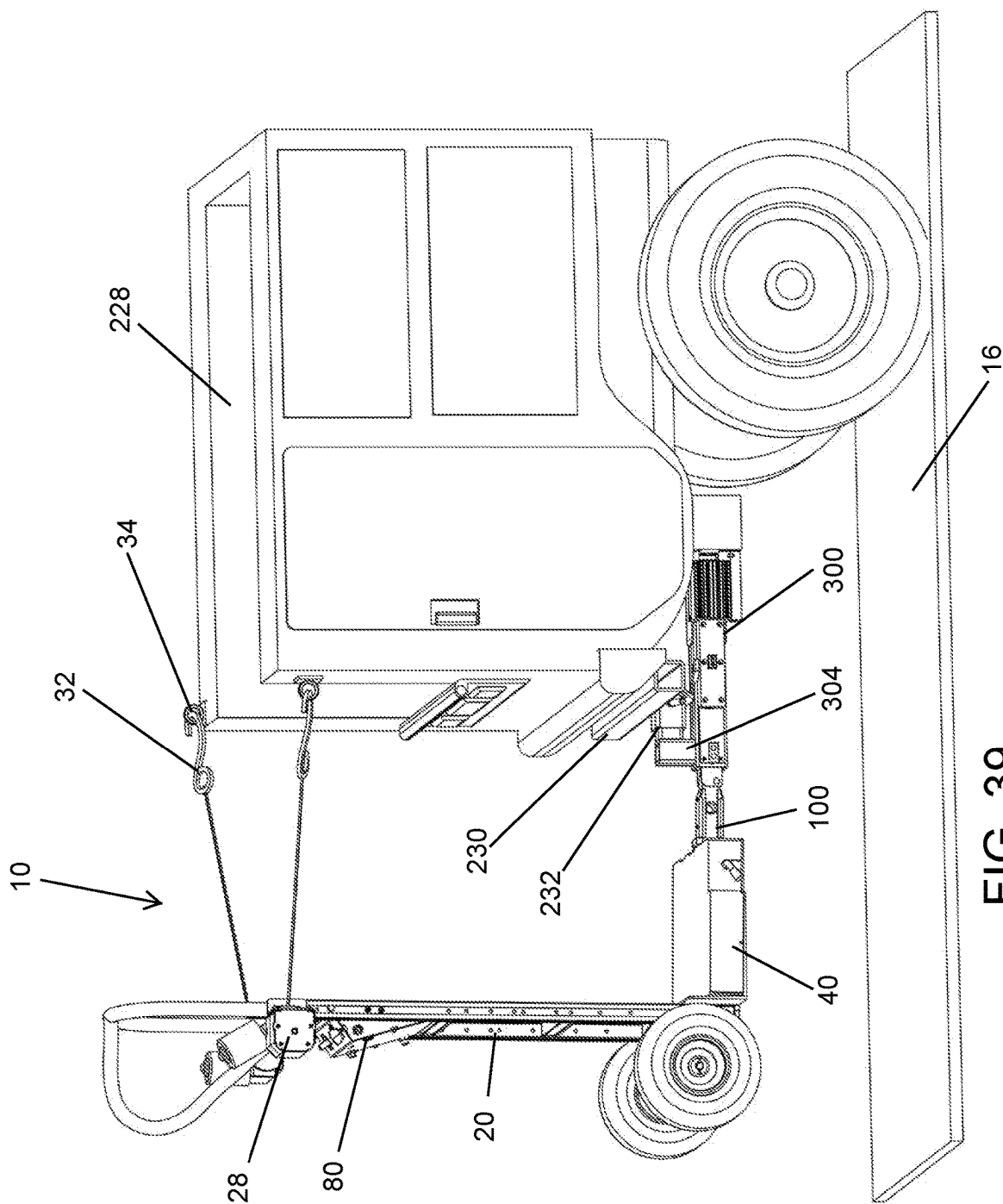


FIG. 38



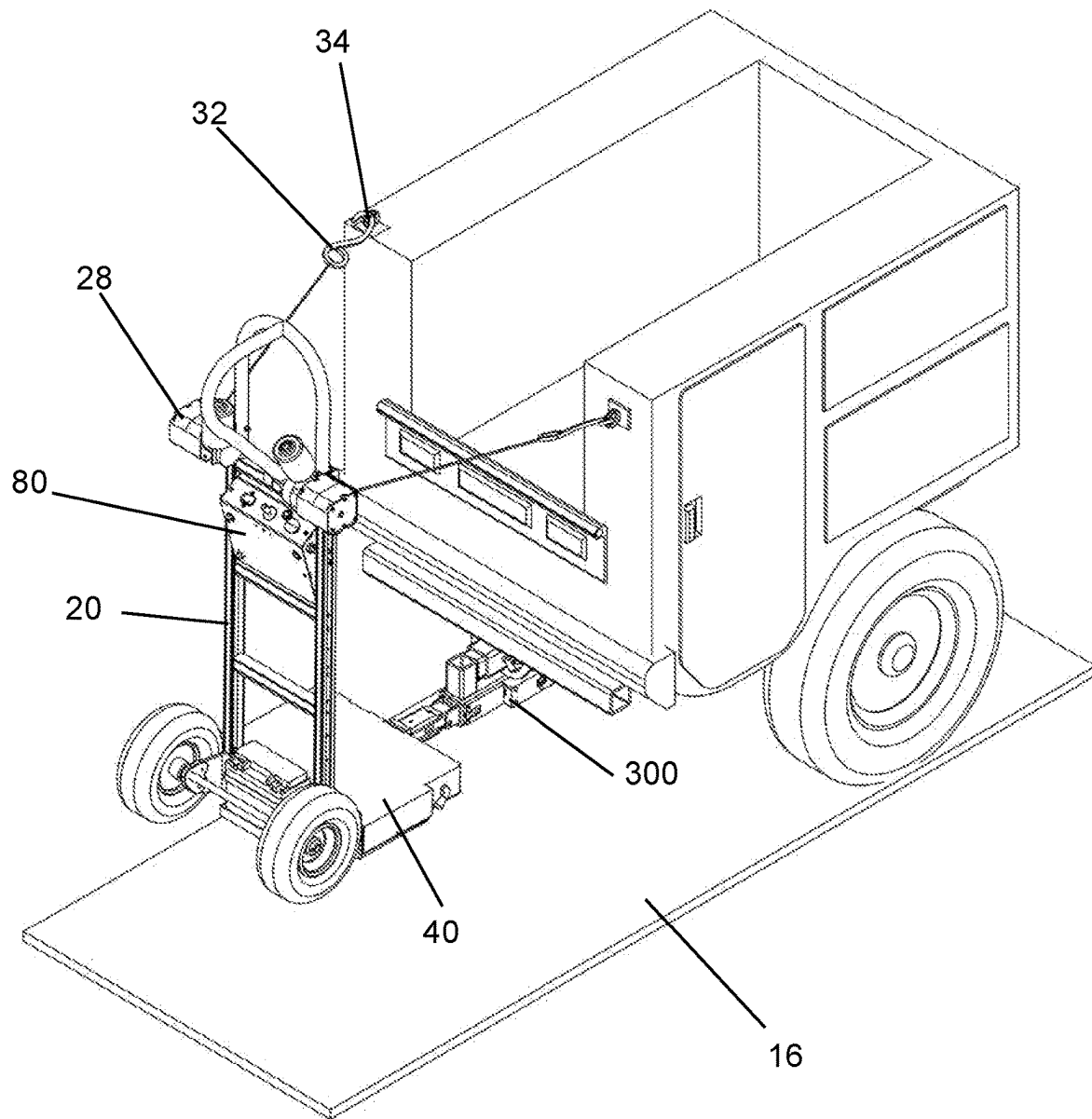
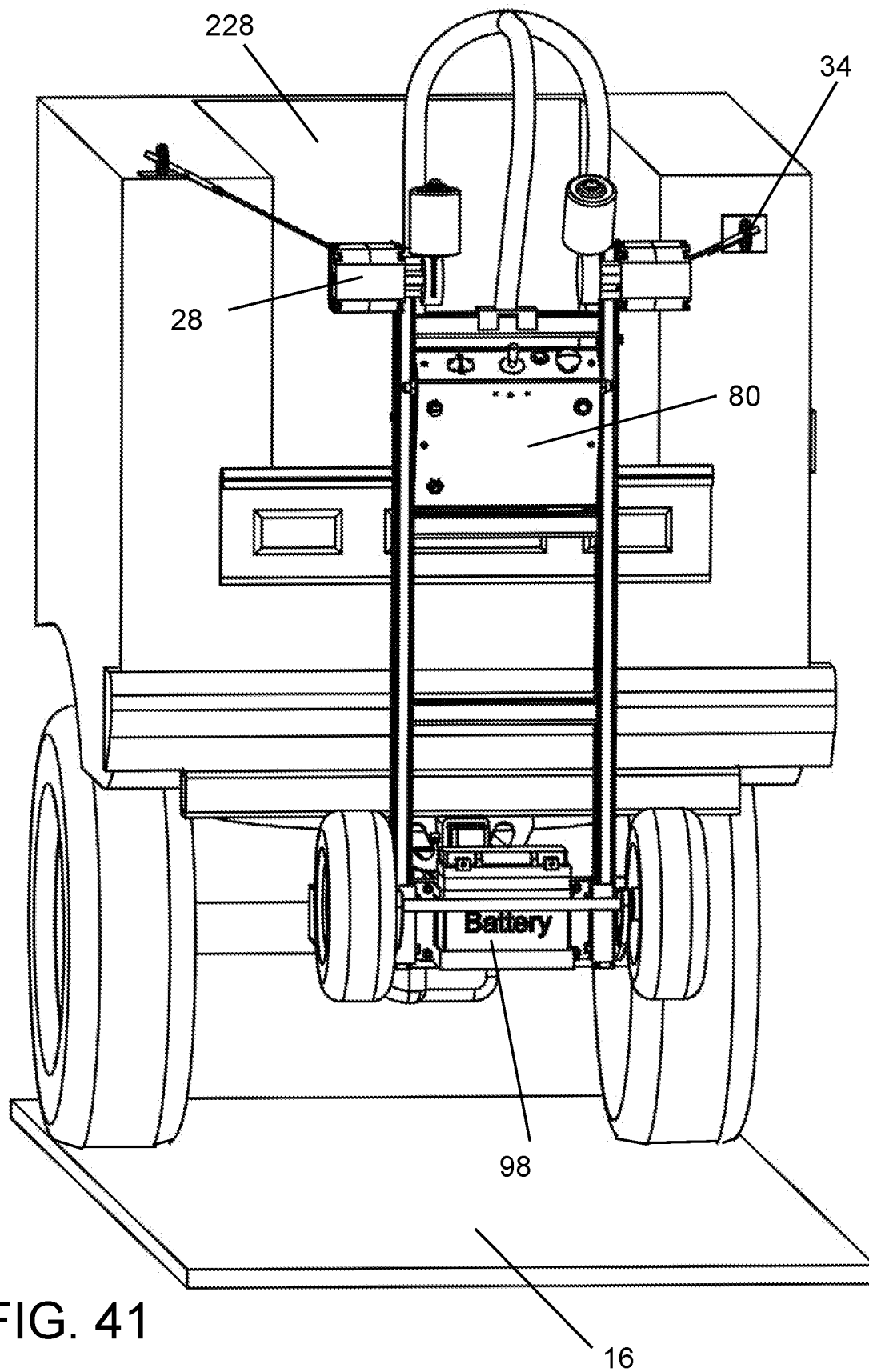


FIG. 40



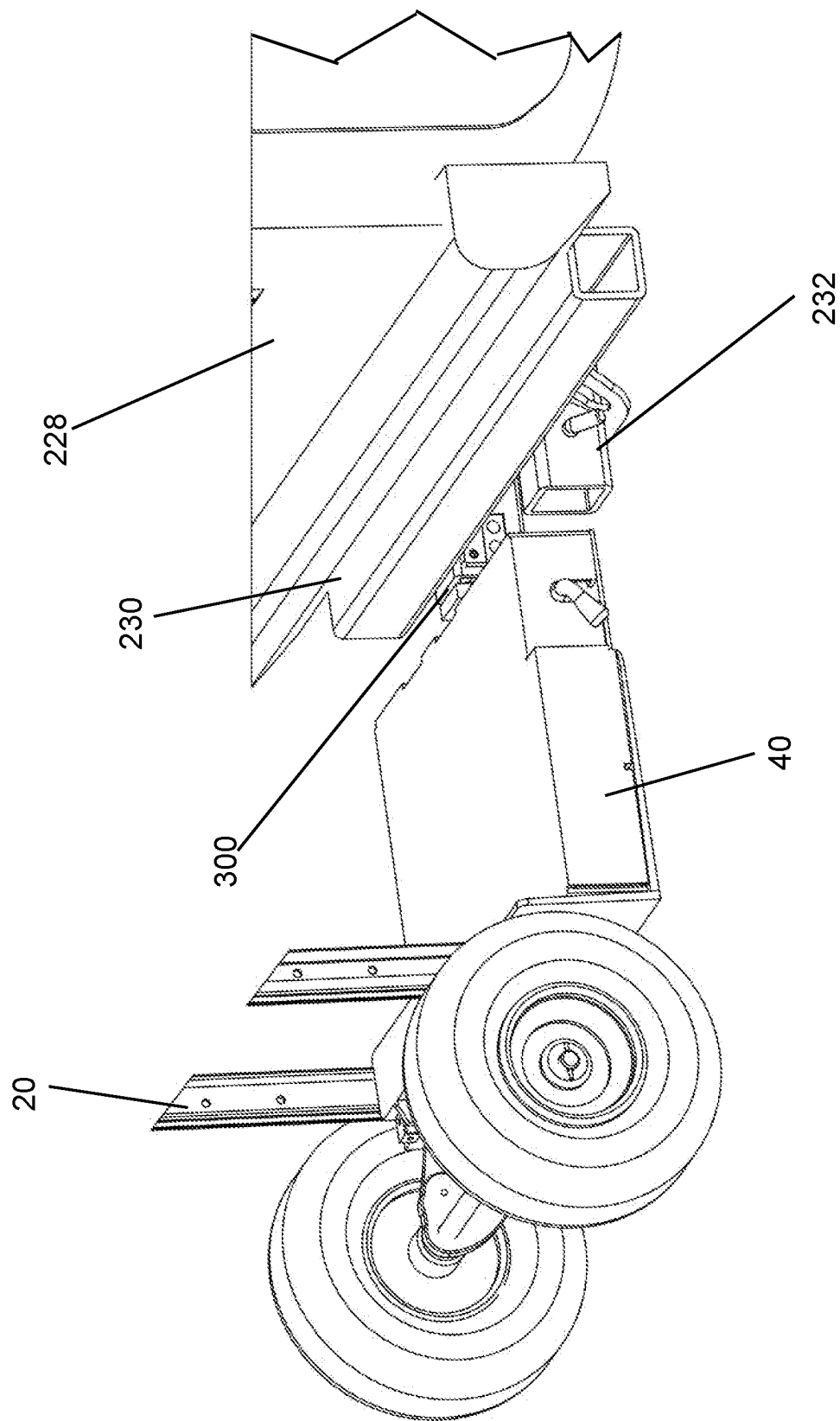


FIG. 42

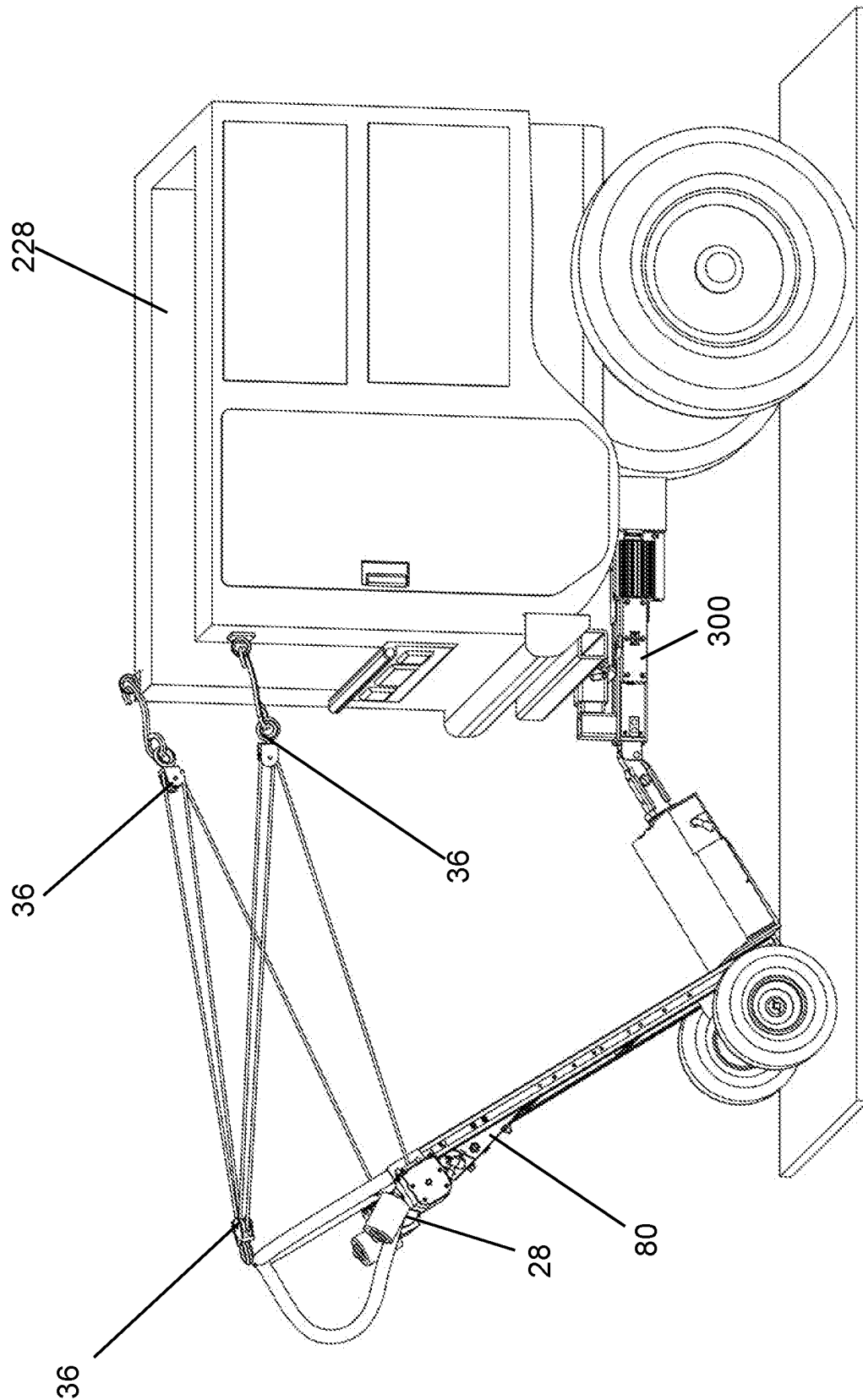


FIG. 43

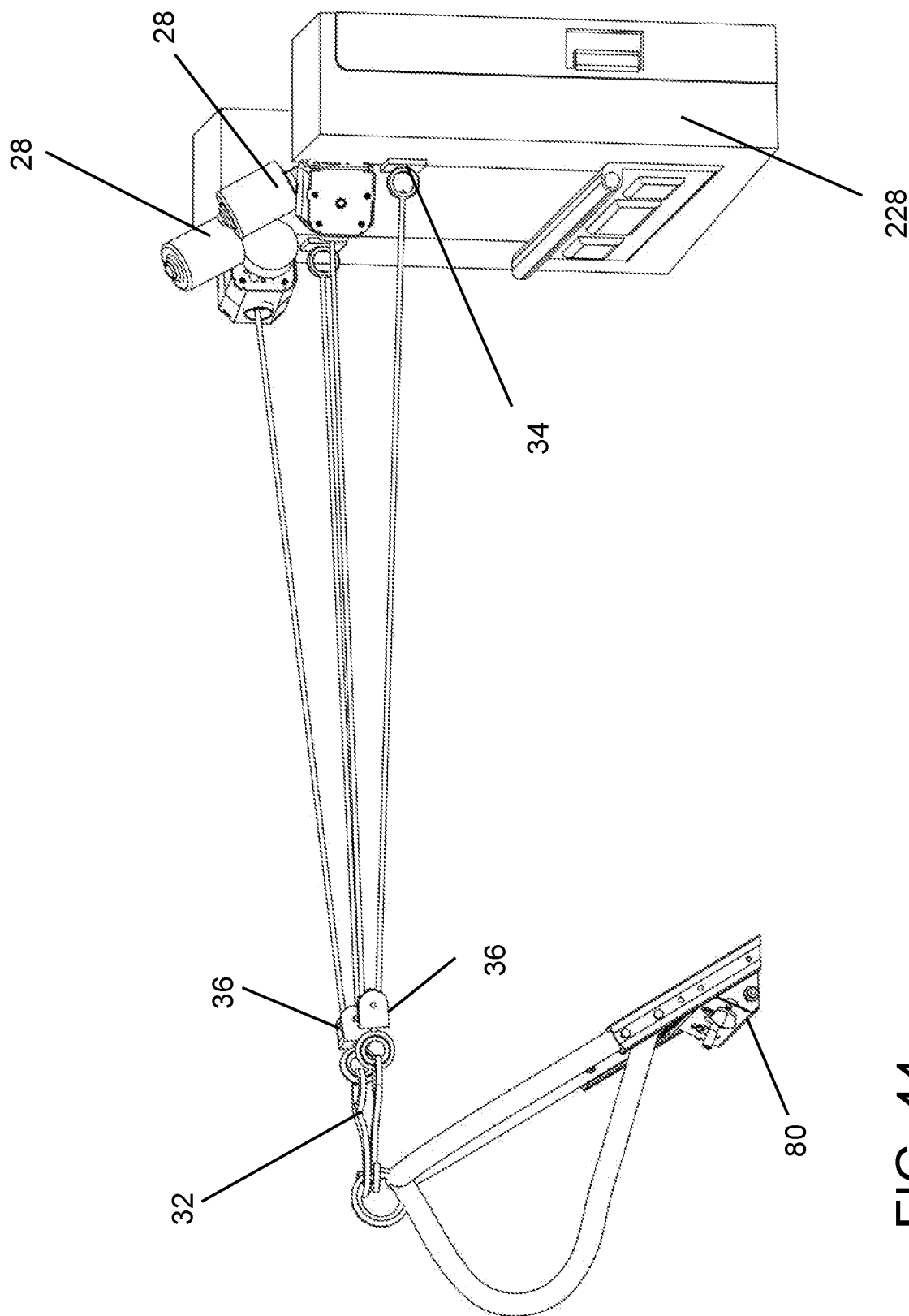


FIG. 44

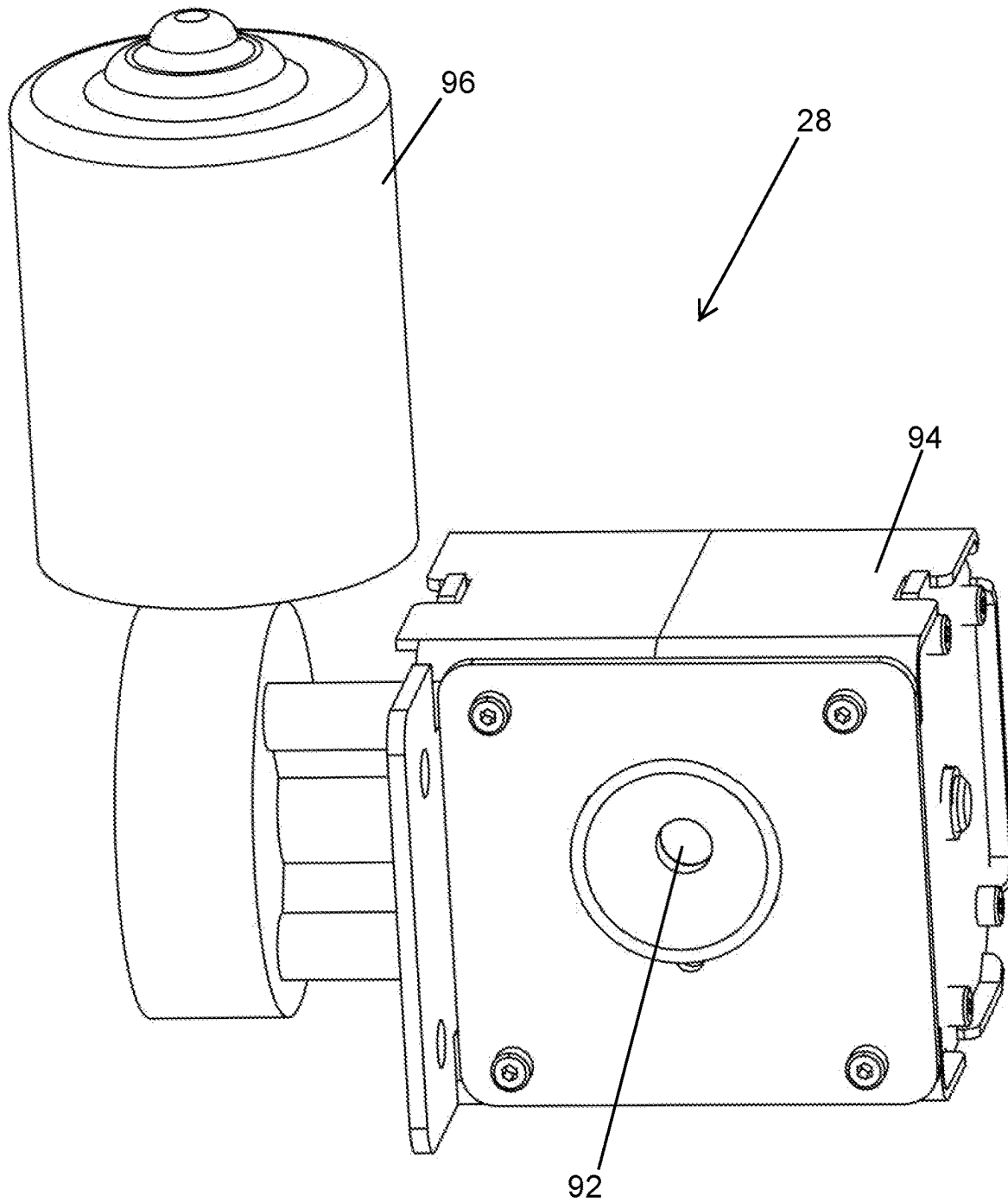


FIG. 45

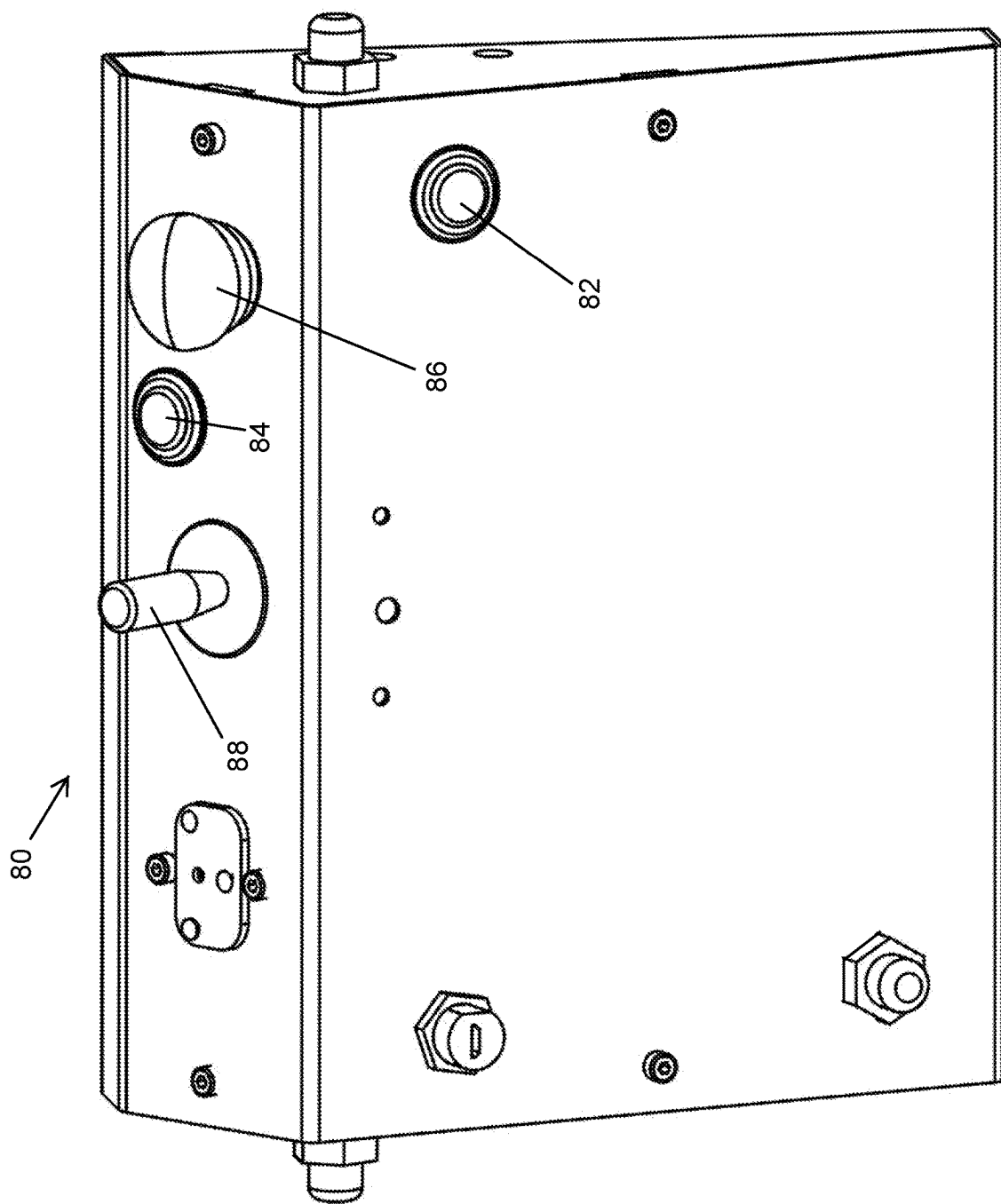


FIG. 46

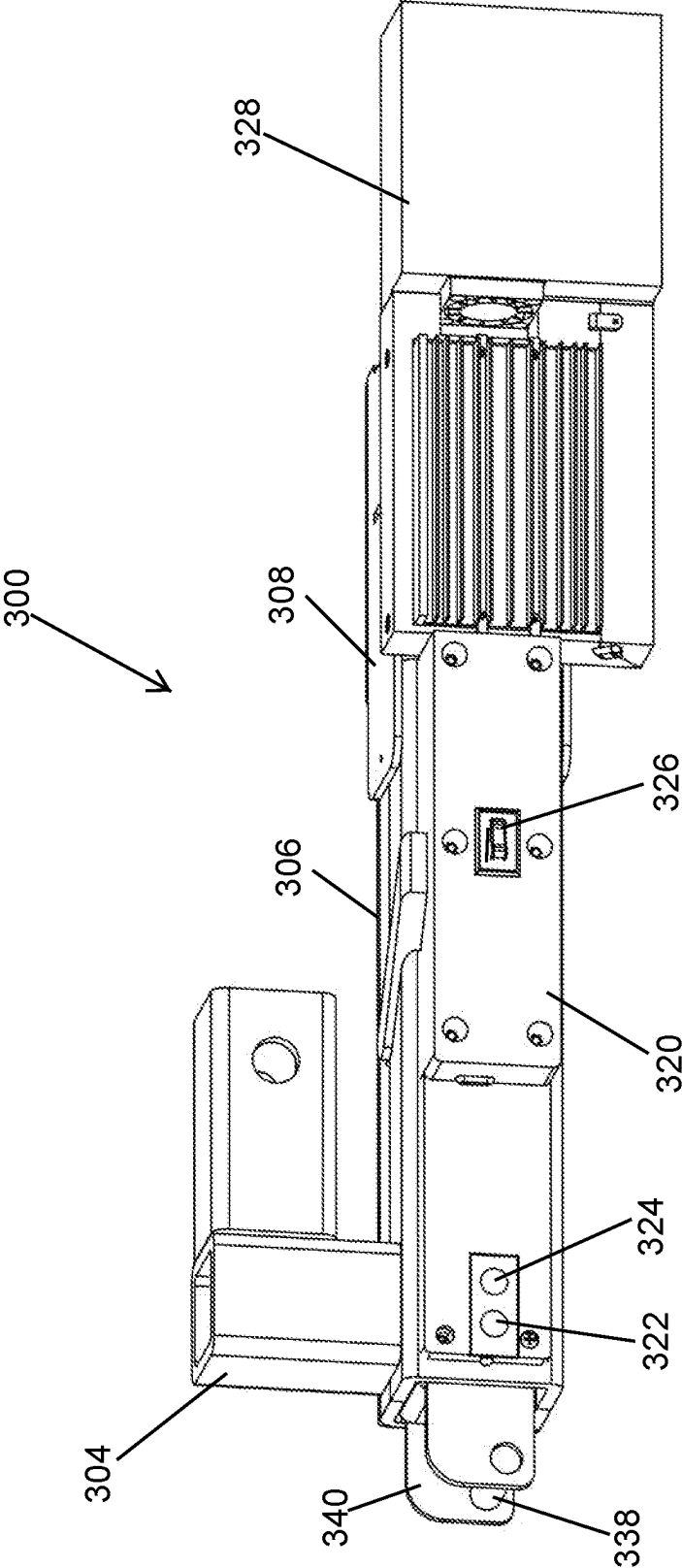


FIG. 47

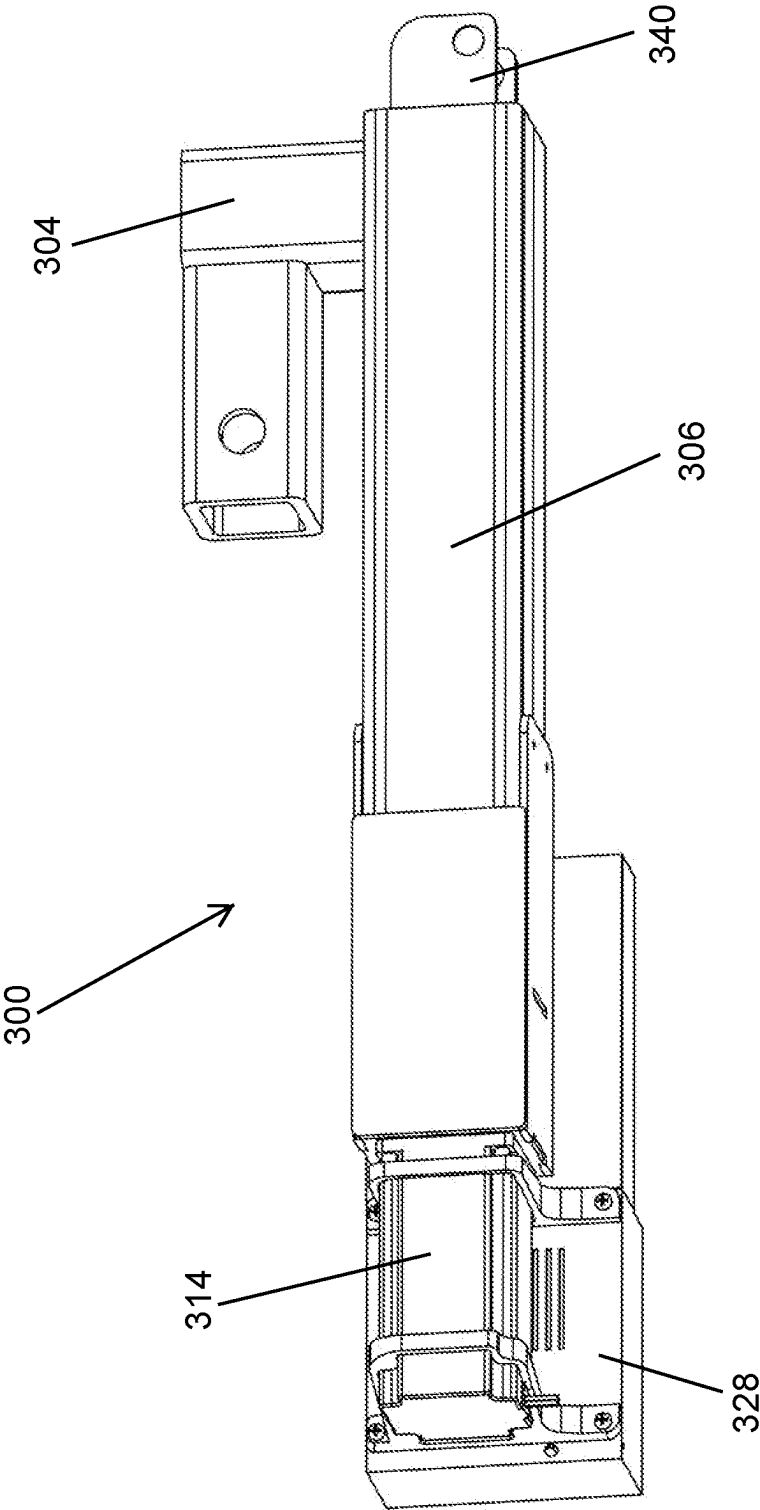


FIG. 48

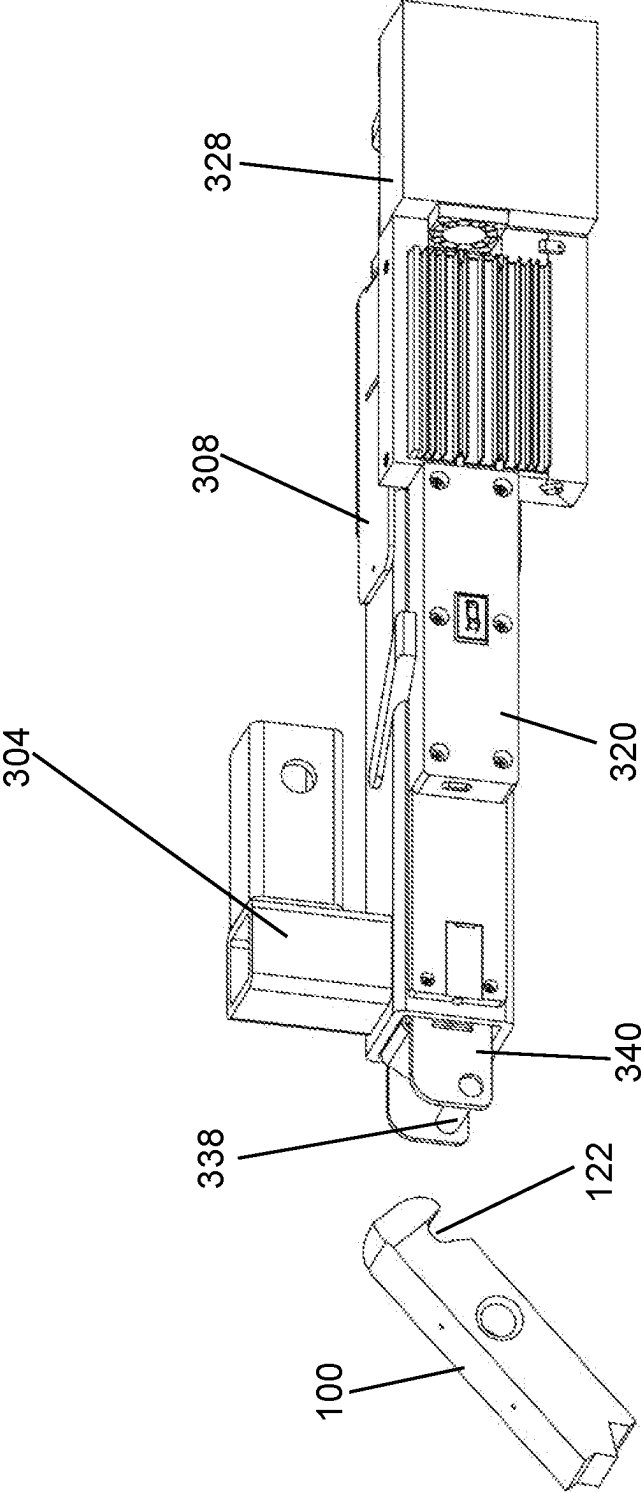


FIG. 49

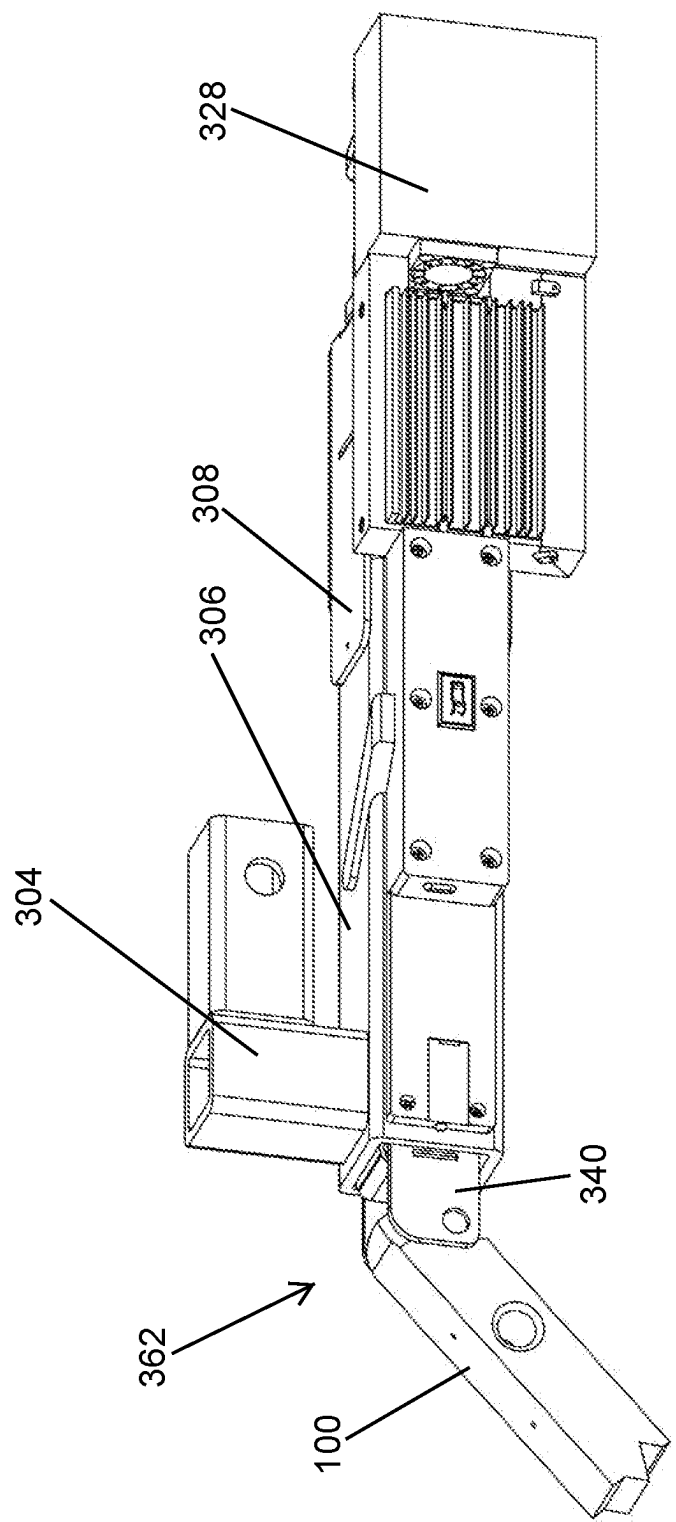


FIG. 50

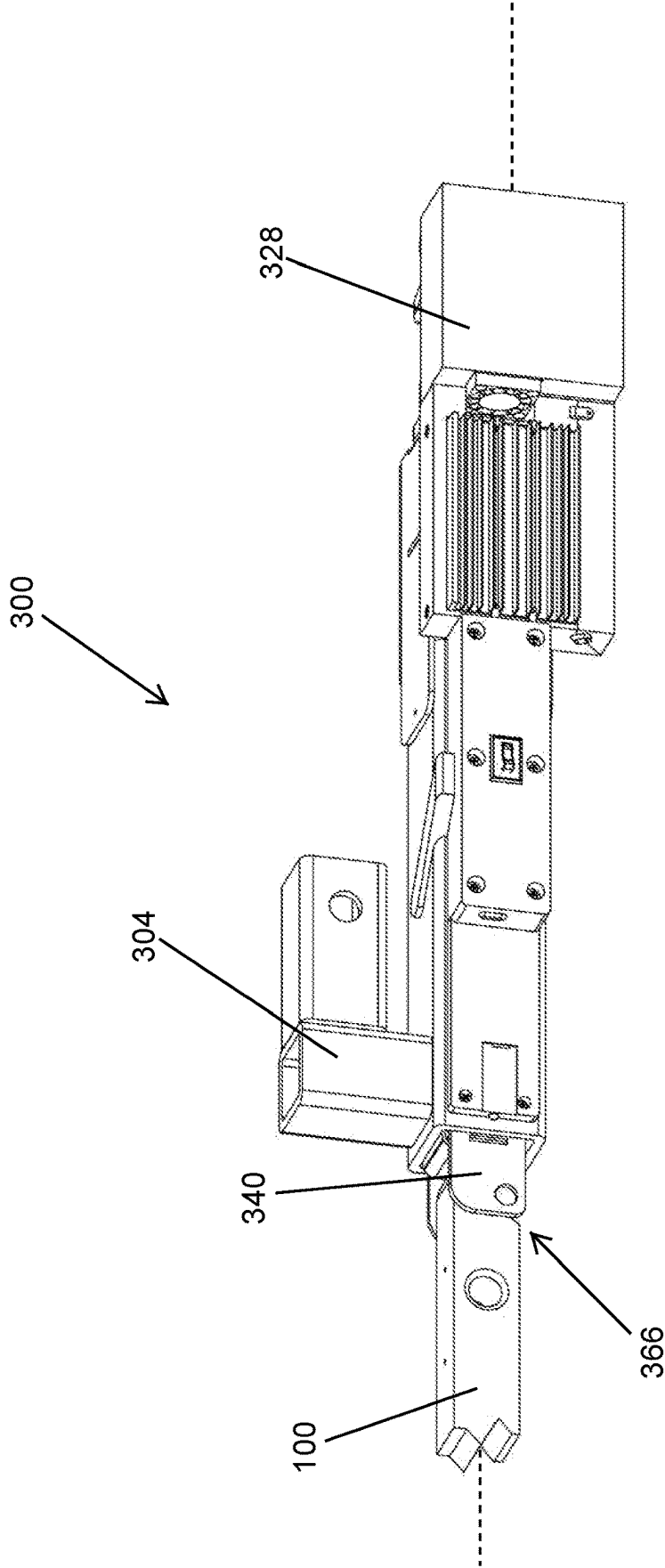
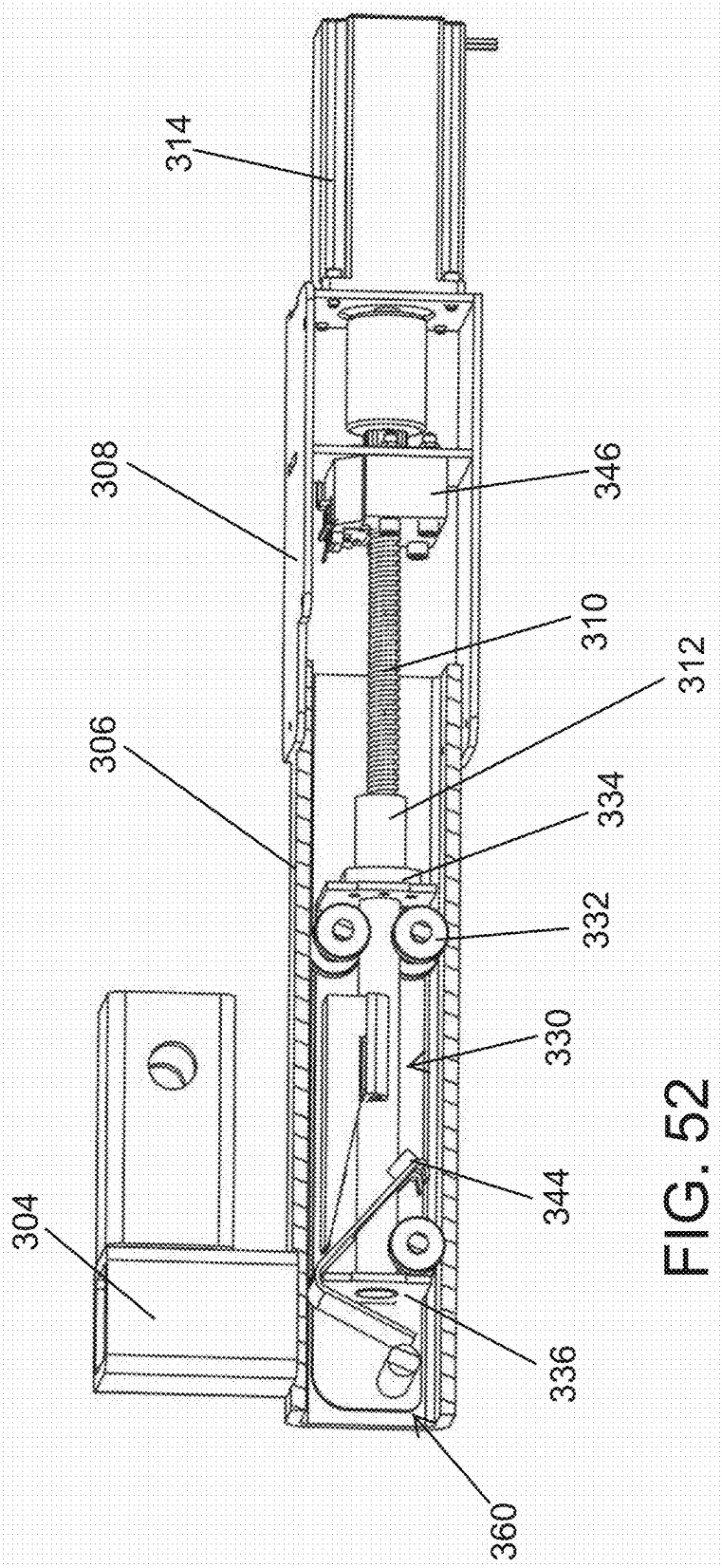


FIG. 51



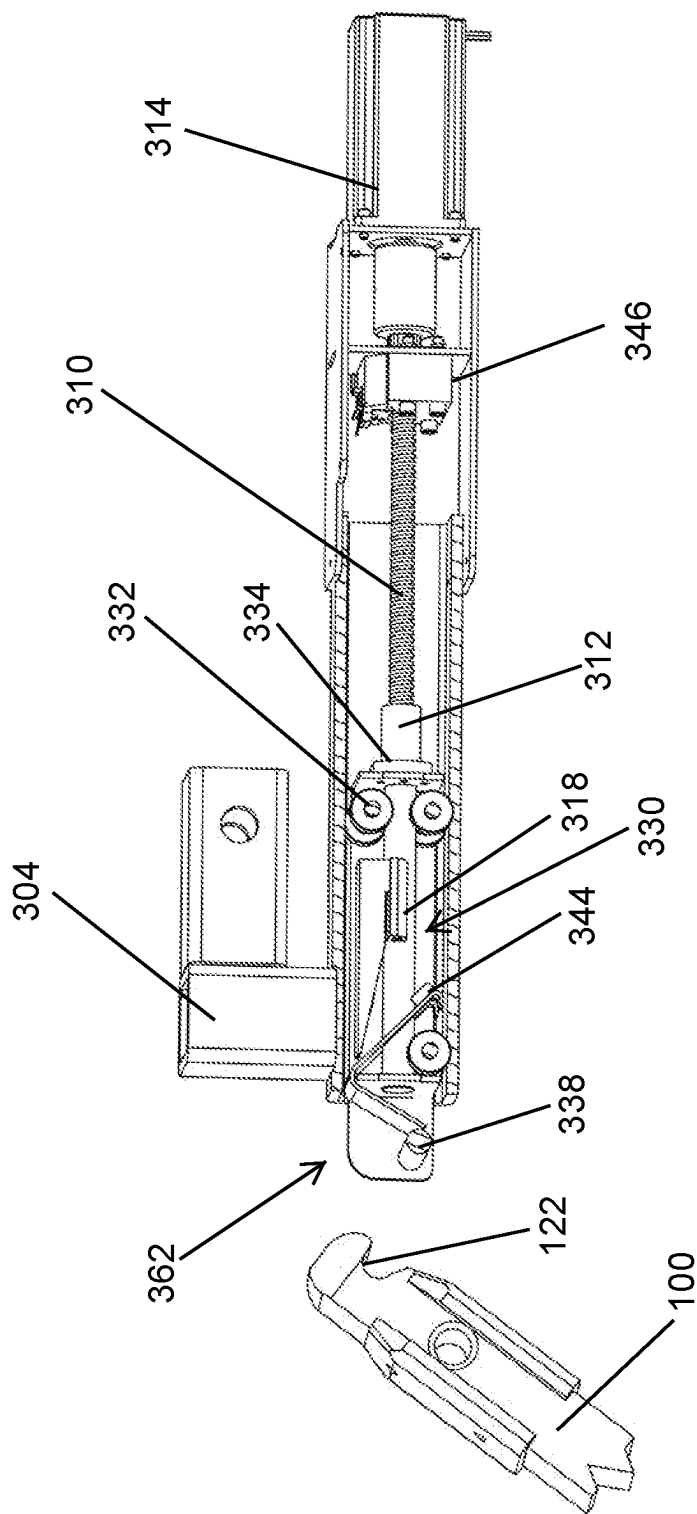


FIG. 53

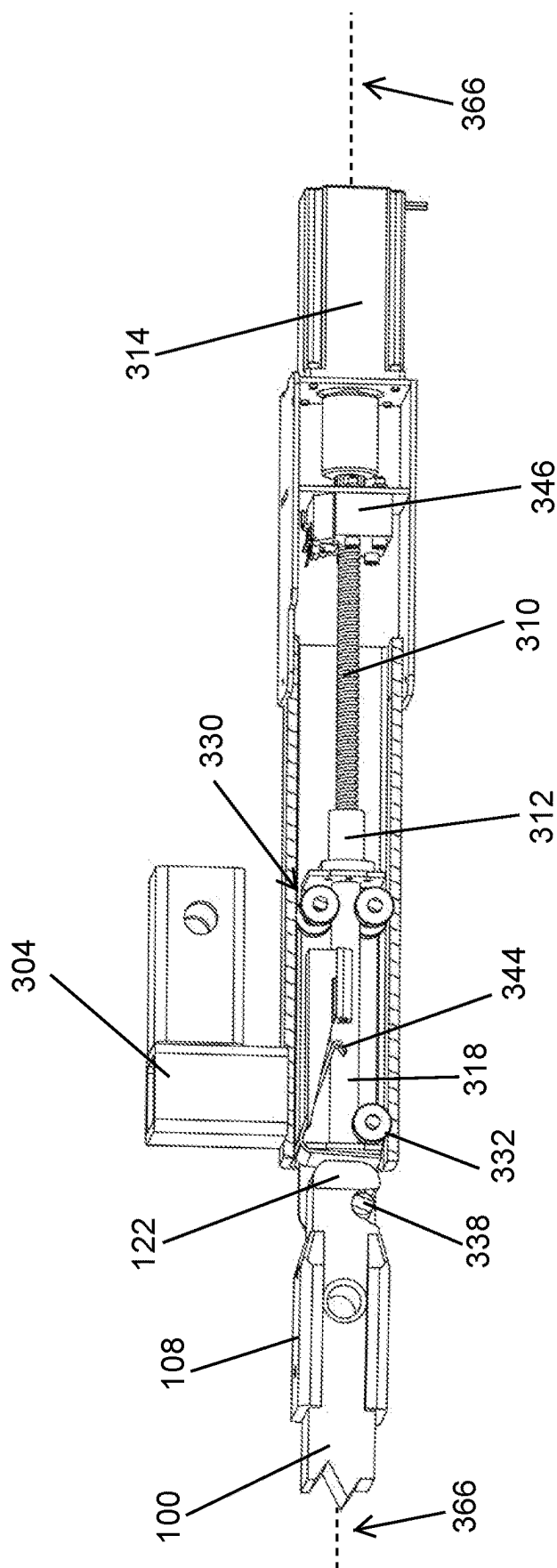
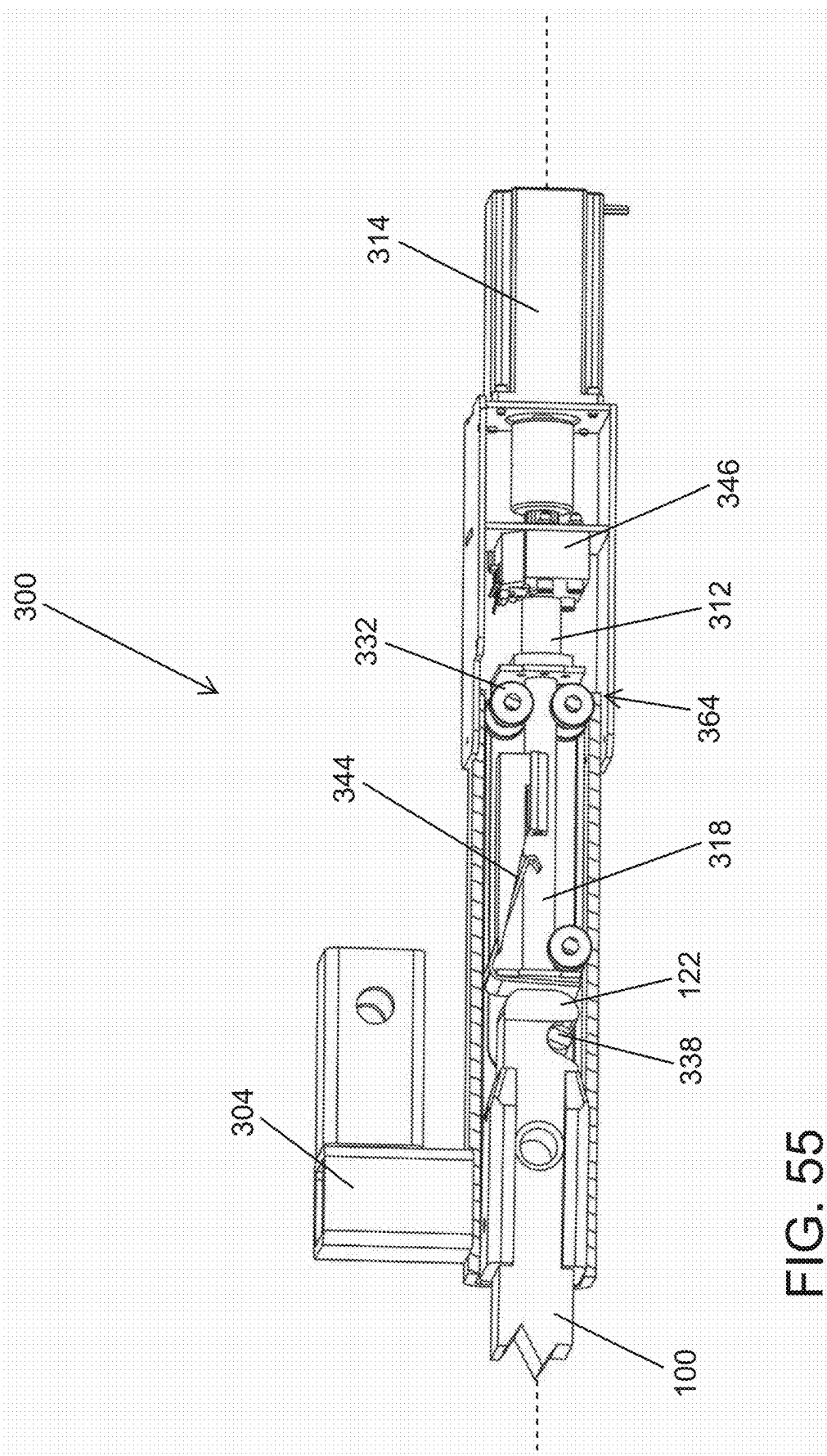


FIG. 54



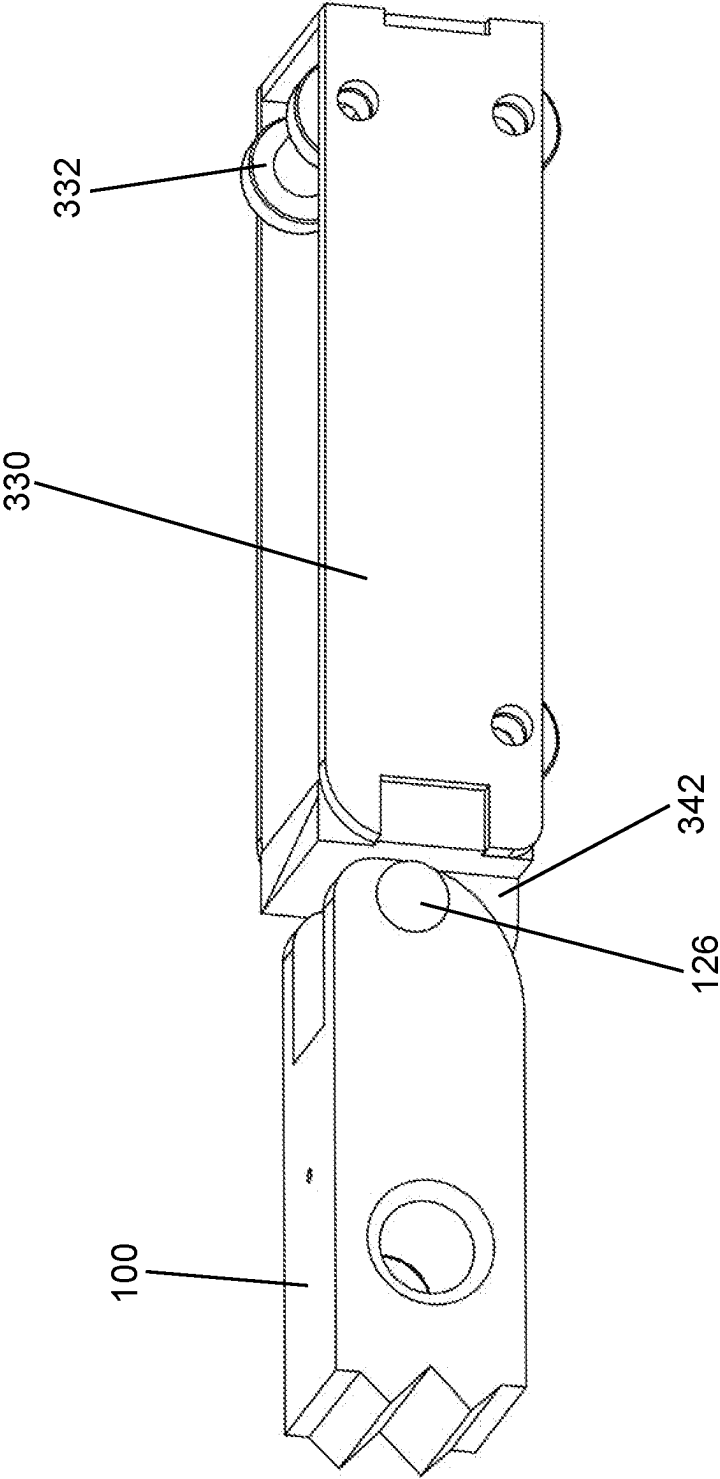


FIG. 56

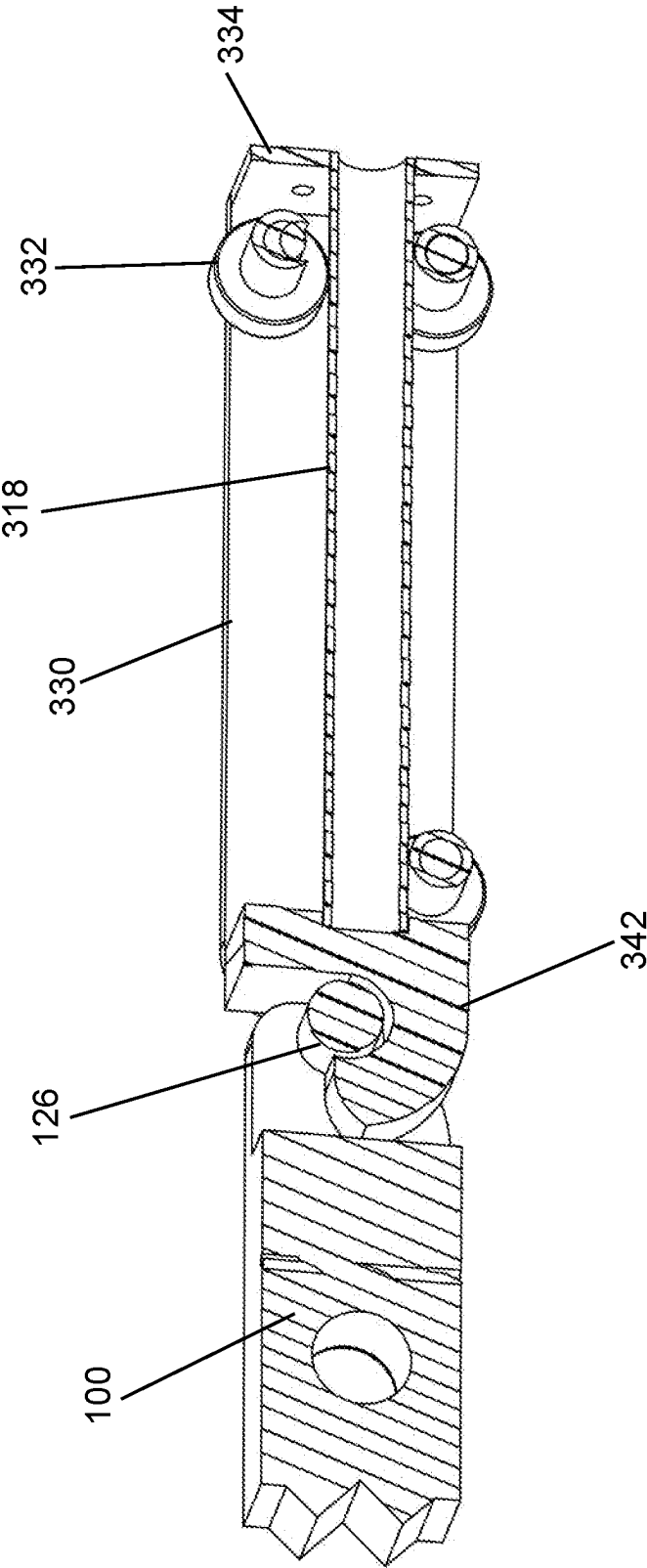


FIG. 57

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**HAND TRUCK AUTOMATED LOAD
UNLOAD****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application claims the filing benefit and priority of U.S. Provisional Patent Application Ser. No. 62/848,021 filed on May 15, 2019 and U.S. Non-Provisional patent application Ser. No. 16/727,981 filed on Dec. 27, 2019 the contents of which are incorporated herein by reference in their entireties.

FEDERAL SPONSORSHIP

Not Applicable

JOINT RESEARCH AGREEMENT

Not Applicable

TECHNICAL FIELD

The present invention relates to an interconnection between a vehicle and a connecting hitch. More particularly, the invention relates to a hitch or receiver and receiver interconnection that is utilized to join a hand truck or dolly to a vehicle. The receiver assembly in accordance with the present invention allows a user to wheel a loaded dolly to a vehicle and secure the dolly to the vehicle receiver for transport, without requiring ramps or a linear vertical lifting of the dolly to a height of the vehicle's hitch receiver. In accordance with certain aspects of embodiments of the invention the receiver and hand truck may include an automated system to stabilize the hand truck, lift it into alignment with the receiver, and then lock it into place relative to the receiver and vehicle. Further, the system in accordance with aspects of the invention may automatically remove the hand truck from locked engagement with the vehicle.

BACKGROUND

Over the years various mechanism have been contrived to ease the burden of moving a heavy load. By way of example, a person may desire to load a piece of heavy goods into a vehicle's rearward compartment, box or bed. To move the heavy goods a person may utilize for example, ramps and a hand truck, cranes, lift gates, multiple people to lift the goods safely to the vehicle, or other hoisting mechanisms. Use of these mechanisms may require additional space within a vehicle to allow a person to load and unload the heavy goods. Further, use of these mechanisms often requires more than one person to load or unload the heavy goods. External cargo racks have been attached to a vehicle receiver, however, when loading the cargo rack vertical lifting of the load is required to load the cargo rack at the height of the vehicle receiver.

Alternatively, a ramp may be required to move a heavy load from the ground to the height of the cargo rack engaged to the vehicle receiver. The present invention provides an apparatus that one person may use without ramps to secure and remove heavy equipment from an exterior of a vehicle while utilizing a hitch receiver attached to the vehicle. Further, the invention allows one user to couple and uncouple a hand truck to a vehicle hitch receiver without the need for ramps, lift gates, or multiple persons. Additionally,

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the invention includes a lock mechanism that ensures that portions of the invention do not unintentionally pull out of the vehicle while unloading the hand truck.

SUMMARY

Embodiments of the invention include a device to allow a single person to be able to load/unload a hand-truck or dolly to/from a receiver or hitch while reducing the required lifting force. Embodiments of the invention further automate the loading and unloading the hand truck to/from the vehicle's receiver. The device remains engaged to the hand truck when the hand truck is coupled to the vehicle receiver. The apparatus attaches to a hand-truck and allows for a "shaft" to reach up to vehicle receiver while the hand truck is at ground level. The vehicle receiver may be incorporated into a trailer hitch receiver or may be attached to a vehicle independent of the vehicle's trailer hitch. The shaft attaches to a "glide" inserted into the receiver allowing the load secured on hand truck to be raised off the ground, then slid into the receiver for secure attachment to the vehicle. The glide may be coupled to a drive mechanism to further assist the user in loading and unloading the hand truck.

According to aspects of the invention the apparatus is well suited for connecting a hand truck or dolly to a receiver attached to a vehicle. The apparatus includes a base, a lever arm and a glide. The base has a hand truck receiving portion to which a hand truck may be attached. The lever arm extends from the base. A free end portion of the lever arm extends outward from the base and terminates in a claw. The free end portion is adapted to be received within the receiver. The glide is separable from the lever arm and receiver, wherein the glide is adapted to be received within the receiver. The glide has a pivot member that is adapted to couple with the claw of the lever arm.

Those skilled in the art will appreciate that the lever arm may instead include the pivot member and the claw incorporated into the glide. A vehicle's trailer hitch receiver may be modified so that the receiver is long enough to contain the glide and a portion of the lever arm when the dolly is connected to the vehicle.

Embodiments according to aspects of the invention may additionally include an end portion of the lever arm that is slidably extendable from the base. Also, a lever lock may be coupled to the base in a manner having two positions: a first locked position that engages the lever arm and a second extending position that disengages the lever arm from the lock. The base may further include a storage slot formed in the base that is well suited and adapted for storing the glide within the base. The glide may further include a button ball release that compresses into the glide and actuates outward from the glide when a compressing force is not present. The button ball release engages a hole formed in the hitch receiver for the hitch pin. When compressed, the glide may be slid further into the hitch receiver. Low friction wear plates may be attached to sides or top and bottom of the glide and lever arm to facilitate smooth sliding of the lever arm and glide within the hitch receiver. The lever arm may further include an aperture extending through a sidewall of the lever arm, wherein the aperture adaptable to receive a hitch pin. The hitch pin may be used to lock base and lever arm to the hitch receiver of the vehicle. In this manner, a hand truck may be locked in place to the back end of a vehicle.

A further embodiment according to aspects of the invention includes a base, lever arm and glide. The base is adapted to couple with a hand truck or dolly. The lever arm is

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extendable from the base, wherein the lever arm has an end portion slidably engaged with the base and a free end portion extending outward from the base. The free end portion includes a claw or pivot wherein the free end portion is adapted to be received within a trailer hitch receiver. The glide is separate from the lever arm and trailer hitch receiver, wherein the glide is adapted to be received within the trailer hitch receiver and has a pivot member or claw that is adapted to couple with the claw or pivot of the lever arm.

Embodiments according to aspects of this invention may further include a lever lock coupled to the base. The lock has a first locked position that engages the lever arm and has a second extending position that disengages the lever arm from the lock. When the lock is in the extending position the lever arm may slide outward from the base. A storage slot may be formed in the base and dimensioned to receive and store the glide within the base. The glide may further include a compressible button ball release that compresses inward into a hollow portion the glide. The button ball release is sized to engage within a hitch pin hole formed in the vehicle receiver. When engaged, the button ball retains the glide in a fixed position within the vehicle receiver. A user may compress the button ball release and slide the glide further into the vehicle receiver. The lever arm includes an aperture extending through a sidewall of the lever arm, wherein the aperture adaptable to receive a hitch pin. The claw of the lever arm may engage the pivot pin of the glide. The base may be pushed such that the lever arm presses against the glide and slides into the vehicle's receiver until the aperture of the lever arm aligns with the hitch pin hole of the receiver.

A still further embodiment according to aspects of the invention includes a base, lever arm and glide that act together to connect a hand truck to the trailer hitch receiver of a vehicle. The base includes a hand truck receiving portion that couples or is fixed to a hand truck. The lever arm slides within a channel formed in the base and may extend between a stowed position and a lift position. The lever arm has a rearward end portion that slides within the base. On the opposite end of the lever arm is a free end portion that extends outward from the base and includes a claw or pivot formed on the end portion. The free end portion and claw or pivot are adapted to be received within the trailer hitch receiver. The glide is separable from both the lever arm and trailer hitch receiver. The glide is adapted to be received within the trailer hitch receiver. The glide has a mating pivot member or claw adapted to couple with the respective claw or pivot of the lever arm.

The glide also includes a compressible button ball release that engages the trailer hitch receiver to lock the glide in the receiver during loading. Additionally, a lever lock is coupled to the base. The lock has a first locked position that engages the lever arm to restrict movement of the lever arm within the base. The lock has a second extending position that disengages the lever arm from the lock to allow the sliding and extension of the lever arm from the base. Low friction wear plates may be attached to sides of the glide and lever arm to reduce the required force to push the glide and lever arm into the receiver.

In accordance with further aspects of the invention the base may be solidly attached (welded) to a hand-truck or may be fastened or otherwise coupled to the tongue portion of a hand truck. The lever arm or shaft protrudes and retracts into base to allow for increased vertical reach when mounting the hand-truck to the vehicle hitch. The glide has ball catches that may be used to temporarily hold the glide in place when attempting to hook up the lever arm or shaft to the glide. The glide is designed to reduce drag when the

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weight of the load on the hand-truck transfers to glide. Also, the lever arm or shaft is designed to reduce drag while sliding in and out of the receiver under load, while maintaining tight clearances for exacting operation. The lever arm or shaft is designed to lock into an extended position while the hand truck is tilted back and engaged to the glide. The lock is spring loaded to allow the lever arm or shaft to retract into body and automatically stop when in mounted or travel ready position.

A compartment or slot is formed in the base allowing the glide to be stored in the body or base in a secure location during storage. The glide slides out of receiver when unloading, stopping at hitch pin hole when the compressible ball catches engage the hitch pin holes. Once the glide is locked the hand cart may be tilted until the wheels reach the ground. Those skilled in the art will appreciate that the apparatus of the invention is not limited to any specific size dolly or hand-cart and the apparatus can be scaled to accommodate different load sizes and weights. The lever arm is fixed to the vehicle receiver with a hitch pin and is not limited to size or hole setback distance. The base and hand cart allow for multiple methods of securing a load to a vehicle.

In accordance with aspects of the invention, an embodiment of the invention includes an apparatus for autonomously connecting and disconnecting a hand truck to a vehicle with minimal user interaction. The apparatus includes a hand truck, a lever arm, a receiver, a glide, and a drive mechanism. The lever arm is coupled to a base of the hand truck. A free end portion of the lever arm extends outward from the hand truck. The receiver is adapted for coupling to a vehicle. By way of example, the receiver may be welded to the vehicle or may couple with a receiver already fixed to the vehicle. The glide is contained within the receiver and slides within the receiver between a storage position, a load position (the glide partially extends from the receiver), and a retracted position (the glide is fully contained within the receiver).

The drive mechanism is fixed or coupled to the glide and actuates to linearly displace the glide within the receiver between the loading position and the retracted position. A free end of the glide and lever arm link together allowing the glide to pull the lever arm into the receiver when the drive mechanism actuates the glide inward. Similarly, when the drive mechanism actuates the glide outward, the glide pushes against the lever arm and forces the lever arm out of the receiver. Without limitation intended, the linkage between the free end of the lever arm and the glide may include a pivot and claw arrangement. Either the glide or lever arm includes a claw and then the other of the glide and lever arm has a pivot member that is adaptable to link or couple with the claw.

In accordance with further aspects of the invention, two spaced apart winches are attached to either the hand truck or the vehicle. Hooks at the end of each winch line are connected to the opposing vehicle or hand truck. When the lever arm is linked to the glide the winches spool in the winch line which cause the hand truck to rotate upwards. A controller is electrically linked with the winches and drive mechanism, wherein the controller provides a signal to control coordinated activation of the winches and drive mechanism. Further, sensors are positioned on the hand truck and receiver, which provide signal outputs to the controller. The signal output is processed within the controller to determine and correlate a three-dimensional orientation of the hand truck and receiver. By way of example and without limitation intended, the device orientation may be determined using device orientation sensors such as

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micromechanical gyroscopes or accelerometers and geo-magnetic field sensors. With output from the device orientation sensors the controller controls activation of the winches to stabilize the hand truck and orient the hand truck to align the length axis of the lever arm with the longitudinal axis of the receiver. In certain embodiments the drive mechanism includes a linear actuator that has a lead screw, lead nut, and stepper motor of known suitable construction.

The accompanying drawings, which are incorporated in and constitute a portion of this specification, illustrate embodiments of the invention and, together with the detailed description, serve to further explain the invention. The embodiments illustrated herein are presently preferred; however, it should be understood, that the invention is not limited to the precise arrangements and instrumentalities shown. For a fuller understanding of the nature and advantages of the invention, reference should be made to the detailed description in conjunction with the accompanying drawings.

DESCRIPTION OF THE DRAWINGS

In the various figures, which are not necessarily drawn to scale, like numerals throughout the figures identify substantially similar components.

FIG. 1 is a top front perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;

FIG. 2 is a left side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;

FIG. 3 is a right side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;

FIG. 4 is a left back perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled and having a portion of a hand truck fixed to the base;

FIG. 5 is a bottom front perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;

FIG. 6 is a bottom left side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;

FIG. 7 is a bottom right side perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base, glide and receiver aligned but uncoupled;

FIG. 8 is a partial section bottom perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base and glide aligned but uncoupled;

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FIG. 9 is a bottom perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle in accordance with an embodiment of the present invention, showing the base tilted and coupled to the glide and showing the glide aligned but uncoupled to the receiver;

FIG. 10 is a partial sectional bottom perspective view of an apparatus for connecting a hand truck to the trailer hitch receiver of a vehicle of the type shown in FIG. 9, showing the base tilted and coupled to the glide and showing the glide aligned but uncoupled to the receiver;

FIG. 11 is a partial exploded top plan view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

FIG. 12 is a partial exploded side perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

FIG. 13 is a partial exploded front bottom perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

FIG. 14 is a partial exploded side view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

FIG. 15 is a partial exploded bottom side perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

FIG. 16 is a partial exploded top side perspective view of an apparatus in accordance with an embodiment of the present invention illustrating a hand cart coupled to a base and aligned with a lever arm, glide and receiver in accordance with aspects of the invention;

FIG. 17 is a perspective view of a hand-truck and lift device in accordance with an embodiment of the present invention shown in a pre load position;

FIG. 18 is a perspective view of the hand-truck and loading device of the type shown in FIG. 17 with the receiver glide inserted into the receiver;

FIG. 19 is a rear perspective view of the hand truck and loading device of the type shown in FIG. 18;

FIG. 20 is a perspective view of the hand truck of the type shown in FIG. 18 and illustrated as tilting back at an approximately 40° angle in preparation to attach the lever arm or shaft to the glide or receiver glide;

FIG. 21 is a side perspective view of the hand truck base of the type illustrated in FIG. 20 and shown tilted back and attached or coupled to the glide device;

FIG. 22 is a rear perspective view of the hand truck of the type illustrated in FIG. 21;

FIG. 23 is a perspective view of the hand truck, base and lever arm in accordance with the present invention and shown lifted up to the same height or level as the receiver;

FIG. 24 is a bottom perspective view of the invention of the type illustrated in FIG. 23;

FIG. 25 is a perspective view of the invention of the type illustrated in FIG. 23 and further showing the lever arm retracted into the body or base of loading device of the present invention;

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FIG. 26 is a bottom perspective view of the device illustrated in FIG. 25 illustrating the hand truck lifted up and the lever arm or shaft retracting into the body or base of loading device;

FIG. 27 is a perspective view of the apparatus in accordance with the present invention showing the lever arm or shaft fully inserted into a vehicle receiver and ready to secure with hitch pins and straps;

FIG. 28 is a bottom perspective view of the invention of the type illustrated in FIG. 27;

FIG. 29 is a side perspective view illustrating a glide sticking out of receiver and having the lever arm or shaft attached to guide;

FIG. 30 is a side perspective view of an apparatus in accordance with aspects of the invention showing the lever arm attached to glide and ready for a load to be raised and slid into vehicle receiver;

FIG. 31 is a side perspective view of an apparatus in accordance with aspects of the invention showing the hand truck coupled to the vehicle receiver (in a loaded position) and secured at a back of vehicle receiver and ready for transport;

FIG. 32 is a back perspective view of a hand truck in accordance with aspects of the invention showing winch lines extending from winches coupled to the hand truck;

FIG. 33 is a side perspective of a drop down style auto lift/load receiver of the present invention;

FIG. 34 is a perspective view showing an auto lift/load receiver of the present invention shown fixed to a rear portion of a vehicle frame;

FIG. 35 is a perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a preload orientation;

FIG. 36 is a side perspective view of a hand truck in accordance with aspects of the invention;

FIG. 37 is a side perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation;

FIG. 38 is a perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation;

FIG. 39 is a side perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated orientation and having the lever arm axis aligned with the longitudinal axis of the receiver;

FIG. 40 is a perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated orientation and having the lever arm axis aligned with the longitudinal axis of the receiver;

FIG. 41 is a rear perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated orientation and having the lever arm axis aligned with the longitudinal axis of the receiver;

FIG. 42 is a partial section perspective view of a hand truck linked with an auto lift/load receiver in accordance with aspects of the invention and shown in a fully stowed or travel orientation;

FIG. 43 is a side perspective view of a hand truck aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation and having a block and tackle configuration incorporated with the winches;

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FIG. 44 is a partial section perspective view of a hand truck in accordance with aspects of the invention and shown having a block and tackle configuration incorporated with the winches;

FIG. 45 is a perspective view of a winch embodiment in accordance with aspects of the invention;

FIG. 46 is a perspective view of a controller embodiment in accordance with aspects of the invention;

FIG. 47 is a right side perspective view of a drop down style auto lift/load receiver of the present invention;

FIG. 48 is a left side perspective view of a drop down style auto lift/load receiver of the present invention;

FIG. 49 is a partial section perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a preload orientation;

FIG. 50 is a partial section perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an initial linked orientation;

FIG. 51 is a partial section perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated and aligned orientation;

FIG. 52 is partial sectional perspective view of a drop down style auto lift/load receiver of the present invention;

FIG. 53 is a partial sectional perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a preload orientation;

FIG. 54 is a partial sectional perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in an elevated and aligned orientation;

FIG. 55 is a partial sectional perspective view of a lever arm aligned with an auto lift/load receiver in accordance with aspects of the invention and shown in a fully stowed or travel orientation;

FIG. 56 is a perspective view of a portion of the lever arm and glide in accordance with aspects of the invention and shown linked together; and

FIG. 57 is a partial section perspective view of a portion of the lever arm and glide in accordance with aspects of the invention and shown linked together.

DETAILED DESCRIPTION

The following description provides detail of various embodiments of the invention, one or more examples of which are set forth below. Each of these embodiments are provided by way of explanation of the invention, and not intended to be a limitation of the invention. Further, those skilled in the art will appreciate that various modifications and variations may be made in the present invention without departing from the scope or spirit of the invention. By way of example, those skilled in the art will recognize that features illustrated or described as part of one embodiment, may be used in another embodiment to yield a still further embodiment. Thus, it is intended that the present invention also cover such modifications and variations that come within the scope of the appended claims and their equivalents.

The hand truck vehicle mounting assembly 10 of the present invention is particularly well suited for externally coupling a hand truck or dolly to a receiver mounted to a vehicle. In some embodiments a vehicle's trailer hitch receiver or a trailer hitch receiver having an extended receiver is suitable for use with the invention. The present

invention is further particularly well suited to allow a single user to couple a heavily loaded dolly to the vehicle hitch without requiring simple vertical lifting of the dolly to the height of the receiver. By way of example, a loaded dolly is tilted backward and then positioned so that a lever arm engages with a glide that is engaged with the vehicle hitch receiver. As the user rotates the dolly upward towards the vehicle, the glide slides into the receiver and the lever arm also slides into the receiver. The lever arm and glide of the present invention reduces the amount of lifting force required to lift the dolly to the height of the vehicle receiver.

Turning attention now to the Figures, embodiments according to aspects of the invention will now be described. Referring first to FIGS. 1-7, the hand truck vehicle mounting assembly 10 generally includes a base 40, lever arm 100, and glide 150 that may couple to a vehicle hitch receiver 200. In an embodiment of the invention the base has a hollow interior 42 with dividing walls 44 extending between the sides of the base to provide stability and support to the base. The top surface 48 of the base is particularly well suited for supporting a load. The bottom of the base includes an attachment flange 46 extending inward from the sidewalls. Base plate 52 may be coupled to the flange 46 with a weld, fasteners, or other known suitable coupling. The base plate may be further engaged and coupled to the tongue of the dolly or may form a portion of the tongue of the dolly. Further, a back side of the base 40 may include a backing plate 50 or, alternatively, a portion of the dolly may form or establish the back side of the base 40. The front side of the base includes an opening into a storage compartment 56 for the glide 150. Also internal to the base is a channel or tube 54 into which the lever arm 100 slides between a stowed position and extended position. Front extension stop 136 is formed on or coupled to lever arm 100. A blocker member 134 is attached to the front of the base at the opening of the channel or tube 54. The lever 100 is slidable through the blocker member 134 until the stop 136 contacts blocker 134. A rear extension stop 132 on the lever arm 100 inhibits the lever arm from being pulled completely out of the channel 54.

The lever arm 100 includes sides 102, side wear guides 106, top wear plate 108 and bottom wear plate 110. Free end 120 of the lever arm tapers to a nose 124. A claw 122 is formed on the lower portion of the free end 120 of the lever arm 100. Multiple hitch pin holes 130 extend from one side of the lever arm to the other. The multiple holes 130 allowing a user to orient the lever arm 100 relative to the base 40 and hitch receiver 200 in a variety of orientations. Independent glide 150 includes sidewalls 152, a hollow interior (not shown), a compression release button 156 and a latch 158 (that is pushed out of the glide via spring 160). An end of the glide 150 includes a pivot pin 170 to which the claw 122 engages. The glide 150 further includes top wear plate 176 and bottom wear plate 178. In another suitable embodiment, the wear plates may be eliminated by making the glide 150 from a solid wear resistant core (such as nylon) and having side plates and the pivot pin mounted to the core.

The wear plates 106, 108, 110, 176 and 178 may be constructed from a variety of known materials having wear resistant characteristics when sliding against metal and further having coefficients of friction that are lower than a metal surface. One such known suitable material is a carbon fiber filled nylon. The hitch receiver 200 includes tubular sidewalls 202 and a hitch pin hole 204 extending through both sides of the sidewalls 202. Throughout many of the Figures a ground or surface level is identified with numeral

16 and the vehicle or addition portions of the vehicle receiver frame are identified with dashed lines and numeral 230. Additionally, the orientation of the base relative to the ground 16 and height of the receiver 200 relative to the ground 16 are depicted in several of the Figures.

Referring next to FIGS. 8-10, lock pin 60 is mounted in base 40 with mounts 62. Lock pin 60 is able to slide back and forth through the mounts 62. An end of the pin 60 extends through lock pin hole 70 formed in the lever arm channel or tube 54 of the base 40. The end of the pin 60 engages and disengages within holes 130 extending through lever arm 100. Stop 66 retains spring 64 which returns the lock to the locked position. Groove 68 extends around the lock pin 60 and may be utilized to temporarily restrict the lock pin in a disengaged position. Lock pin cover or plate 72 is mounted or fastened to the bottom flange 46 of the base 40 to enclose the lock mechanism.

FIGS. 11-16 illustrates a linear alignment of the lever arm channel or tube 54 of the base 40, the lever arm 100, the glide 150 and the vehicle hitch 200. The lever arm 100 is shown completely removed from the base to further illustrate components of the lever arm 100 that are contained within the base 40. A portion of the lever arm 100 slides within and is contained by the lever arm channel or tube 54. When the lever arm is installed within the lever arm channel 54, a stop 132 having a perimeter larger than a perimeter of the hole extending through the blocker 134 restricts the lever arm from being removed completely from the channel. The blocker 134 may be attached to or formed as a part of the base 40. A front extension stop prevents or restricts the lever arm from sliding into the base further than where the stop 136 is positioned on the lever arm 100. Although two holes 130 are shown extending through the lever arm between the rear extension stop 132 and the front extension stop 136, those skilled in the art will appreciate that if desired more holes may be created in the lever arm. The holes 130 between the two stop are utilized in conjunction with the lock pin 60 to lock the lever arm into position having a desired length extending from the base. When two holes 130 are provided between the stops, the rearward hole may be utilized to lock the lever arm 100 in the extended or loading position and the forward hole 130 may be utilized to lock the lever arm in the retracted or travelling position. The hole 130 near the free end 120 of the lever arm 100 may be aligned with the hitch pin hole 204 of the hitch receiver 200. In this manner a hitch pin may be utilized to lock the lever arm 100 to the receiver 200.

FIGS. 17-28 illustrates the use of the assembly to mount a load of a hand truck 20 to a vehicle hitch receiver 200. The hand truck 20 includes a loading handle or grip 22 to assist a user when pushing or pulling on the hand truck to slide the lever arm 100 and glide 150 fore and aft. FIG. 18 illustrates a hand truck 20 coupled to the base 40 together with a glide 150 both of which are positioned near a vehicle hitch 100. When a user desires to connect the dolly to a vehicle, a user then inserts the glide into the vehicle hitch and pushes the glide 150 into the receiver 200. As the glide 150 is pushed into the receiver latch 158 actuates inward against the compression force of spring 160. As the glide is pushed inward into the receiver 200 the latch engages with hitch pin hole 204. The latch includes an angled end portion so that as the user continues to push the glide inward the latch pin 158 will push against the compression force of spring 160 and recede into the glide. The user continues to push the glide 150 inward until the compression buttons 156 engage within the holes 204 of the hitch receiver 200 (see FIGS. 19 and 20). Next the user tilts the hand truck 20 backwards (FIG.

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21) and moves the hand truck 20 towards the vehicle hitch 200. The dolly 20 is tilted enough so that the claw 122 is higher than the pivot pin 170 of the glide 150. Once the claw 122 is positioned over the pivot pin 170 the dolly is tilted forward until the claw engages the pivot pin (FIGS. 22 and 23). The dolly 20 is then pivoted or rotated upwards from the ground 16 until the lever arm 100 is in linear alignment with the glide 150 and hitch receiver 200 (FIGS. 24 and 25). Once in linear alignment the lock pin 60 is disengaged the user may apply a force against the hand truck to cause the lever arm 100 to slide into the base 40 (FIG. 26), the free end 120 slides into the receiver until the front stop 136 engages the receiver (FIG. 27) and the glide slides further into the receiver 200. As the slide 150 slides further into the receiver 200 the buttons 156 compress from the sliding force. The compression button 156 disengages from the hitch hole 204 and the glide slides further into the receiver until the front extension stop 136 of the lever arm 100 engages with the hitch receiver 200 (FIGS. 27 and 28). A hitch pin may then be inserted through hitch pin hole 204 and lever arm hole 130 to secure the hand truck vehicle mount assembly 10 to the vehicle.

When the user wishes to remove the dolly from the vehicle the hitch pin is removed and the user pulls the dolly away from the vehicle. As the glide is pulled from the receiver the compression buttons 156 remain compressed inward against the sidewalls and compress as they pass through the hitch pin hole 204. The glide continues to pull from the receiver until latch 158 engages the hitch pin hole 204 so that the glide does not pull completely from the receiver 200. The lock pin 60 on the base 40 is disengaged and the dolly 20 is further pulled outward and away from the vehicle so that the lever arm 100 extends to its extended position from the base 40. The dolly 20 and base 40 may then pivot downward until the wheels contact the ground 16. The dolly is then tilted further to disengage the claw 122 from the pivot pin 170.

FIGS. 29-31 further illustrates the use of the assembly to mount a load of a hand truck to a vehicle hitch receiver in conjunction with adjustable straps. Although a load on the base 40 and dolly 20 are not illustrated, those skilled in the art can appreciate how a load may be positioned on the base 40 and secured to the dolly 20 with straps 30 and hooks 32. Alternatively, ratchet straps may be used to assist a user when rotating the dolly and base upwards.

With reference to FIGS. 32-57 additional embodiments according to aspects of the invention will be described. With reference first to FIGS. 32-34 the hand truck and glide include additional features that significantly automated the lifting and loading of the hand truck onto the vehicle. The hand truck 20 includes spaced apart winches 28, a controller 80 and battery or power supply 98 fixed to the hand truck. The winches include load straps or cable 30 wound on a spool contained within a spool housing 94. The spool is rotated by a winch motor 96 to thereby extend or retract the cable through a cable outlet 92 formed in the housing 94. FIG. 45 further illustrates components of the winch 28.

The winch is powered by battery 98 and operation of the winches may be achieved with controller 80. The controller 80 includes a controller housing 90 in which a processor and circuitry are contained. The controller 80 is electrically coupled to the winches 28 and power supply 98. The controller includes a joystick 88 for user controlled operation of the lifting and loading of the hand truck to the vehicle 228. The controller also includes power switch 82, auto load activation switch 84, and quick stop switch 86. FIG. 46 further illustrates components of the controller 80.

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The lever arm 100 of the hand truck 20 couples to the auto glide or auto lift/load receiver 300 of the present invention. The lift load receiver 300 generally includes a main receiver housing 306, linear actuator or motor 314, glide carriage 330, and electronic controls 320. In certain embodiments according to aspects of the invention the lift load receiver 300 may include a drop down receiver 304 which couples to a receiver 232 of a vehicle 228 or may mount directly to the vehicle receiver frame 230 (as illustrated in FIG. 34). The electronic controls 320 includes a power switch 320, stop switch 322, and glide actuator switch 324. Voltage amplifier 328 is electrically coupled to the linear actuator 314 and electronic controls 320. FIGS. 47 and 48 further illustrate components of the lift load receiver 300.

With reference to FIGS. 35-42 the auto load sequence will next be described. The controller 80 on the hand truck 20 and electronic control 320 coupled to the auto lift/load receiver 300 are wirelessly synced to coordinate the lift and load operation. A user may activate the system with a suitable cell phone app or by activating switches on the controllers. Without limitation intended a bluetooth connection between the hand truck and the auto lift/load receiver uniquely communicates between single units avoiding cross talk from other nearby systems. When a user activates the load switch on the hand truck, the auto receiver will automatically load the hand truck to the back of a vehicle. That same single activation process will also unload the device from the back of the vehicle. Those skilled in the art will appreciate that the controllers may include multiple sensors and other electrical components such as proximity switches, piezo sensors, magnetic reed switches, device orientation sensors (for example, micromechanical gyroscopes or accelerometers and geomagnetic field sensors) to perform the various functions herein described. For example, output from sensors are processed to determine and align the 3-axis orientation of the hand truck with the 3-axis orientation of the lift load receiver attached to a vehicle. When the vehicle is on a slope, the controller compensates for both device orientations to align them together.

The user positions the hand truck 20 so that a free end of the lever arm 100 is below the pivot 338 of the receiver 300 (See FIGS. 36 and 36). The hand truck 20 is then tilted backward and moved forward so that the claw 122 of the lever arm 100 engages the pivot pin 338 of the glide carriage 330 of the receiver 300. The user spools out enough cable from each winch 28 so that each hook 32 may be engaged to the d-loops or other fixtures 34 fixed to the vehicle 228 (see FIGS. 37 and 38). The joystick 88 is then used to retract cables 30 so that a slight tension is on the cable. With the power switches 82 and 326 activated, the user may then initiate the load sequence by activating the auto load switch 84 on the controller 80. The controller 80 activates the winches 28 and retracts the cable in a controlled manner such that the hand truck remains in an upright orientation. As the cables 30 retract, claw 112 rotates about pivot 338 thereby raising the attached hand truck 20 into the air. The controller 100 and winches 28 continues to retract the cables 30 until the longitudinal axis of the lever arm is linearly aligned with the longitudinal axis of the lift load receiver 300 (see FIG. 39-40). Once the axes are aligned 366 the winches 28 are again activated while the linear actuator 314 is also activated. As will be described in greater detail below, activation of the linear actuator 314 pulls the hand truck 20 into locked engagement position 364 with the vehicle 228 (see FIGS. 41 and 42). The cable 30 may be further tightened so that the load is tight to the back of the vehicle. To unload the hand truck the controller controls the winches and linear

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actuator in reverse. Once the cart is removed from the glide, the movable glide retracts into the receiver to a storage position. When user wants to load the device, the user presses a button on glide to extend the glide from the 'storage' position 360 to an 'extended' position 362.

FIGS. 43 and 44 illustrate the use of block and tackle 36 and an alternative mounting of the winches 28 and d-loops 34 without departing from the scope of the present invention. FIGS. 49-51 further illustrates the orientation and alignment of the lever arm 100 relative to the lift/load receiver 300 during a load or unload sequence. During preload, the hand truck is tilted backwards. This causes an angled orientation of the lever arm 100 relative to the lift load receiver 300 (See FIG. 49). The lever arm includes a claw 122 that engages with the pivot pin 338 coupled to the glide carriage 330. Guide plates 340 assist a user to align the claw with the pin (See FIG. 50). Electronic controller communicates with the hand truck controller 80 and receives power from the voltage amplifier 328. Dust cover 308 restricts dust from entering the interior of the main receiver housing 308. Once the hand truck is lift or elevated into the load position the axis of the lever arm 80 and receiver 300 are aligned (see FIG. 51).

Referring next to FIGS. 52-55, the partial cross section of the receiver 300 cross section of autoloading sequence further illustrates the orientation and alignment of the lever arm 100 relative to the lift/load receiver 300 during a load or unload sequence. The receiver 300 includes main receiver housing 306. Glide carriage 330 slides or rolls within the receiver housing 306 along roller bearings 332. The glide carriage 330 is actuated within the receiver via linear actuator or stepper motor 314. The linear actuator is coupled to a lead drive screw 310. An enclosed end 334 of the glide carriage is coupled to a drive screw lead nut 312. Without limitation intended, the drive screw extends into a screw tube 318 to thereby allow approximately 6 inches of travel by the glide carriage 330 within the receiver housing 306. As the drive screw 310 turns within the lead nut 312 the glide is pushed or pulled within the housing. The electronic controller controls actuation of the linear actuator 314. Stop switch 322 allows a user to stop the actuation as desired. The actuate glide switch 324 may be activated to orient the glide so that the pivot pin 338 at the lever arm end 336 of the glide 330 extends slightly from the receiver housing 306. An engagement switch 344 is activated when the lever arm claw 122 engages the pivot pin 338. Sensor 346 further provides feedback to the controller regarding the operation of the lead screw 310. The glide 330 actuates between a storage position 36 (see FIG. 52), a load/unload position 362 (see FIGS. 53 and 54) and a locked and loaded position 364 (see FIG. 55).

Those skilled in the art will appreciate that the pin and claw arrangement may be reversed such that the free end of the lever arm 100 includes a pivot pin 126 and a free end 336 of the glide carriage 330 includes claw 342 (see FIGS. 56-57). Additional modifications to the various components of the invention may be altered without departing from the intended scope of the invention.

These and various other aspects and features of the invention are described with the intent to be illustrative, and not restrictive. This invention has been described herein with detail in order to comply with the patent statutes and to provide those skilled in the art with information needed to apply the novel principles and to construct and use such specialized components as are required. Further, those skilled in the art will recognize the components described and illustrated and will further appreciate other suitable

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components to perform the above-described features of the invention. It is to be understood, however, that the invention can be carried out by specifically different constructions, and that various modifications, both as to the construction and operating procedures, can be accomplished without departing from the scope of the invention. Further, in the appended claims, the transitional terms comprising and including are used in the open-ended sense in that elements in addition to those enumerated may also be present. Other examples will be apparent to those of skill in the art upon reviewing this document.

What is claimed is:

1. An apparatus for connecting a hand truck to a receiver mounted to a vehicle, the apparatus comprising:
 - a lever arm coupled to a hand truck and having a free end portion extending outward from the hand truck; the free end portion is adapted to be received within a receiver;
 - a glide that is separable from the lever arm and receiver, wherein the glide is adapted to be received within the receiver; and
 - wherein one of the glide and the lever arm includes a claw and wherein an other of the glide and the lever arm has a pivot member adapted to couple with the claw.
2. The apparatus as recited in claim 1, further including a base engaged to the hand truck wherein the lever arm engages the base, and further including an end portion of the lever arm that is slidably extendable from the base.
3. The apparatus as recited in claim 2, further including a lever lock coupled to the base and having a first locked position that engages the lever arm and having a second retracted position that disengages the lever arm from the lock.
4. The apparatus as recited in claim 1, further including wear plates attached to sides of the glide.
5. The apparatus as recited in claim 1, further including wear plates attached to sides of the lever arm.
6. The apparatus as recited in claim 1, wherein the glide further includes an actuatable latch having an end portion extending outward from a side of the glide.
7. An apparatus for connecting a hand truck to a receiver of a vehicle, the apparatus comprising:
 - a lever arm coupled to a base of a hand truck, the lever arm having a free end portion extending outward from the hand truck;
 - a receiver adapted for coupling to a vehicle;
 - a glide slidably engaged within the receiver;
 - a drive mechanism coupled to the glide to actuate the glide within the receiver between a loading position and a retracted position; and
 - wherein one of the glide and the lever arm includes a claw and wherein an other of the glide and the lever arm has a pivot member adapted to couple with the claw.
8. The apparatus as recited in claim 7, further including at least two spaced apart winches coupled to at least one of the hand truck and the vehicle.
9. The apparatus as recited in claim 8, further including a controller electrically linked with the winches and drive mechanism, wherein the controller provides a signal to control coordinated activation of the winches and drive mechanism.
10. The apparatus as recited in claim 9, further including sensors positioned on the hand truck and the receiver.
11. The apparatus as recited in claim 10, wherein the sensors provide an output that is capable of being used to correlate a three-dimensional orientation of the hand truck and the receiver relative to gravity.

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12. The apparatus as recited in claim **11**, wherein the output from the sensors is transmitted to the controller.

13. The apparatus as recited in claim **7**, wherein the drive mechanism includes a linear actuator.

14. The apparatus as recited in claim **13**, wherein the linear actuator includes a lead screw, lead nut, and stepper motor.

15. The apparatus as recited in claim **7**, wherein the glide includes roller bearings coupled to the glide.

16. An apparatus for connecting a hand truck to a receiver of a vehicle, the apparatus comprising:

a lever arm coupled to a base of a hand truck, the lever arm having a free end portion extending outward from the hand truck;

a receiver adapted for coupling to a vehicle;

a glide slidably engaged within the receiver;

a drive mechanism coupled to the glide to actuate the glide within the receiver between a loading position and a retracted position;

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wherein one of the glide and the lever arm includes a claw and wherein an other of the glide and the lever arm has a pivot member adapted to couple with the claw; at least two spaced apart winches coupled to at least one of the hand truck and the vehicle; and

a controller electrically linked with the winches and drive mechanism, wherein the controller provides a signal to control coordinated activation of the winches and the drive mechanism.

17. The apparatus as recited in claim **16**, further including sensors positioned on the hand truck and the receiver.

18. The apparatus as recited in claim **17**, wherein the sensors provide an output that is capable of being used to correlate a three-dimensional orientation of the hand truck and the receiver relative to gravity.

19. The apparatus as recited in claim **18**, wherein the output from the sensors is transmitted to the controller.

20. The apparatus as recited in claim **16**, wherein the drive mechanism includes a linear actuator.

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